CHEMICAL BONDING

Chemical bonding

Telegram app.
https://t.me/alk_sir

→ chemical bond is the force of attraction between

CHEMICAL BONDING

→ chemical bond is the force of attraction between constituent particles in a compound.

CHEMICAL BONDING

between constituent particles in a compound.

Types of chemical bond

- Ionic bond
- covalent / coordinate bond
- Metallic bond.

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Reason of bond formation :

① Lowering of P.E. of system :

→ when two atoms combine to form chemical bond, potential energy decreases and hence

bond, potential energy decreases and hence stability increases.

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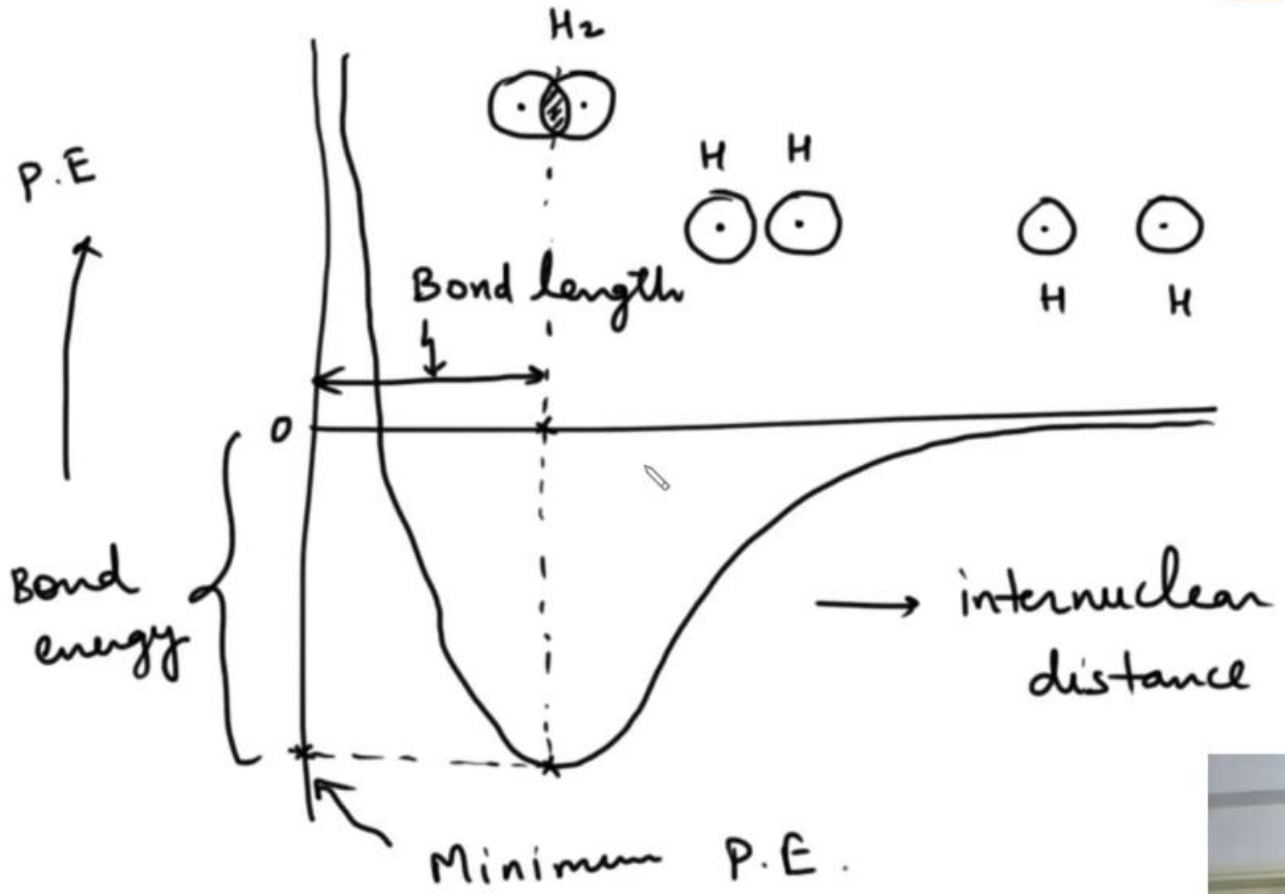

(2) To attain stable electronic configuration :

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(e.g Octet)

→ Elements lose, gain / share electrons to to attain stable E.C. similar to noble gases (octet)

Ionic bond (or Electrovalent bond):

→ Generally formed between a metal & non-metal

→ It constitutes ions occupying lattice points in the ionic crystal.

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→ Generally formed between a metal & non-metal

→ It constitutes ions occupying lattice points in the ionic crystal.

→ It is formed by losing e^- and by gaining electrons.

In the ionic crystal

→ Na^+ is formed by losing e^- and by gaining electrons.

e.g Na^+Cl^-

$Na \longrightarrow Na^+$

2,8,1 → 2,8 (Ne)

$Cl \longrightarrow Cl^-$

2,8,7 → 2,8,8 (Ar)

$Na \longrightarrow Na^{+}$

2,8,1 → 2,8 (Ne)

2,8,7

2,8,8 (A_2)

Covalent bond :

→ It is formed by sharing one, two or 3 pair of electrons

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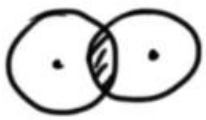
→ It is formed by sharing one, two or 3 pair of electrons.

→ It is generally formed between non metal and non metal.

e.g 

H_2

H—H

↑ single bond

or O=O

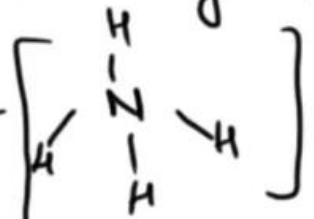
↑ double bond

CHEMICAL BONDING

⇒ **Coordinate bond**: (Dative bond)

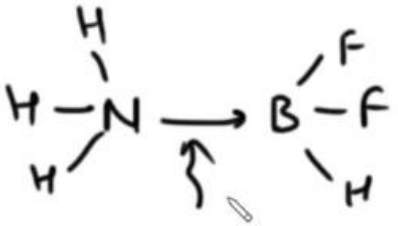
→ 9+ is formed by sharing a pair of electron between two atoms, but the pair of $\bar{e}$ shared is contributed by one species only.

e.g $H_3N: + H^+ \rightarrow NH_4^+$ or



donor (L.B) acceptor (L.A) ammonium ion

→ Lone pair donor is also called Lewis b
and L.P. acceptor is called Lewis acid.

e.g. $H_3N:$ + BF_3 ⟶ 

Donor (Lewis base)

Acceptor (Lewis acid)

Coordinate bond

→ Unshared pair of electron is also called Lone pair.

Donor (Lewis base) Acceptor (Lewis acid) Coordinate bond

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→ There is no difference in covalent bond and Coordinate bond wrt their bond properties (bond length & bond energy).

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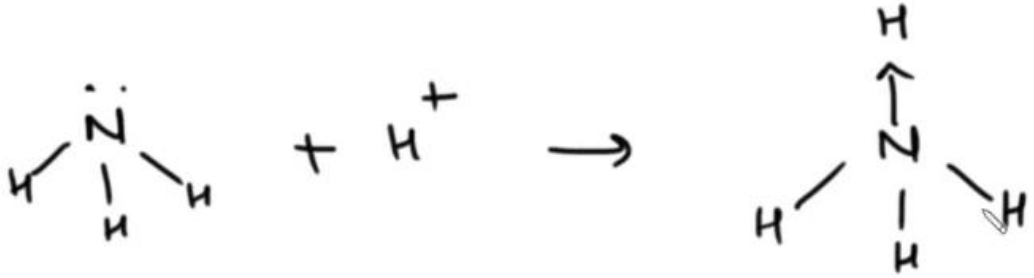
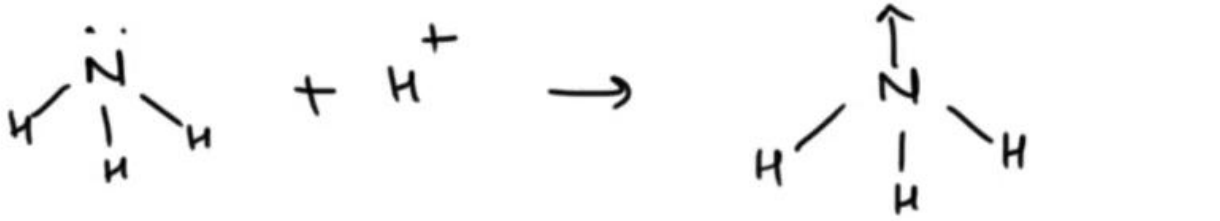
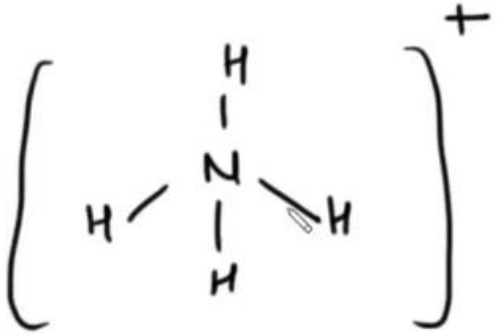




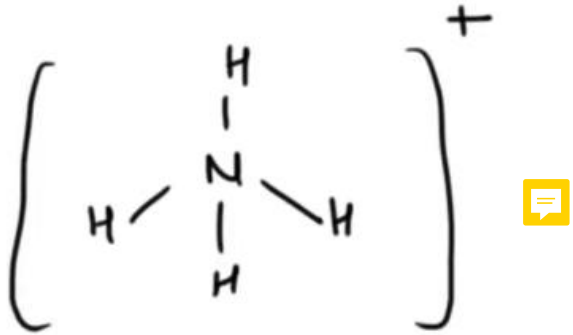
OR



Ammonium ion

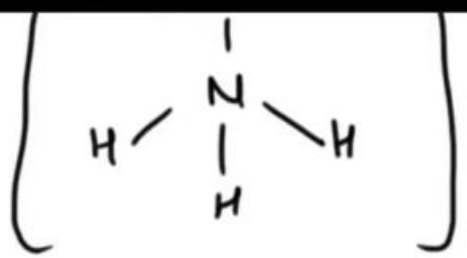
OR

* In NH_4^+, spectroscopic study suggest that all the 4 bonds (N–H) in NH_4^+ are equivalent wrt



Ammonium ion

Study suggest that all the 4 bonds (N-H) in NH_4^+ are equivalent wrt bond length and bond energy.



Ammonium ion

bond length and bond energy.

④ Metallic bond :

→ Form between metal and metal.

→ metal atoms occupy lattice points in crystal structure of metal and it is

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→ metal atoms occupy lattice points in crystal structure of metal and it is the electrostatic force of attraction between metal kernel and valence electrons (or mobile or free electrons)

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e.g. Na : $1s^2, 2s^2 2p^6, 3s^1$

This e^- is valence e^-.

mobile or free electrons)

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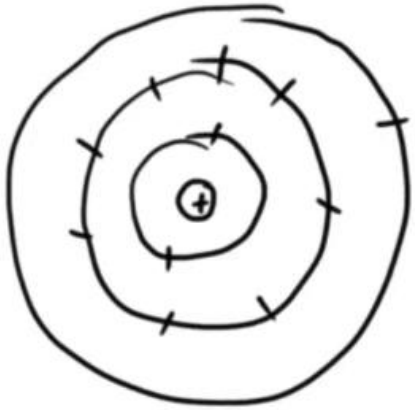
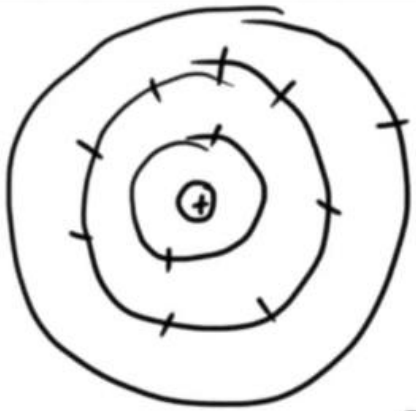


valence e^-

Na^+ is kernel



This e^- is valence e^-.

valence e^-

Na^+ is kernel

consisting of sodium nucleus and 10 inner electrons

e.g Mg : 2, 8, 2

two valence e^-.

Ionic bond (or Electrovalent bond)

→ formed between metals and non-metals by complete transfer of 1, 2 or 3 electrons forming ions which occupy lattice point in ionic crystal.

crystal.

→ It is the Electrostatic force of attraction between cations and anions in ionic lattice.

crystal.

→ It is the Electrostatic force of attraction between cations and anions in ionic lattice.

→ It is strong bond.

→ It is strong bond.

e.g. $K \longrightarrow K^+ + e^-$

$Cl + e^- \rightarrow Cl^-$

$K^+ Cl^-$

$Cl + e \rightarrow Cl^-$

$K^+ Cl^-$

→ Favourable conditions of formation of ~~bond~~ ionic bond :

→ Favourable conditions of formation of ~~co~~ ionic bond :

⇒ one of the element should be electropositive (i.e. having low ionisation energy i.e. metals)

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⇒ other element should be highly electro negative (i.e. high electronegativity i.e. non-metals)

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⇒ should have high EN difference between combining atoms.

non-metals)

⇒ should have high EN difference between combining atoms.

⇒ should have high lattice energy.

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e.g. of ionic compounds : NaCl, KCl, $BaSO_4$, $BaCl_2$, Na_2CO_3, Na_2SO_4 etc.

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⇒ Valency → it is the no. of bonds formed by an element in a compound.

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→ Electrovalency → i.e. no. of e^- lost or gained in formation of ionic solid.

an element in a compound.

→ Electrovalency → i.e. no. of e^- lost or gained.
in formation of ionic solid.

e.g. Na^+Cl^-

electrovalency of Na = 1
" " Cl = 1

e.g $Na^{+}Cl^{-}$

electrovalency of Na = 1

" " Cl = 1

e.g $BaSO_4$

2 2

Electrovalency

e.g Na^+Cl^-

electrovalency of Na = 1

" " Cl = 1

e.g $Al_2(SO_4)_3$

3 2

Al^{3+} SO_4^{2-}

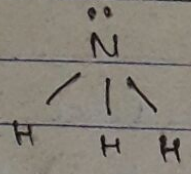
e.g $BaSO_4$

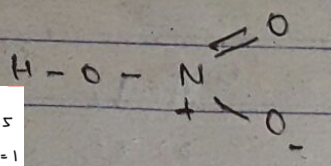
2 2

Electrovalency

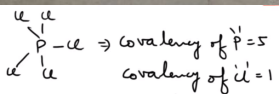
Covalency :-

- Applies for Covalent Compounds
- It is the no. of covalent bonds formed by an element in a covalent molecule.

eg. NH_3 ⇒  Covalency of 'N' = 3

HNO_3 ⇒  Covalency of 'N' = 4

O.N. of N = +5

eg. PCl_5 ⇒  ⇒ covalency of 'P' = 5

covalency of 'Cl' = 1

⇒ O.N of 'P' = +5

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* General properties of ionic Compounds :

1. Ionic bond is non-directional
2. Ionic Compounds have high M.P and B.P **(ionic bond is a strong bond)**
3. Ionic Compounds are brittle and hard.
4. Ionic compounds conduct electricity in molten state (or pure liquid form) or in solution phase.
5. Ionic Compounds are highly Soluble (in general) in polar solvents like water.

⇒ General Rule of solubility in a solvent,
" Like dissolves like"

⇒ Highly polar Compounds and ionic compounds dissolve better in polar solvents.

⇒ Non-polar compounds are better soluble in non-polar solvents like benzene, CCl_4, CS_2, ether etc.

eg. NH_3, HCl, NaCl, KCl, Na_2CO_3 etc are highly soluble in water (due to high di-electric constant of water)

⇒ di-electric constant of solvent ↑, solubility of polar or ionic compounds ↑.

⇒ water is one of the best solvent for ionic compounds as its di-electric constant is nearly 78.

eg. Na^+Cl^-, $F = \frac{K q_1 q_2}{D \times r^2}$

D = di-electric constant of medium.
= 1 for vacuum or air.
= 78 for water.

almost non-polar

⇒ eg. hydrocarbons (CH_4, C_2H_6, C_3H_8) Br_2, I_2 etc are better soluble in non-polar solvents like benzene or ether.

(6) Reactions of ionic compounds in solution phase are very fast (eg. acid-base neutralization, precipitation reaction)

(7) Ionic compounds do not show isomerism.
Infact they exhibit isomorphism and polymorphism

* Polymorphism :
When an ionic solid can exist in more than one crystal structures. Such structures are called polymorphism.

polymorphs

thus, CaCO3 and ZnS exhibit polymorphism

eg. ZnS → Zinc blende, Wurtzite

$CaCO_3$ → Calcite, Aragonite.

* Isomorphism:
Different Compounds showing same crystal structure.

eg. $FeSO_4 \cdot 7H_2O$, $MgSO_4 \cdot 7H_2O$, $ZnSO_4 \cdot 7H_2O$

Green ~~Blue~~ Vitriol — Epsom Salt — White Vitriol.

⇒ Vitriol is sulphate.
⇒ H_2SO_4 is called oil of vitriol.
eg. Blue Vitriol → $CuSO_4 \cdot 5H_2O$

* Lattice Enthalpy ($\Delta H_{L.E}$): or Lattice energy

It is the enthalpy change (or energy required) to convert 1 mol of ionic solid into its constituent ions.

eg. $NaCl(s) \longrightarrow Na^+(g) + Cl^-(g)$, $\Delta H_{L.E}(NaCl)$
1 mol

eg. $MgCl_2(s) \longrightarrow Mg^{2+}(g) + 2Cl^-(g)$, $\Delta H_{L.E}(MgCl_2)$

→ Factors affecting lattice Energy :-

$$L.E = P.E = \frac{k q_1 q_2}{r}$$

$$\Delta H_{L.E} \propto q_1 \times q_2$$

$$\propto \frac{1}{(r_+) + (r_-)}$$

eg. $LiCl > NaCl > KCl > RbCl > CsCl$

decreasing L.E →

eg. $MgCl_2 > CaCl_2 > SrCl_2 > BaCl_2$

decreasing L.E →

eg. $NaCl < MgCl_2 < AlCl_3$

increasing L.E →

⇒ lattice energy is indirectly determined using enthalpy of formation data.

* Standard enthalpy of formation :— ($\Delta H^\circ f$)

It is the enthalpy change when 1 mol of a substance is formed from its elements in their standard state at given temperature.

x Standard State :-

for an element is generally the most stable state (except phosphorous) at 1 bar pressure and specified temperature.

eg.	Element	Standard State at 298K
→	Iron	$Fe(s)$
→	Chlorine	$Cl_2(g)$
→	Hydrogen	$H_2(g)$
→	Oxygen	$O_2(g)$
→	Bromine	$Br_2(l)$
→	Iodine	$I_2(s)$
→	Carbon	C(graph)
→	sulphur	S(rhom)

→ Phosphorus — P(white)*

(Note that P(white) is not the most stable allotropic form.)

$P_{White} < P_{Red} < P_{Black}$

Increasing order of stability.

→ ΔH_f° of all elements in standard state is assigned zero.

⇒ $\Delta H_f^\circ [C(graph)] = 0$, $\Delta H_f^\circ [Cl_2(g)] = 0$

$\Delta H_f^\circ [Na(s)] = 0$, $\Delta H_f^\circ [Br_2(l)] = 0$

* Hess's law :-

If a process is performed in more than one ways, then enthalpy change in both the ways will be equal.

eg: $C(s) + O_2(g) \longrightarrow CO_2(g)$, $\Delta H = x$

$C(s) + \frac{1}{2} O_2(g) \longrightarrow CO(g)$, $\Delta H = x_1$

$CO(g) + \frac{1}{2} O_2(g) \longrightarrow CO_2(g)$, $\Delta H = x_2$

$C(s) + \square\, O_2(g) \longrightarrow CO_2(g)$, $\Delta H = (x_1 + x_2)$

from Hess's law $x = (x_1 + x_2)$

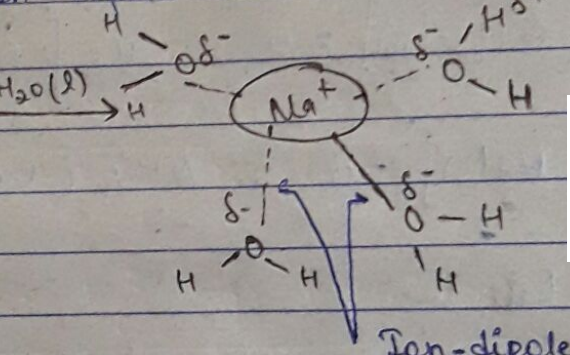
⇒ Parallel path alternate to convert reactants into products is called Born - Haber Cycle.

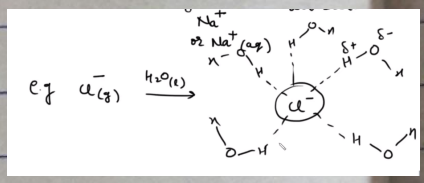
(*) Hydration enthalpy or hydration energy ($\Delta H_{H.E}$)
{for Ions}

→ It is the enthalpy change when 1 mol of gaseous ions are dissolved in excess of water to form aqueous (hydrated) ions. (always exothermic process).

→ When an ion (gaseous) is dissolved in polar solvent like water, water molecules cling around ion and solvate (or hydrates) it through ion-dipole attractions (exothermic process)

eg. $Na^+(g) \xrightarrow{H_2O(l)} Na^+(aq)$ or hydrated Na^+





→ The process is called Solvation if solvent is other than water.

* Factors affecting enthalpy of Solvation :-

for same charge comparison

(1.) Size of gaseous ion ↓, extent of hydration ↑. $|\Delta H_{H.E}|$ ↑

(2) Charge on gas ion ↑, extent of hydration ↑. $|\Delta H_{H.E}|$ ↑

(3) di-electric constant of solvent ↑, $\Delta H_{H.E}$ ↑ (i.e more energy released)

Application of enthalpy of hydration :

1. Size of hydrated ion :-

→ Size of hydrated ion $\propto \frac{1}{\text{size of gaseous ion}}$

→ Size of hydrated ion $\propto$ Charge on ion.

- $Li^+(g) < Na^+(g) < K^+(g) < Rb^+(g) < Cs^+(g)$

  ⟶ increasing gaseous size.

- $Li^+(aq) > Na^+(aq) > \ldots\ldots\ldots > Cs^+(aq)$

  ⟶ decreasing hydrated radius (or size)

→ $Be^{2+}(aq) > Mg^{2+}(aq) > \ldots\ldots\ldots Ba^{2+}(aq)$

⟶ decreasing hydrated size (or radius).

→ $Al^{3+}(aq) > Mg^{2+}(aq) > Na^+(aq)$

⟶ decreasing hydrated size

→ $F^-(aq) > Cl^-(aq) > Br^-(aq) > I^-(aq)$

⟶ decreasing hydrated size.

(2) Mobility of ion :-
It is the speed acquired by an ion in solvent under unit potential gradient.

$$\text{Mobility of ion} \propto \frac{1}{\text{hydrated size}}$$

- $Li^+(aq) < Na^+(aq) < K^+(aq) < Rb^+(aq) < Cs^+(aq)$
  (increasing mobility →)

- $Be^{2+}(aq) < Mg^{2+} < \ldots\ldots\ldots < Ba^{2+}(aq)$
  (increasing mobility →)

- $Al^{3+}(aq) < Mg^{2+}(aq) < Na^+(aq)$
  (increasing mobility →)

(3) Electrical Conductance of Solution :-
Electrical Conductance contributed by an 'ion' $\propto$ charge

$$\text{ion} \propto \frac{1}{\text{Hydrated size}}$$

→ For same charge, Hydrated radius ↓, Electrical Conductance ↑.

eg. $NaCl_{(aq)} < KCl_{(aq)}$ (↑ greater electrical Conductance.)

- $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
  (increasing ionic conductivity in water →)

Reducing Power of Lithium :-

→ Lithium is most powerful reducing metal (Not Caesium metal)

$Li(s) \longrightarrow Li^+(aq) + e^-$, $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

more -ve is ΔG° for a process or reaction, greater will be spontaneity or tendency to move in forward dir^n

$M(s) \longrightarrow M^+(aq) + e^-$, $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$M(s) \xrightarrow{\Delta H^\circ_{sub}} M(g) \xrightarrow{\Delta H^\circ_{IE_1}} M^+(g) \xrightarrow{\Delta H^\circ_{H.E}\,(M^+(g))} M^+(aq)$

$\Delta H^\circ = \Delta H^\circ_{sub} + \Delta H^\circ_{IE_1} + \Delta H^\circ_{H.E}\,(M^+(g))$

→ For Caesium (Cs)

$(\Delta H^\circ)_{Cs} = (\Delta H^\circ_{sub})_{Cs} + (\Delta H^\circ_{IE_1})_{Cs} + \Delta H^\circ_{H.E}\,(Cs^+(g))$

endo endo exo

→ For lithium (Li)

$(\Delta H^\circ)_{Li} = (\Delta H^\circ_{sub})_{Li} + (\Delta H^\circ_{IE_1})_{Li} + \Delta H^\circ_{H.E}\,(Li^+(g))$

→ $Li > Na > K > Rb > Cs$ (decreasing ΔH°_{sub})

$Li > Na > K > Rb > Cs$ (decreasing $\Delta H^\circ_{IE_1}$)

→ Both Sublimation energy and Ionisation energy support Cs to be more powerful reducing element. But hydration enthalpy of $Li^+(g)$ is very highly -ve (more -ve than that of $Cs^+(g)$) which is responsible for greater reducing power of Li.

$$\left|(\Delta H^\circ)\right|_{Li} > \left|\Delta H^\circ\right|_{Cs}$$

$$\Rightarrow \left|\Delta G^\circ\right|_{Li} > \left|\Delta G^\circ\right|_{Cs}$$

$$\Delta G^\circ = -2.303\,RT \log_{10}(K_{eq})$$

(ΔG°: more -ve; K_{eq}: higher value.)

⇒ Conclusion :-

Li is most powerful reducing metal (more than Cs) due to very high hydration energy (released) of small size Li^+ ion.

⇒ $Li > Cs > Rb > K > Na$

(decreasing order of reducing power)

 # Oxidising power of Fluorine:

$$\frac{1}{2}F_2(g) + e^- \longrightarrow F^-(aq) \; ; \; \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$\frac{1}{2} \times \Delta H_{B.E}(F-F)$

$F(g)$

ΔH_{eg_1}

$F^-(g)$

$\Delta H_{H.E}(F^-(g))$

Hess's law,

$$\Delta H^\circ = \frac{1}{2} \times B.E_{F-F} + \Delta H_{eg_1\,(F_{g1})} + \Delta H_{H.E\,(F^-(g))}$$

(-ve) is more desirable

Endo — exo — exo

→ ΔH_{eg_1} (Cl) is more -ve than that of F(g)

⇒ B.E of F_2 and H.E of $F^-(g)$ support for greater oxidizing power of F_2 (in comparison to Cl_2), therefore F_2 is most powerful oxidizing element in entire periodic Table.

⇒ In general, Bond energy $\propto \frac{1}{\text{Bond length}}$

exceptional:

Note that,

$$B.E_{Cl-Cl} > B.E_{Br-Br} > B.E_{F-F} > B.E_{I-I}$$

→ decreasing Bond Energy.

⇒ Greater oxidizing power of F_2 than Cl_2 is due to

(i) high H.E (liberated) of small F(g) ion.
(ii) low Bond dissociation energy of F_2.

(*) Solubility of Salts in Water :—

Solubility : It is the molar Concentration of a substance in water at Saturation equilibrium, at a given temperature.

Solubility is governed by —

(1) Relative variation in lattice energy and hydration energy (for salts of alkali and alkaline earth metals except halides of Be).

(2) Extent of Polarisation (Fajan's Rule)
(Salts of p-block, d-block and halides of Be)

Solubility of alkali and alkaline earth metal Salts :
(governed by relative variation of L.E and H.E)

⇒ High L.E indicates the strong ionic bond between ionic solid and hence lesser solubility.

$$L.E = P.E = \frac{k q_1 q_2}{r} \quad \Big| \quad L.E \propto \frac{1}{(r_+ + r_-)}$$

⇒ High H.E (released) of ions of ionic solid indicate greater solubility in water.

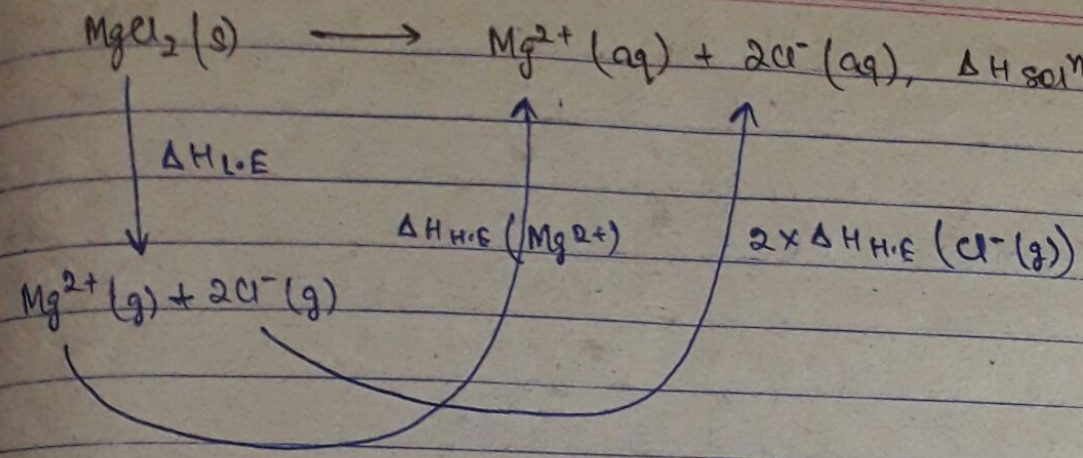
eg. $Na^+ Cl^-$

$$(H.E)_{Na^+} \propto \frac{1}{r_+}$$

$$(H.E)_{Cl^-} \propto \frac{1}{r_-}$$

(H.E) ionic Compound $\propto \frac{1}{r_+} + \frac{1}{r_-}$

Example



$\Delta H_{soln} = \Delta H_{L.E}(MgCl_2) + \Delta H_{H.E}(Mg^{2+}) + 2 \times \Delta H_{H.E}(Cl^-(g))$

-ve is desirable

endo — exo — exo

$\Delta G = \Delta H - T\Delta S$

$-ve = -ve + ve$

⇒ If a salt is soluble in water, then we generally say that
$L.E < H.E$

⇒ If salt is sparingly soluble (or almost insoluble) then
$L.E > H.E$

Case-I Keeping anion to be same, (varying metal ions). Also taking large size difference between ions (eg $r_- >>> r_+$ or $r_+ >>> r_-$)

eg. $r_- >>> r_+$ (large anions like I^-, SO_4^{2-}, CrO_4^{2-}, ClO_4^- etc)

$H.E \propto \frac{1}{r_+} + \frac{1}{r_-}$ | $L.E \propto \frac{1}{r_+ + r_-}$

Model Calculation :- In iodides of alkali metals LiI, NaI, KI, RbI, CsI

↑ highest L.E & H.E

eg. I^- or $r_- = 10A^\circ$

$r_+ = 1A^\circ \longrightarrow 2A^\circ$

% change in $\upsilon E = \dfrac{\frac{1}{1+10} - \frac{1}{2+10}}{\frac{1}{1+10}} \times 100$

$= \dfrac{100}{12} = 8.3\%$.

% change in H.E $= \dfrac{\left(\frac{1}{1} + \frac{1}{10}\right) - \left(\frac{1}{2} + \frac{1}{10}\right)}{\left(\frac{1}{1} + \frac{1}{10}\right)} \times 100$

$= \dfrac{1/2}{11/10} \times 100 = \dfrac{500}{11} = 44\%$ (approx)

⇒ In such case ($r_+ << r_-$ or vice versa)

→ decrease in H.E is more dominant (sharp) than the L.E

⇒ Solubility ↓ from Top to Bottom.

Case-II If size difference between cation and anion is small ($r_- \simeq r_+$)

eg. anion fixed (F^-)

Cations of alkali metals

LiF NaF KF

Model Calculation : $r_- = 1A^\circ$

$r_+ = 1A^\circ \longrightarrow 2A^\circ$

Change in L.E $= \frac{\left(\frac{1}{1+1} - \frac{1}{1+2}\right)}{\left(\frac{1}{1+1}\right)} \times 100 =$ 33.3%

Change in H.E $= \frac{\left(\frac{1}{1} + \frac{1}{1}\right) - \left(\frac{1}{1} + \frac{1}{2}\right)}{\left(\frac{1}{1} + \frac{1}{1}\right)} \times 100$

$= 25\%$

⇒ In such case (when size difference is small) ~~decreasing~~ decrease in L.E is more sharp and hence solubility ↑ from Top to Bottom.

In General, :
for salts of alkali and alkaline earth metals

⇒ Solubility is high if size difference is very large & solubility is low.
If size difference is small,

$NaI > KI > Rb > CSI$
(decreasing solubility →)

$BeSO_4 > MgSO_4 > CaSO_4 > SrSO_4 > BaSO_4$
(decreasing solubility →)

$LiF < NaF < KF$
(increasing solubility →)

* SUMMARY (of solubility of alkali and alkaline earth metals & salts)

(a) For alkali metal salts :-

(i) When $r_+ << r_-$: (i.e large anions)

SO_4^{2-}, ClO_4^-, BO_3^-, CrO_4^{2-}, I^- etc.

Solubility ↓ from top to Bottom.

$NaI > KI > RbI > CsI$

→ decreasing solubility

$LiClO_4 > NaClO_4 > KClO_4 > RbClO_4 > CsClO_4$

→

decreasing Solubility

Water soluble — Sparingly soluble.

(ii) When $r_+ \approx r_-$ (For small anion

eg F^-, OH^-, CO_3^{2-}, HCO_3^-)

In such case, Solubility ↑ from Top to Bottom.

$LiF < NaF < KF$

→

Increasing Solubility

Increasing Solubility

$LiOH < NaOH < KOH < RbOH < CsOH$

→

$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$

→

$LiHCO_3 < NaHCO_3 < KHCO_3 < RHCO_3 < CsHCO_3$

→

(b) For alkaline earth metal salts :-

(i) When $r_- >> r_+$ (i.e large anion)
(almost all anions except F^- or OH^-)

eg. SO_4^{2-}, SO_3^{2-}, ClO_4^-, CrO_4^{2-}, CO_3^{2-} etc.

Solubility ↓ from Top to Bottom.

$BeSO_4 > MgSO_4 > \ldots\ldots\ldots > BaSO_4$ ppt

$BeSO_3 > MgSO_3 > \ldots\ldots\ldots > BaSO_3$

$BeCrO_4 > MgCrO_4 > \ldots\ldots\ldots > BaCrO_4$ ppt

$BeCO_3 > MgCO_3 > CaCO_3 \ldots\ldots > BaCO_3$ ppt

↳ decreasing Solubility.

(ii) When $r_+ \cong r_-$ (i.e Small anions F^- and OH^-)

In such case (F^-, OH^-), solubility increase (↑) from Top to Bottom.

$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

$BeF_2 < MgF_2 < CaF_2 < SrF_2 < BaF_2 < BeF_2$

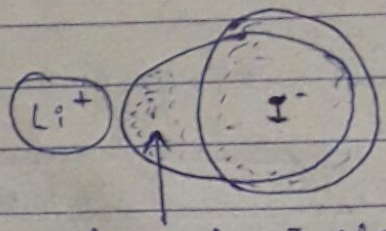
↳ Increasing Solubility.

Exception.

* BeF_2 is highly soluble in water, probably due to high Hydration Energy of small sized Be^{2+}.

(*) Polarisation in Ionic Compounds: (Fajan's Rule)

eg. LiI



distorted e^- cloud of large I^-

(covalent character is introduced in this ionic bond)

→ Due to small size of cation, Z_{eff} is considerably high (as in Li^+) and polarises (or disorts) outer e^- cloud of large anion, introducing covalent character in such ionic bond. This is called Fajan's Rule.

Extent of Polarisation depends on → Polarising Power of Cation
→ Polaraisability.

(a) Polarising Power of Cation :-

It is the ability of cation to polarise electron cloud of anion.

→ P

$$\text{P.P of cation} \propto \text{charge on cation}$$
$$\propto \frac{1}{\text{size of cation}}$$

cation

→ P.P also depends on electronic configuration. cations with pseudo inert conf^n. have greater P.P than noble gas conf^n. with nearly same size and same charge

Noble gas conf^n : $ns^2 np^6$ (eg. Na^+, Ca^{2+} etc)

Pseudo inert conf^n : $(n-1)s^2\ (n-1)p^6\ (n-1)d^{10}$
(eg. Cu^{2+}, Zn^{2+}, Ag^+, Cd^+, Hg^{2+} etc)

e.g Zn^{2+} : $3s^2 3p^6 3d^{10} \underline{4s^0}$
$(n-1)s^2 (n-1)p^6 (n-1)d^{10}$

Reason : Due to poor shielding of d and f - electron the cations with pseudo inert conf^n. have greater Z_{eff} & hence greater polarising power.
(provided charge & sizes are comparable).

Examples

$Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$ (→)

$Be^{2+} > Mg^{2+} > Ca^{2+} > Sr^{2+} > Ba^{2+}$ (→)

} decreasing polarising power.

$Al^{3+} > Mg^{2+} > Na^+$

$Ag^+ > K^+$ & $Ag^+ > Na^+$

$Hg^{2+} > Ca^{2+}$

due to pseudo inert conf^n.

(b) Polarisability of anion :-
Ability of anion to get easily polarised.

Polarisability of anion $\propto$ magnitude of charge on anion
$\propto$ size.

$I^- > Br^- > Cl^- > F^-$

$S^{2-} > O^{2-}$

$P^{3-} > S^{2-} > Cl^-$

$N^{3-} > O^{2-} > F^-$

} decreasing polarisability of anion.

* Application of Fajan's Rule :-

(1) Covalent character in ionic compound :

→ Extent of Polarisation ↑, Covalent Nature ↑

$LiCl < NaCl < KCl < RbCl < CsCl$ →

$BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$ →

} increasing ionic character

essentially covalent.

$AlCl_3 < MgCl_2 < NaCl$ →

↑ enough covalent

$AgCl > NaCl$ →

$HgCl_2 > CaCl_2$ →

$ZnSO_4 > CaSO_4$ →

} decreasing covalent Nature

more covalent

$AgF < AgCl < AgBr < AgI$ →

increasing covalent ~~character~~.

$MgF_2 < MgO < Mg_3N_2$ →

increasing covalent nature.

* Fajan's Rule

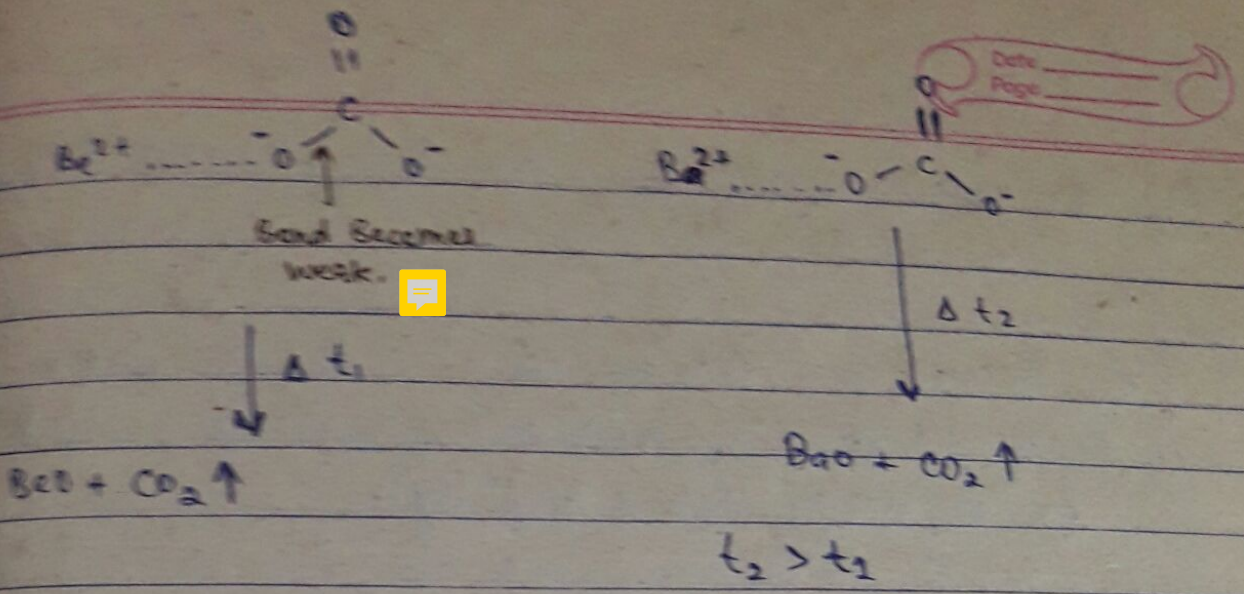
(2) Thermal Stability of ionic compounds :-

(CO_3^{2-}, HCO_3^-, SO_4^{2-}, OH^-, $C_2O_4^{2-}$, ClO_4^-, O_2^{2-} etc)

(a) for salt (ionic compounds) having polyatomic anions :

⇒ Extent of polarisation & hence covalent character is deciding factor.

⇒ Extent of Polarisation ↑, Covalent character ↑, Thermal stability ↓



$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$

increasing thermal stability

$Li_2CO_3 < Na_2CO_3 < K_2CO_3 < Rb_2CO_3 < Cs_2CO_3$

$Li_2O + CO_2 \uparrow$

do not appreciably liberate CO_2 on heating, rather they melt.

$Be(OH)_2 < Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$

$BeSO_4 < MgSO_4 < CaSO_4 < SrSO_4 < BaSO_4$

$LiHCO_3 < NaHCO_3 < KHCO_3 < RbHCO_3 < CsHCO_3$

Increasing thermal stability.

$BeC_2O_4 < MgC_2O_4 < CaC_2O_4 < SrC_2O_4 < BaC_2O_4$

Increasing thermal stability

$MgSO_4 \xrightarrow{\Delta} MgO + SO_3$

$SO_3 \xrightarrow{t > 800^\circ C} SO_2 + O_2 \uparrow$

(b) for ionic compounds of metals having monoatomic anion :-
(F^-, O^{2-}, H^- etc.)

⇒ In such case, lattice energy is deciding factor.
(Note that extent of polarisation is not very much significant for considering thermal stability).

⇒ L.E ↑, thermal stability ↑

Examples.

$LiF > NaF > KF > RbF > CsF$ →

$LiH > NaH > KH > RbH > CsH$ →

$MgO > CaO > SrO > BaO$ →

$Mg_3N_2 > Ca_3N_2 > Sr_3N_2 > Ba_3N_2$ →

} decreasing order of thermal stability

* Fajan's Rule

(3) Solubility of p-block | d-block salts and halides of Be :

⇒ In such salts, extent of polarisation is considerably high and will be the deciding factor for solubility in water.

⇒ Extent of Polarisation ↑, Covalent Character ↑, Solubility in Water ↓ , solubility in non-polar solvents will be high

$BeF_2 > BeCl_2 > BeBr_2 > BeI_2$ →

$AgF > AgCl > AgBr > AgI$ →

water Soluble — Sparingly Soluble.

$PbF_2 > PbCl_2 > PbBr_2 > PbI_2$

} decreasing order of Solubility

$HgCl_2 > HgBr_2 > HgI_2$ →

$NaCl > MgCl_2 > AlCl_3$ →

decreasing solubility in water

$Ag_2O > Ag_2S$
↑
More soluble

Note: However note that order of solubility in organic solvents (or in general non-polar eg. ether, benzene, CCl_4 solvents) is opposite.

∴ Covalent Nature ↑, Solubility in Ether / C_6H_6 etc ↑

eg LiCl > NaCl
↳ greater solubility in ether.

eg. $AlCl_3$ is soluble in ether also but NaCl is not.

④ Melting Point of metal Salts (mostly metal halides):

⇒ Melting point of ionic solids depend upon
(i) lattice energy
(ii) Extent of Polarisation (or covalent nature)

Ionic Solids (metal - non metals)

→ for F^- of s-block metals (except BeF_2) and Al^{3+} are considered very dominantly ionic and L.E is deciding factor

M.P ∝ L.E

for Cl^-, Br^-, I^- of Li^+ and all alkaline earth metals and Al^{3+} extent of polarisation is high and will be the deciding factor for M.P.

$$M.P \propto \frac{1}{\text{extent of polarisation.}}$$

Be^{2+} is extremely small and has very high polarising power.

only Be2+ is able to polarise F- to a large extent

Be^{2+} 0.31Å Al^{3+} 0.5Å

→ Cl^-, Br^-, I^- of Na^+, K^+, Rb^+, Cs^+ are ionic (very low P.P of such large cations)

thus, L.E will decide the melting point

eg. $LiF > LiCl > LiBr > LiI$
decreasing M.P →

eg. $AlF_3 > MgF_2 > NaF$ → But, $AlCl_3 < MgCl_2 < NaCl$ →
decreasing M.P by L.E
increasing M.P by polarisation

eg. $BeCl_2 < MgCl_2 < CaCl_2 < SrCl_2 < BaCl_2$
increasing M.P →

eg. $BeBr_2$ $MgBr_2$ $CaBr_2$ $SrBr_2$ $BaBr_2$ →
increasing M.P

Imp eg. LiCl NaCl KCl RbCl CsCl
IIT JEE MAIN
Polarisation. deciding factor is L.E

⇒ $NaCl > KCl > RbCl > CsCl > LiCl$
decreasing M.P →

Exception LiF NaF KF RbF CsF

⇒ $NaF > KF > LiF > RbF > CsF$
decreasing M.P →

Exception: Oxides of alkaline Earth metals :-

$MgO > CaO > BeO > SrO > BaO$
decreasing M.P →

extremely high M.P = 2850°C and therefore used as refractory material to make inner lining of blast furnace

Colour of ionic compounds :

⇒ Ionic compounds with high extent of polarisation (or high covalent character) are found to be deep coloured.

eg	$AgCl$	– white	$SnCl_4$	– white
	$AgBr$	– Pale Yellow	SnI_4	– Orange.
	AgI	– Bright Yellow		

ZnS	–	white	$PbCl_2$	– white
CdS	–	Yellow	PbI_2	– Yellow
HgS	–	Red and black		

(cinnabar)

Reason :

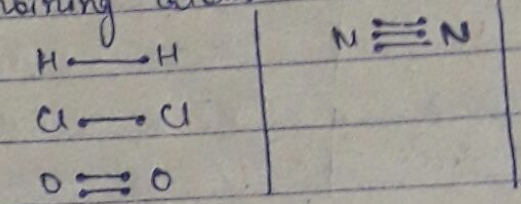
e.g AgI

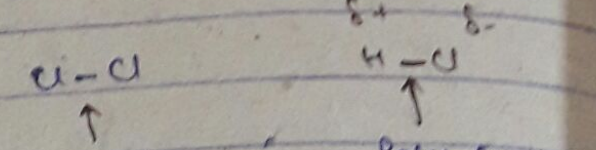
This loosely held $\bar{e}$ cloud of I^- is polarised by Ag^+ ion and becomes further away from I^- nucleus. This polarised $\bar{e}$ cloud absorb light of visible region (4000 Å to 7500 Å) and accordingly give colour.

Covalent Bond :

Covalent Bond is formed by sharing one, two, three pair of electrons between combining atoms.

Combining atoms.



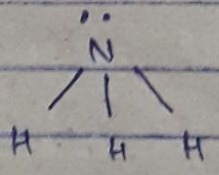


⇒ Favourable conditions for formation of covalent Bond;-

→ generally formed between non-metal and non-metal
→ Both the combining atoms should have high ionisation energy.
→ low or nil difference in EN.

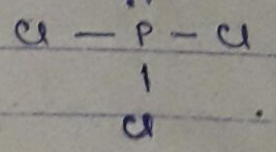
* Covalency : It is the number of covalent bonds formed by an atom in a given species.

eg. NH_3 ⇒



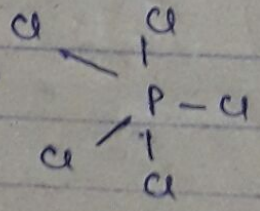
Covalency of N = 3
" " H = 1

eg. PCl_3



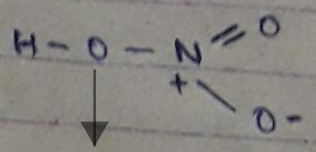
Covalency of P = 3

PCl_5



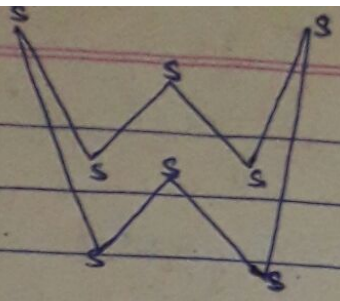
Covalency of P = 5

eg. HNO_3,



Covalency of N = 4

covalency of O = '2'



Covalency of 's' = 2

O.N of 's' = 0

* General characteristics (or properties) of covalent Compounds :-

(1) Covalent Bond is directional. It is formed by overlapping of atomic orbitals (or hybrid orbitals) which are directional. (eg. p_x, p_y, p_z have their electron density oriented in x, y and z direction respectively).

(2) Covalent compounds generally have low M.P and B.P except the covalently bonded network solids which have exceptionally high M.P and B.P

eg. CO_2 ⇒ O = C = O ..... O = C = O

($\delta-$ on O, $\delta+$ on C)

weak vanderwaal's forces are broken when we melt or boil a covalent compound.

(No breakage of covalent bond).

→ Covalently bonded network solids

eg. diamond, graphite, SiO_2, SiC etc.

exceptionally high MP & BP

because here we will have to break all covalent bonds to melt or to boil.

(3) Reactions involving Covalent Compounds are relatively slow.

(4) Covalent compounds are generally bad conductor of electricity.

Exception: Graphite conducts electricity along the sheet due to delocalisation of π - electrons.

(5) Covalent Compounds are more soluble in non-polar solvents (eg. Hydrocarbons etc are more soluble in benzene, CCl_4, CS_2 ether etc).

(6) Covalent compounds may show isomerism.

(*) Lewis Dot structure :- (LDS)

→ It represents the distribution of valence electrons of various atoms in a molecule in the form of covalent bonds / ionic bonds and non-bonded electrons.

→ LDS does not give any information about geometry (or shape) of any compound.

* Guidelines to draw LDS of simple molecules :-

(1) Valence e^- should be shown as bonds and non-bonded electrons as dots.

eg. an ionic compound,

eg. NaCl ⇒ Na^+ $:\ddot{Cl}:^-$

L.D.S of NaCl

eg. BaF_2 ⇒ $:\ddot{F}:^-$ Ba^{2+} $:\ddot{F}:^-$ (L.D.S of BaF_2)

* L.D.S of covalent Molecules :-

→ Covalent bonds shall be shown by solid lines and non-bonded e^- as dots.

– (single bond)
= (double bond)
≡ (triple bond)

– (sing ≡ (triple bond)

≡ (triple bond)

→ Knowledge of covalency is important.

Procedure : (for polyatomic covalent molecules)

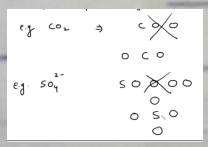
(i) Identify the central atom. Usually central atom is the least EN atom with covalency greater than 1.
⇒ 'F' does not act as central atom (happens to be at terminal positions).

* 'H' also usually does not act as central atom, except in some cases where 3c-2e⁻ bonds form (Banana Bonding).

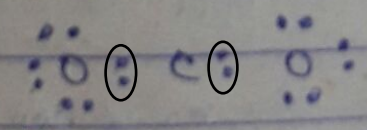
eg. H_2SO_4, HNO_3, CO_2, SO_2, PCl_3

* There may be more than one central atom in a molecule.
eg. $H_2S_2O_7$, $H_4P_2O_7$, $Na_2S_2O_3$, $Na_2S_4O_6$

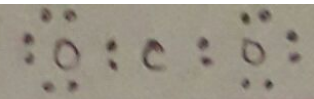
(ii) Guess the skeleton. Usually most symmetrical skeleton is the correct one, however it has to be verified experimentally



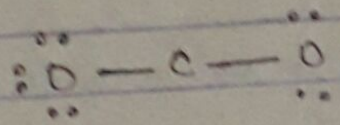
(iii) Count the total no. of valence electrons. Distribute one pair of e⁻ between each atom pair involving central atom.
eg. CO_2 ⇒ Total valence e⁻ = 4 + 12 = 16
− 4
12



(iv) Complete the octet of terminal atoms and place extra remaining e⁻ (if any) on central atom.



This gives one of the possible LDS



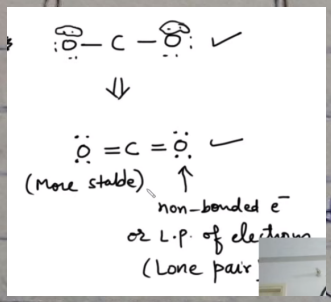
(A possible LDS of CO_2)

Unstable

(v) Try to complete octet of central atom too by making multiple bonds (double / triple bonds) if required. Out of various possible structures, the most plausible (or most suitable or stable) structure is the one in which formal charges are closest to zero.

To satisfy formal charges closest to zero, central atom may have incomplete octet or expanded octet.

Imp Note: Octet can be expanded only in case of 3rd period and higher period elements. { Remember, 2nd period elements can't expand their octet as they have only 4 orbitals in valence shell.)



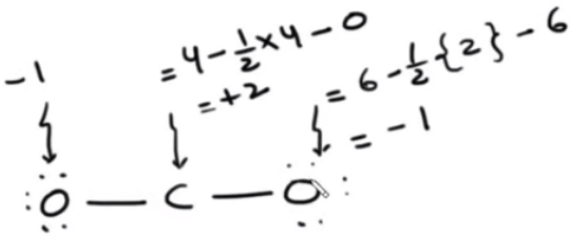
(More stable)

e⁻ or L.P of Lone Pair)

Formal Charge

On an atom in a species = No of Valence e⁻ in free atom
$- \frac{1}{2}$ (total no. of e⁻ involved in bonds)
- no. of non-bonded valence e⁻

→ Sum of formal charges on all the atoms in a species is equal to the total actual charge present on the species.



I (Unstable)

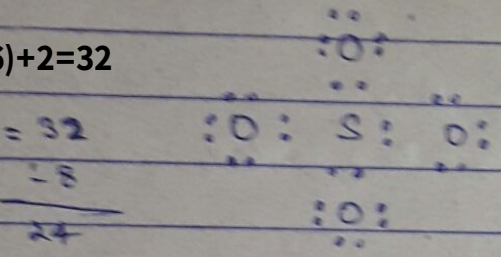
II (Most plausible LDS of CO_2)

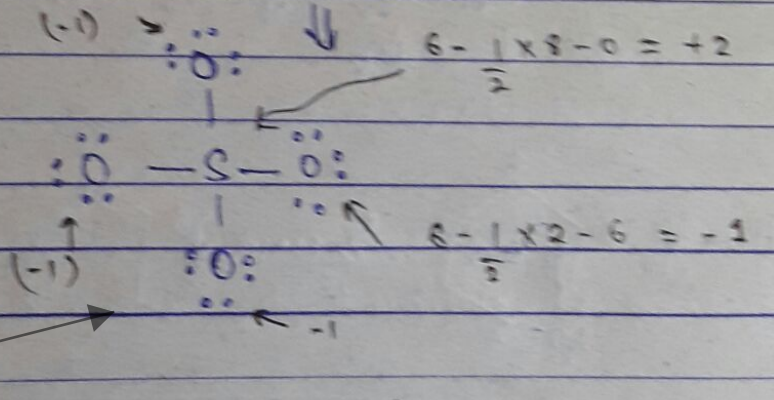
(H_2SO_4) Sulphuric acid. | SO_4^{2-} (oxoanion) (Sulphate ion)

Take SO_4^{2-}

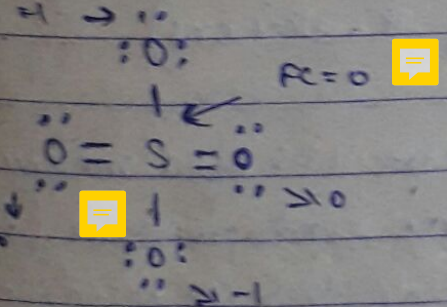
5(6)+2=32

Total valence e^- = 32
-8
24



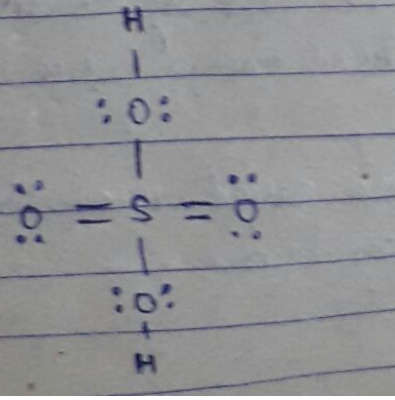


I. (a valid LDS but Unstable)

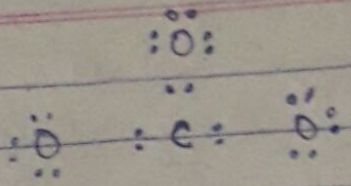


(Most plausible LDS of SO_4^{2-})

eg. H_2SO_4



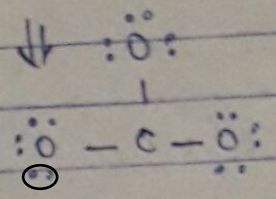
eg. CO_3^{2-}

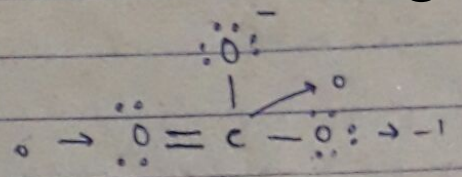


Total valence e^-

$= 4 + 18 + 2$

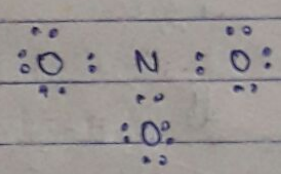
$= 24$

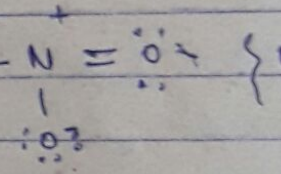




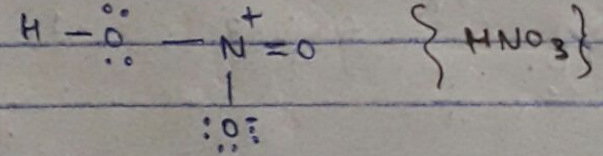
(Most stable L.D.S of CO_3^{2-})

eg. $HNO_3 \rightarrow NO_3^-$ Total valence e^- = 24 − 6 = 18

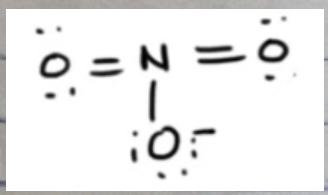




{Most stable str. of NO_3^-)



{HNO_3}



* Resonance: (Delocalisation of π - electrons)

⇒ When we can draw two or more LDS of a molecule / ion and none of the LDS properly explains bond properties (i.e bond length and bond energy), we say there is resonance.

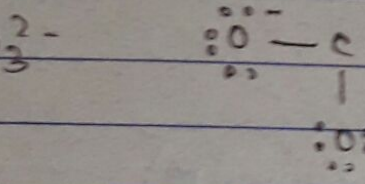
⇒ These LDS are called resonating Structures (RS) or Canonical Structures.

⇒ Actual species may be considered the hybrid of these resonating structures and is viewed as Resonance hybrid (RH).

⇒ Resonance is due to delocalisation of π electrons between 3 or more atoms. π - electrons are relatively loosely held with the nucleus and hence easily delocalised.

⇒ Most stable Resonating Structures contributes maximum towards resonance hybrid (RH)

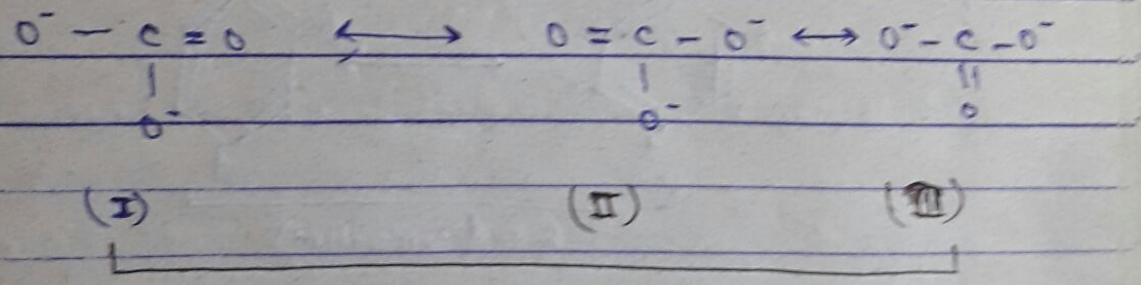
eg: CO_3^{2-}



(LDS)

⇒ Experimentally all the three C-O bonds in CO_3^{2-} ion are found to be identical w.r.t bond length and bond energy.

⇒ So above LDS does not explain the properties of CO_3^{2-}

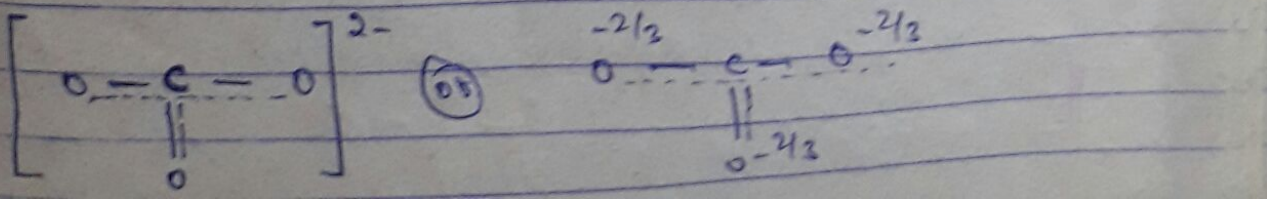


(I) (II) (III)

Three R.S of carbonate ion.

(All three R.S are equivalent or degenerate R.S)

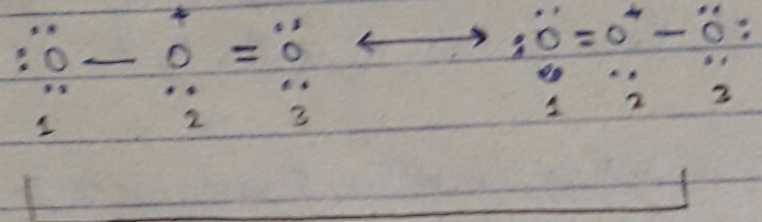
⇒ All three R.S contribute equally towards R.H



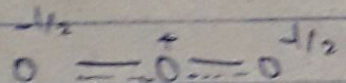
resonance hybrid

This resonance hybrid of CO_3^{2-} shows all three C-O bonds being identical (and thus actually is found for CO_3^{2-})

eg. Ozone (O_3):



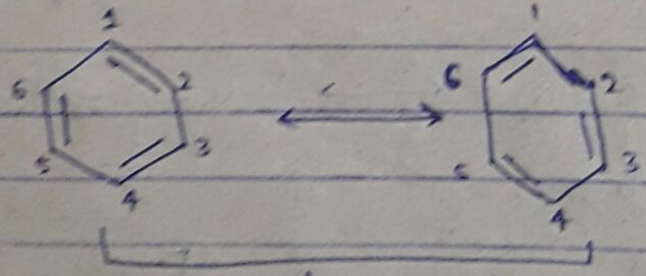
{ Degenerate R.S }



R.H (Both O-O bonds are equivalent)

eg. Benzene (C_6H_6)

- Kekule Structure -



R.S (Equivalent)

Resonance hybrid =

⇒ Delocalisation of π-electrons occur when

(1) Molecule should be linear planar or nearly planar.

(2) π-bonds in conjugation undergo delocalisation.

Bond Order → provides the feel of no. of bonds between two atoms.

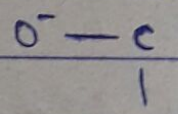
→ Single Bond : Bond order = 1
Double Bond : BO = 2
Triple Bond : Bo = 3

Bond order $\propto \frac{1}{\text{Bond length}}$

Bond Order = $\frac{\text{Total no. of bonds}}{\text{No. of atoms pairs between which these bonds are distributed.}}$ (In degenerate R.S)

Applies when dealing with degenerate R.S,
= total no. of bonds/ no. of equivalent R.S

eg. CO_3^{2-}

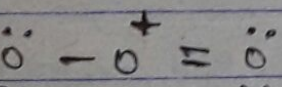


R.S

BO (in CO_3^{2-}) = $\frac{4}{3} = 1.33$

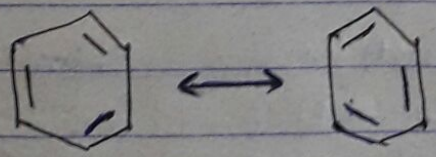
⇒ In CO_3^{2-} C-O bond is stronger than single bond but weaker than the double bond.

eg O_3 :



B.O = $\frac{3}{2} = 1.5$

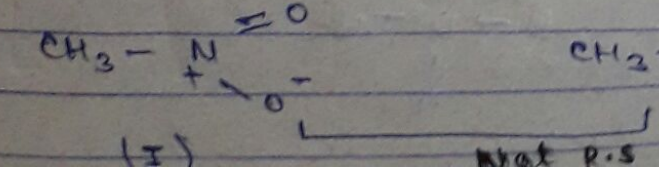
eg. C_6H_6



B.O of C_6H_6 = $\frac{9}{6} = 1.5$

→ Valid R.S should have same connectivity of atoms

eg.



(I) not R.S

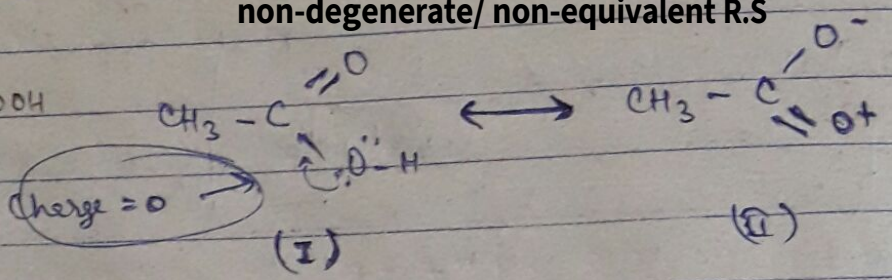
not R.S (since connectivity of atoms is not same)
these are functional isomers

→ for valid R.S, total no. of valence e⁻ and total no of odd e⁻ (or unpaired e⁻) must be same.

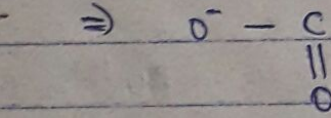
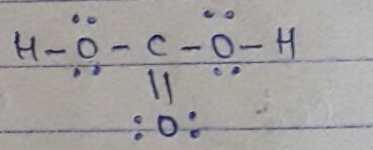
→ Out of various valid R.S, the one where formal charge are closest to zero is most stable and hence contributing R.S towards RH

non-degenerate/ non-equivalent R.S

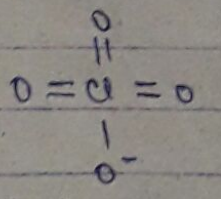
eg. CH_3COOH



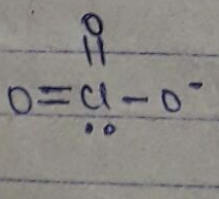
(I) is most contributing R.S for CH_3COOH

eg. CO_3^{2-} ⇒  ⇒ 

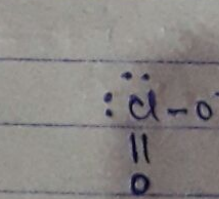
Carbonic Acid (H_2CO_3)

• $HClO_4$ → ClO_4^- 

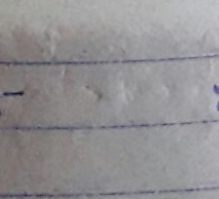
Perchlorate ion

• $HClO_3$ → ClO_3^- 

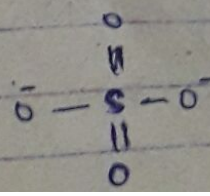
Chlorate ion

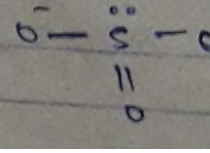
• $HClO_2$ → ClO_2^- 

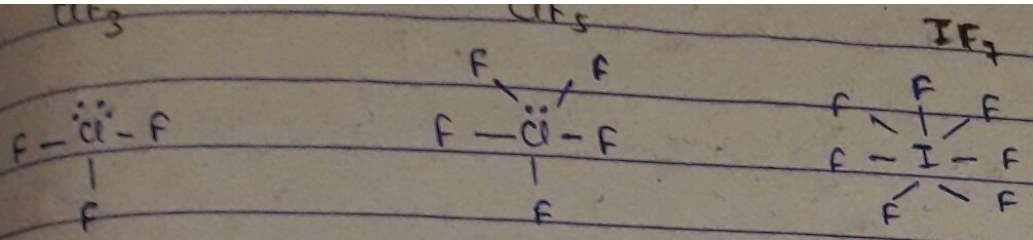
Chlorite ion

• HClO → ClO^- 

Hypochlorite ion

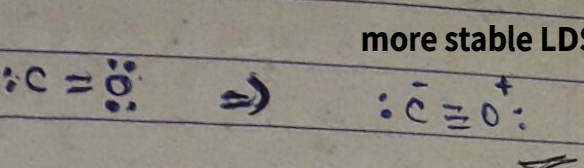
eg. $H_2SO_4 \rightarrow SO_4^{2-}$ 

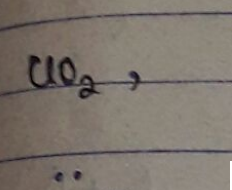
$H_2SO_3 \rightarrow SO_3^{2-}$ 



more stable LDS

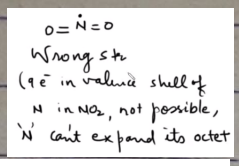
CO (carbon monoxide)



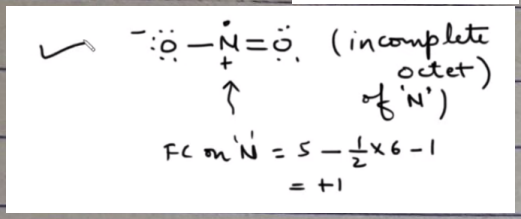


11 e- in valence shell of Cl

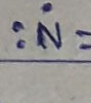
NO_2



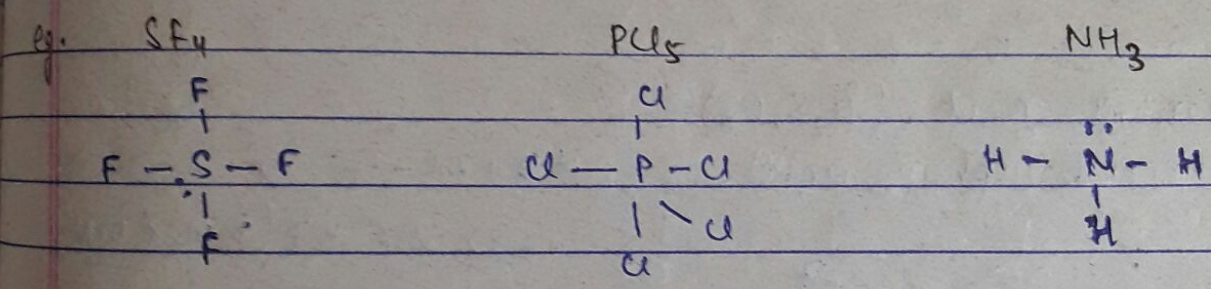
Wrong str. (9 e⁻ in valence shell of N in NO_2, not possible, 'N' can't expand its octet)



NO { odd electron molecules (Paramagnetic) }



(Incomplete octet of 'N')



① L.D.S of oxyacids with 2 or more central atoms :-

If calculated O.N < Max O.N of central atom

↳ X – X (where X → central atom)

↳ exception in case of oxyacids of 'P'.

If calculated O.N = Max O.N of central atom

X – O – X

If calculated O.N > Max. O.N of central atom

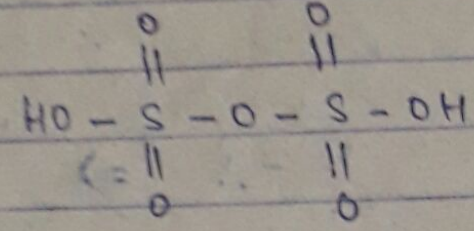
X – O – O – X (for 2 central atoms)

X – O – O⁻ (for 1 central atom)

peroxy linkage

eg. Pyrosulphuric acid ($H_2S_2O_7$ (olium))

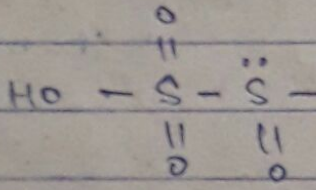
$+2 + 2x - 14 = 0 \Rightarrow x = +6$ (= Max O.N of 'S')

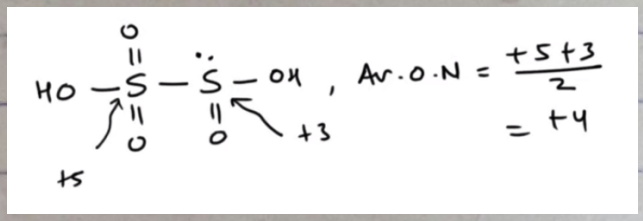


(Dibasic acid)

eg. Pyrosulphurous acid ($H_2S_2O_5$)

$+2 + 2x - 10 = 0 \Rightarrow x = +4$ (< Max. O.N of 'S')

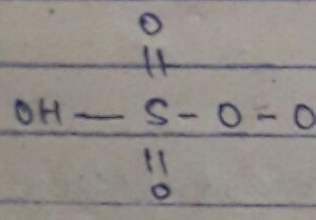




eg. Marshall's acid ($H_2S_2O_8$)
(Peroxydisulphuric acid)

$+2 + 2x - 16 = 0$

$\Rightarrow x = +7$ (> Max O.N)

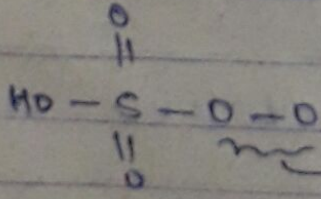


(a powerful oxidising agent)

eg. Caro's acid (H_2SO_5)
(Peroxymonosulphuric acid)

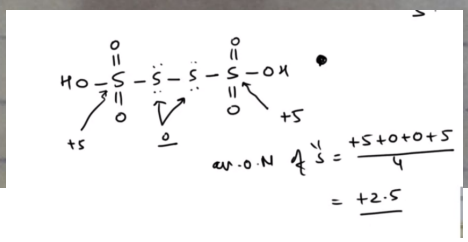
$+2 + x - 10 = 0$

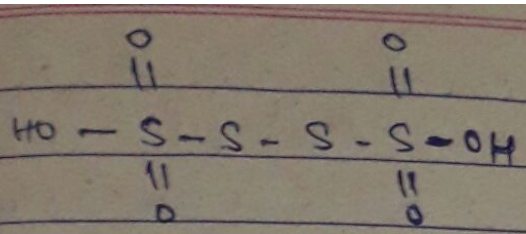
$\Rightarrow x = +8$ (> Max O.N of S)



eg. → Polythionic acid ($H_2S_{n+2}O_6$)

• Tetrathionic acid ($H_2S_4O_6$)

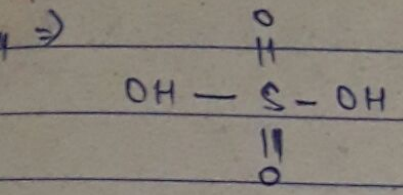


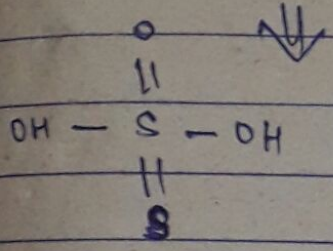


R–OH	Alcohol
R–SH	Thioalcohol
R–OR	Ether
R–S–R	Thioether

eg. Thiosulphuric acid ($H_2S_2O_3$)

H_2SO_4 ⇒





thiosulphuric acid

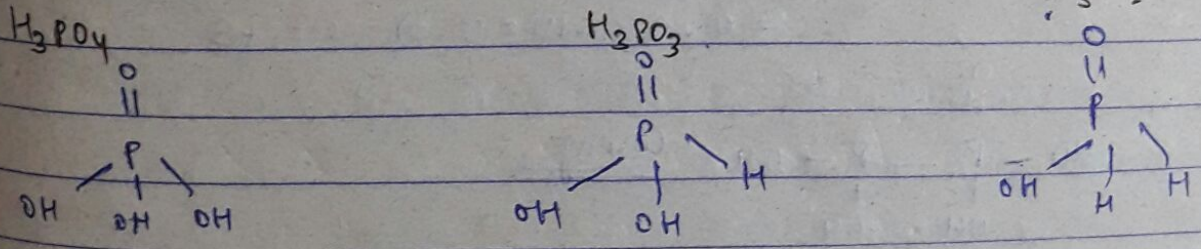
(*) Oxyacids of Phosphorus (P):

H_3PO_4 H_3PO_3 H_3PO_2

⇒ In LDS of oxyacids of 'P'

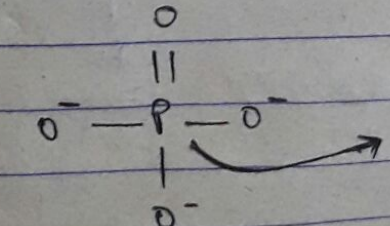
→ Coordination No. or Steric No. of 'P' can't be more than '4'

→ Due to this one P=O must be present.



Tribasic acid Dibasic acid Monobasic acid.

eg. Pg. PO_4^{3-} ⇒

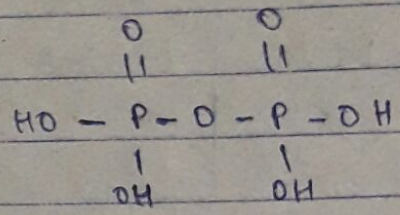


$B.O = \frac{5}{4} = 1.25$

eg. Pyrophosphoric acid or diphosphoric acid

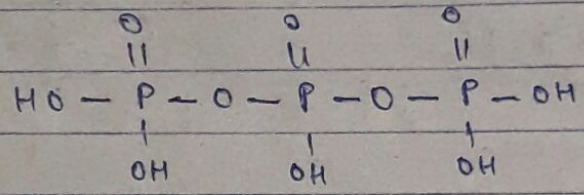
$(H_4P_2O_7)$

$4 + 2x - 14 = 0 \Rightarrow x = +5$ (= Max O.N of 'P')

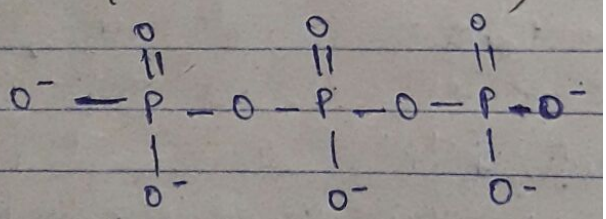


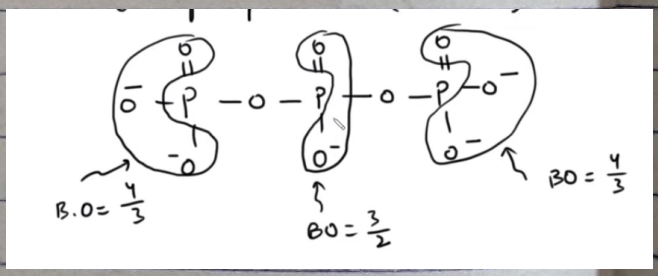
Tetrabasic acid

eg. Triphosphoric acid $(H_5P_3O_{10})$



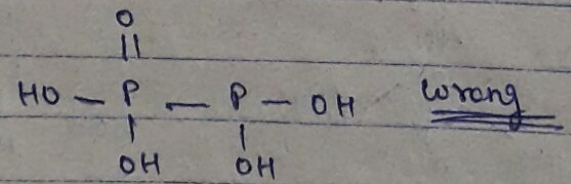
Triphosphate ion $(P_3O_{10}^{5-})$





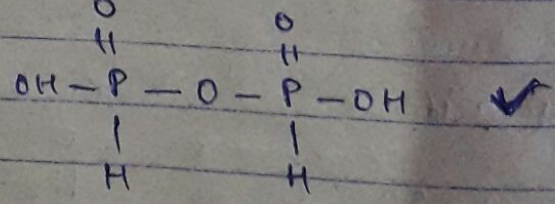
Exceptions: Pyrophosphorous acid :- $(H_4P_2O_5)$

$+4 + 2x - 10 = 0 \Rightarrow x = +3$

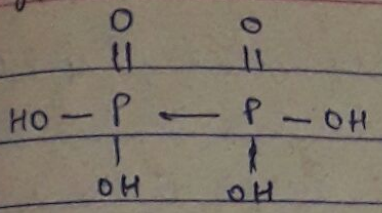


wrong

Correct structure

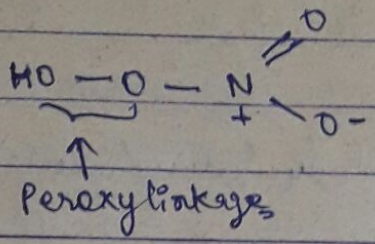


⑥ Hypophosphoric acid : ($H_4P_2O_6$)



⑦ Pernitric acid (HNO_4)

$+1+x-8=0 \Rightarrow x=+7$ (> Max O.N of N)



⑧ Theories of Covalent Bond formation :-

1. Octet Rule
2. Valence Bond Theory (VBT)
3. VBT + Hybridisation. (VSEPR)
4. Molecular orbital theory (MOT)

① Octet Rule : Elements try to chase electronic configuration of noble gases by sharing 1,2,3 electrons to have octet in valence shell.

eg. Compounds which follow octet Rule :

NH_3, PH_3, H_2S, HCl, NCl_3, PCl_3, CH_4, CCl_4, $SiCl_4$ etc.

Exceptions :

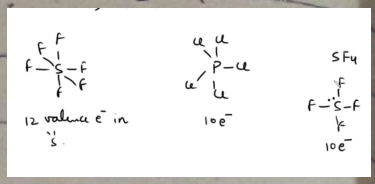
① Incomplete octet : odd e- molecule (NO, NO2)

eg. BeF_2, $BeCl_2$, BCl_3, BF_3, NO, NO_2 etc.

⇒ All these molecules where octet is incomplete, are called hypovalent Molecules.

② Expanded Octet : (Hypervalent Molecules) / Super Octet

eg. SF_6, PCl_5, IF_7, SF_4, ClF_3, ClO_2, ClF_5, H_2SO_4, H_3PO_4 etc.



 # Odd e- molecule :

⇒ NO, NO_2 form dimer on cooling.

$ON\cdot + \cdot NO \longrightarrow O=\ddot{N}-\ddot{N}=O$

(Paramagnetic) — dimer of NO (diamagnetic)

$\dot{N}O_2 + \dot{N}O_2 \longrightarrow N_2O_4$ (diamagnetic)

(Paramagnetic)

⇒ ClO_2 does not form dimer probably because odd e^- of Cl is delocalised in d-orbital Cl-atom.

Note: Expanded octet can occur only in 3rd & higher period elements.

② Valence Bond Theory (VBT)

→ According to this theory, covalent bond formation occurs when appropriately oriented atomic orbitals of combining atoms undergo overlapping.

→ Greater is the extent of overlapping, greater will be the strength of bond formed.

* The atomic orbitals undergoing should have same or nearly same energy and 1 e^- each with opposite spins.

* A coordinate bond is formed by overlapping of atomic orbitals in which one is fully filled and the other is vacant.

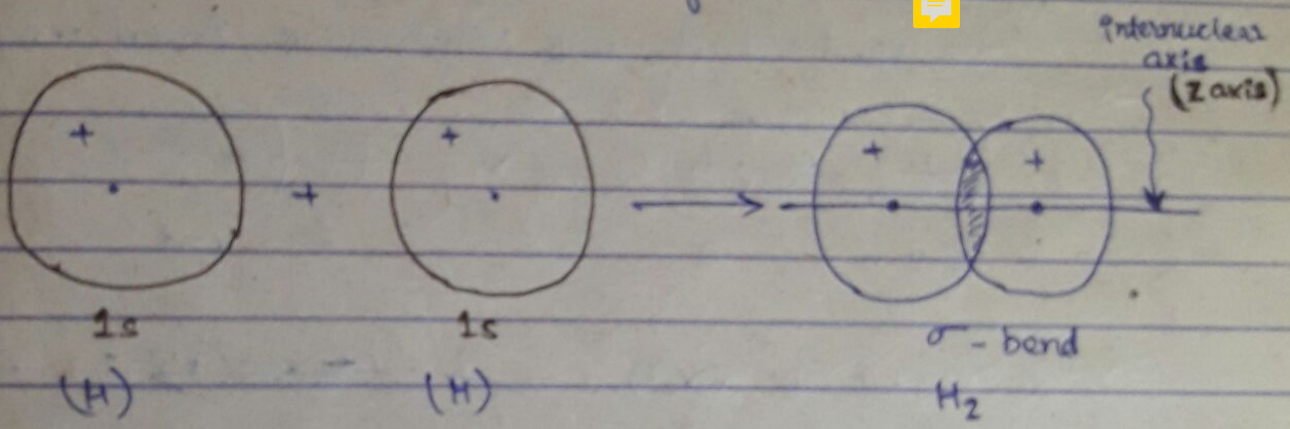
{ Note that, properties wise, there is difference between covalent and coordinate bonds. }

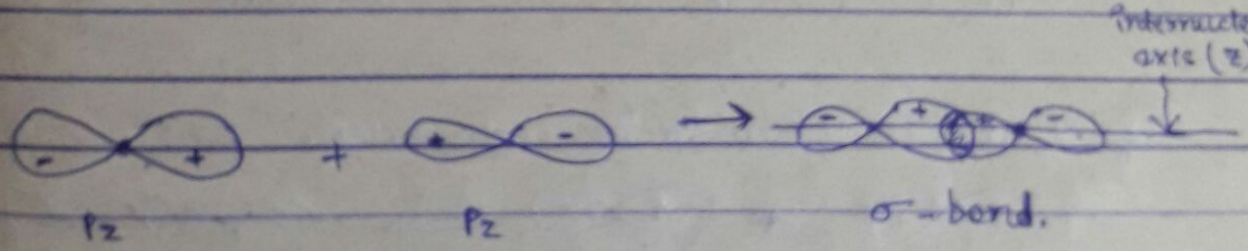
(*) Types of overlapping –

1. Axial or Head-on overlapping :

In this, overlapped e^- cloud intersects the internuclear axis. This results into σ-bond formation.

eg:





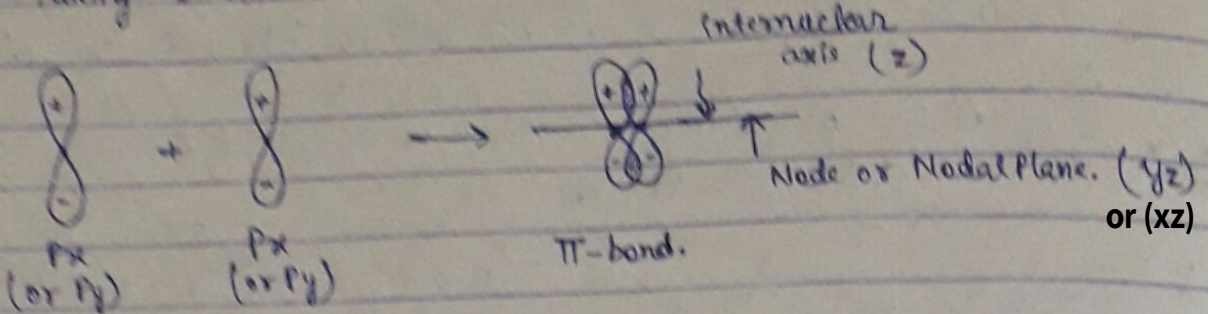
* These sign (+, -) represent sign of wave function (Ψ)

2. Lateral (or Sideways) overlapping :

In this, overlapped e^- cloud does not intersect the internuclear axis, rather e^- cloud (overlapped becomes parallel to internuclear axis.

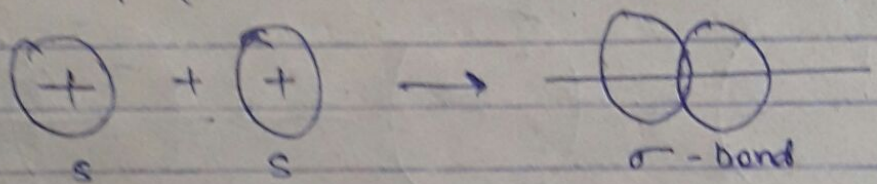
This results into π-bond formation.

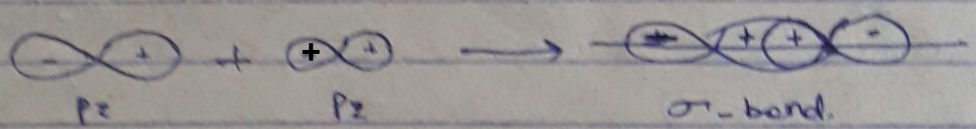
eg. Taking z-axis to be internuclear axis.



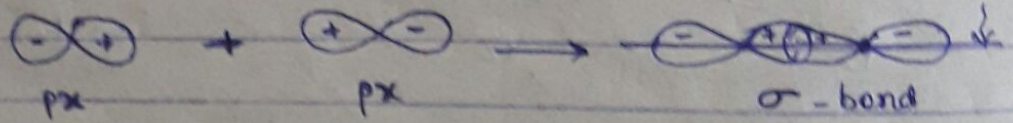
Note:

(1) Only positive overlapping results into bond formation.
→ +ve overlapping means same sign wave functions should overlap. (results into constructive Interference) (CI)



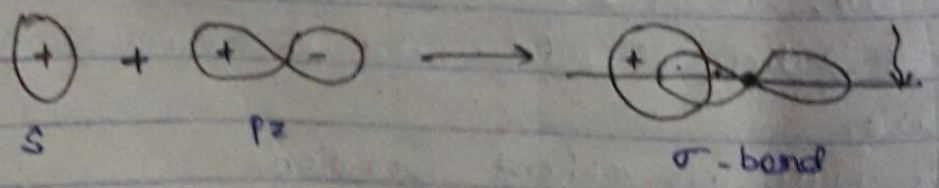


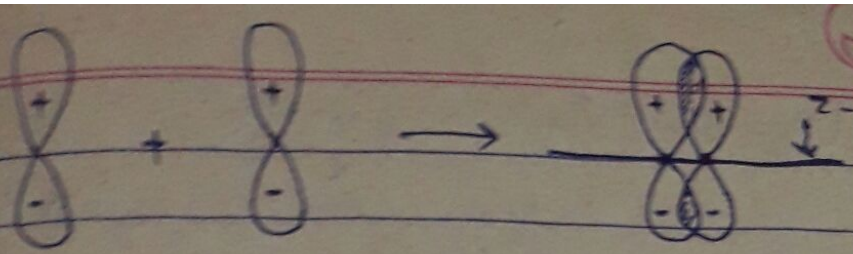
If x-axis is chosen to be internuclear axis,



→ In such case (x-axis is internuclear axis), py - py and pz - pz will form π - bond.

eg of +ve overlapping.

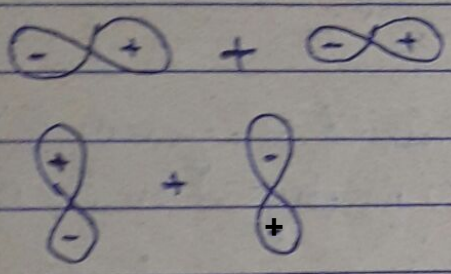




p_x (or p_y) p_x (or p_y) π-bond

i) Negative and zero overlapping do not result into the bond formation.

eg. of -ve overlapping (when opposite sign wave function overlap)



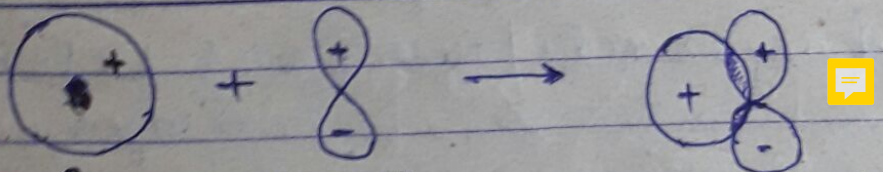
eg. → No Bond

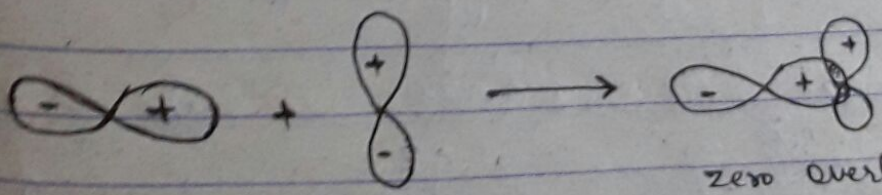
eg. → No Bond.

} destructive Interference. => {No Bond formation}

eg. of zero overlapping
(i.e no net constructive interference)

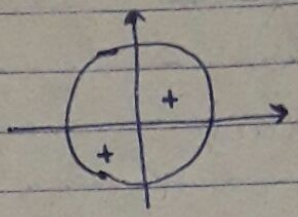
eg.





Shape of Orbitals :-

S - orbital (non-directional) {spherical}

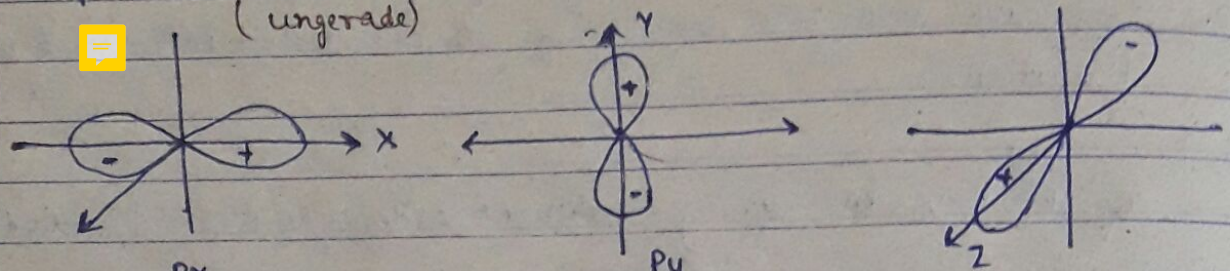


No Nodal plane or angular node (gerade)

↑

S - orbital

P - orbital : directional {dumbelled shape}
(ungerade)



Px — yz - nodal plane

Py — xz - np

xy - np

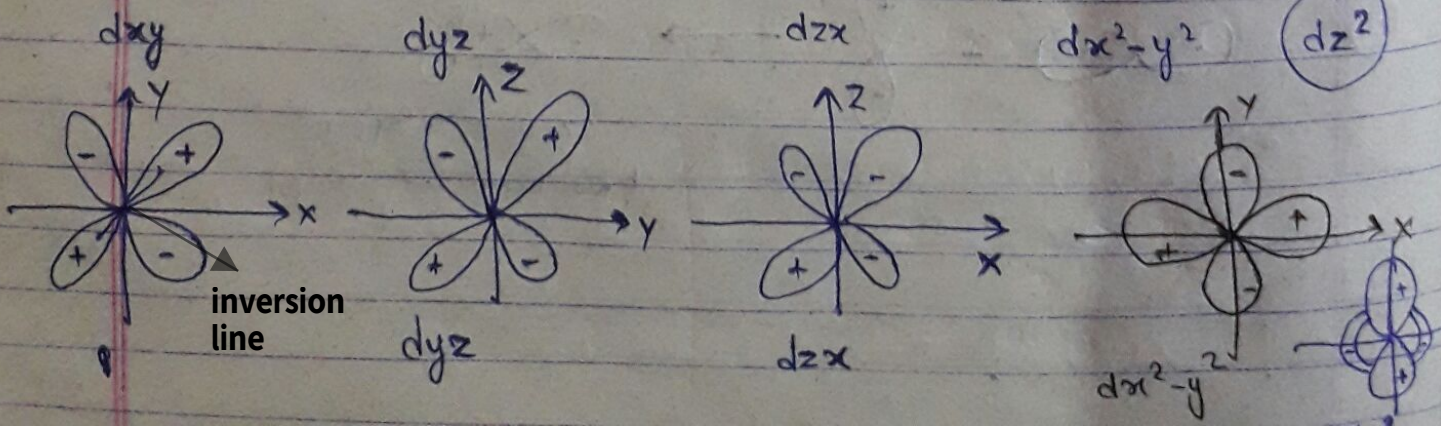
→ Atomic orbitals can be classified as gerade (g) or ungerade (u). If orbital has centre of symmetry, then it is 'g' or else it is 'u'.

→ If no. of angular nodes / Nodal planes = 0, 2 (even)
gerade
(eg. S and d)

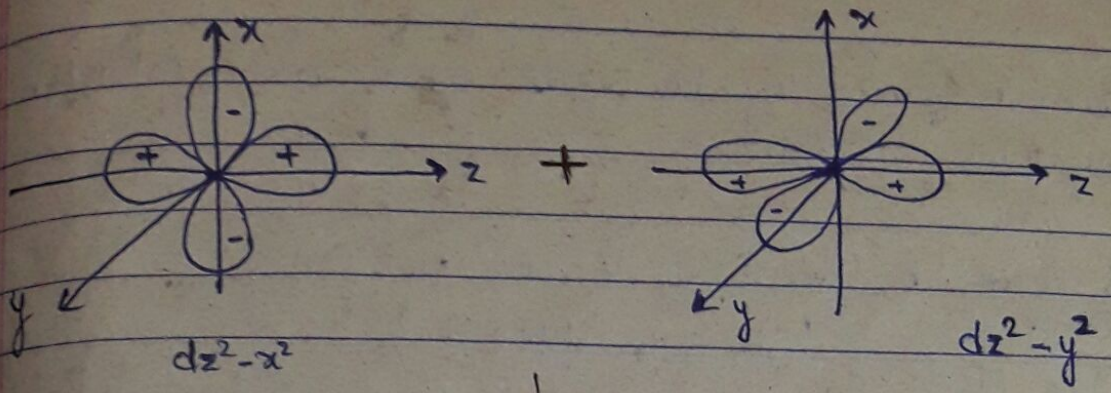
If no. of angular nodes / Nodal planes = 1, 3 (odd)
↳ ungerade eg (P, f)

* Shape of d - orbitals (Gerade)

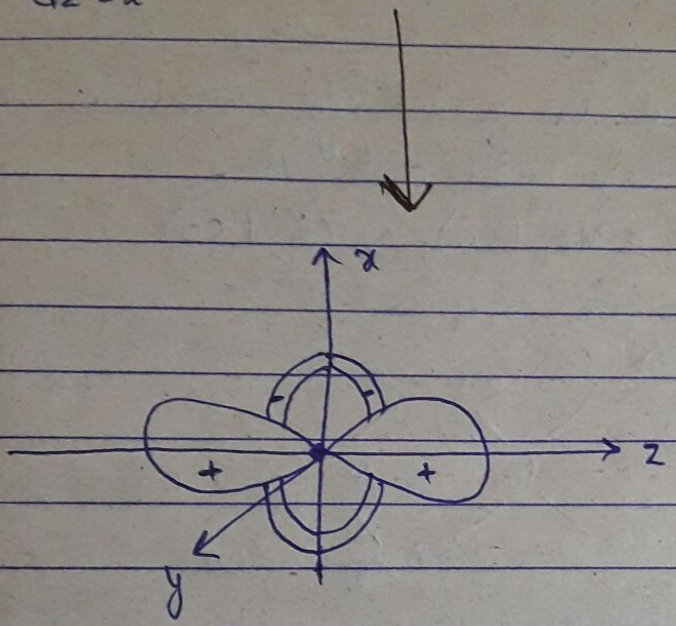
donut shape



In fact d_{z^2} orbital is the combination of $d_{z^2-x^2}$ and $d_{z^2-y^2}$



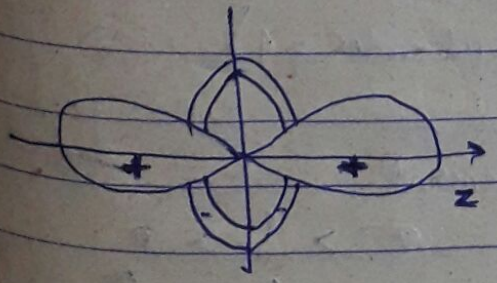
$d_{z^2-x^2}$ $d_{z^2-y^2}$



(*) Bond formed by d-orbitals :-

Considering z-axis to be internuclear axis :

$d_{z^2} + s/p_z/d_{z^2} \longrightarrow$ Sigma (σ bond)



$+ \; s/p_z/d_{z^2} \longrightarrow \sigma$-bond.

σ bond formed by overlapping of one lobe.

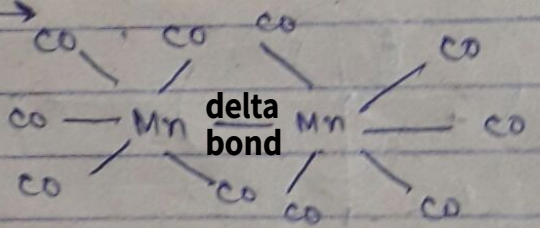
* σ-bond formed by overlapping of one lobe of each combining orbital (Head - on overlapping)

* π-bond is formed by two lobes overlapping (lateral or Sideways overlapping)

* δ-bond (delta bond) is formed by 4 lobes overlapping. (Sideways overlapping).

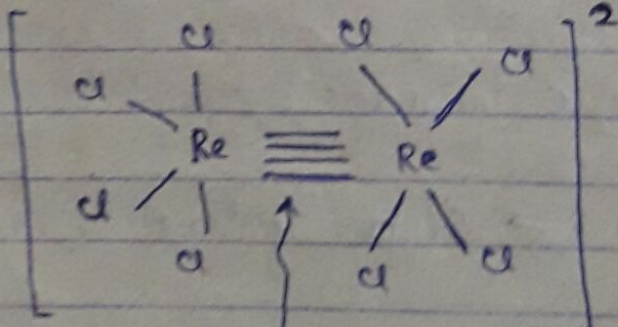
* delta (δ) bond are formed by transition metals.
eg. in polynuclear metal carbonyl
eg. $Fe_2(CO)_4$, $Mn_2(CO)_{10}$, $Co_2(CO)$

eg $Mn_2(CO)_{10}$ →



M — M
↑ δ-bond

eg: Metal clusters like $Re_2Cl_8^{2-}$



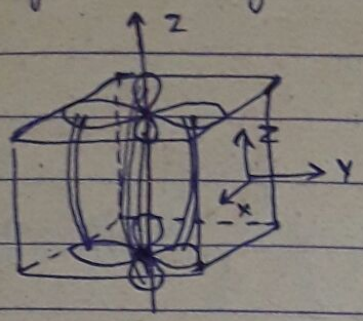
Quadruple bond (Bond order = 4)

1σ
2π
1δ
} Bond order = 4.

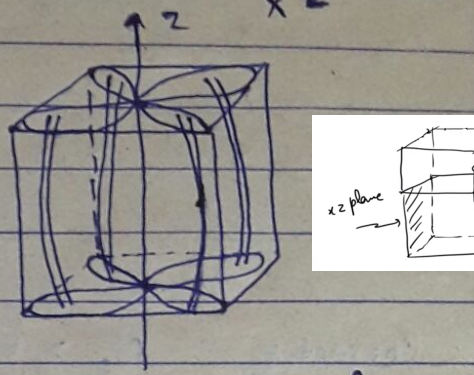
In general
σ > δ > π
↑ 4 lobes ↑ 2 lobes

$dxz + px / dxz \longrightarrow \pi$ - bond.
$dyz + py / dyz \longrightarrow \pi$ - bond.

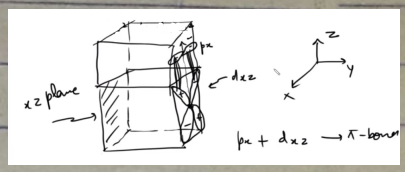
$dx^2-y^2 + dx^2-y^2 \longrightarrow \delta$ - bond
$dxy + dxy \longrightarrow \delta$ - bond.



$dx^2-y^2 + dx^2-y^2$



$dx-y + dxy \longrightarrow \delta$ - bond

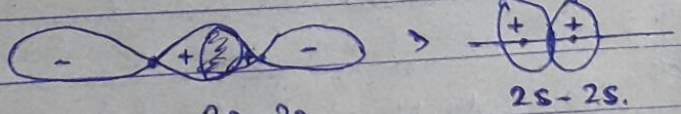


* Comparison of Bond Strength :-

→ Bond Strength or Extent of overlapping.

(a) → Greater is the directional nature (as in p-orbitals relative to 's' orbital) greater is the extent of overlapping and hence greater is the bond energy.
{Provided internuclear distance is Same}. not possible exactly, while comparing.

$2p-2p > 2s-2s$



$F_2 > Li_2$ 105kJ/mol
154 kJ/mol. ↑ exist in gas phase.

(2p - 2p greater extent of overlapping) due to the pointed/directional nature of the p orbital

(b) $B.E \propto \frac{1}{\text{Bond length}}$ { Provided L·P are not present }

$1s-1s > 2s-2s > 3s-3s$ etc.

$H_2 > Li_2 > Na_2 > K_2 > Rb_2 > Cs_2$ →

decreasing Bond Energy.

Exception ⇒ $F_2 > Cl_2 > Br_2 > I_2$ (excepted decreasing order of B.E)

Imp ⇒

Correct : $Cl_2 > Br_2 > F_2 > I_2$ →

decreasing B.E

$F_2 < Cl_2 < Br_2 < I_2$ →

increasing X-X Bond length

→ F-F bond energy is exceptionally low due to heavy repulsion between 3 lone pair on each 'F' (small size) atoms.

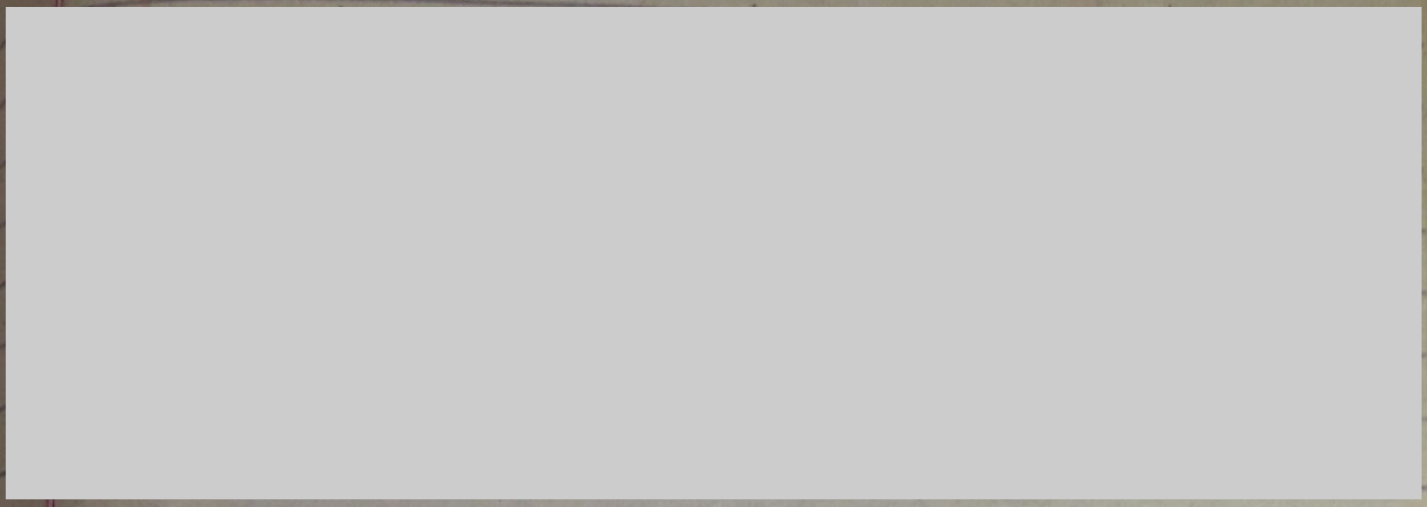
→ L·P repulsion in Cl_2 are eased due to larger size of Cl- atom.

V. Imp ⇒ N-N single bond is weaker than P-P single bond

⇒ O-O single bond is weaker than S-S single bond

⇒ F-F single bond is weaker than Cl-Cl singl bond

Reason: due to heavy repulsion between lone pairs on small atoms of second period.



⇒ Among isotopes, heavier isotope forms slightly stronger bond than the lighter isotope.

$$H-O < D-O < T-O$$

$\longrightarrow$ increasing Bond Energy.

General Points : As per V.B.T

→ First bond is σ-bond (strong bond) between two atoms. Subsequent bonds will be π and δ-bonds.

→ Shape (or Geometry) of a molecule is decided by σ-bonds π-bonds have no role in geometry of a molecule.

→ There is no free rotation of atoms around π-bonds and this gives rise to the geometrical isomerism.

→ Free rotation can occur around σ-bond (with very low energy barrier) and this gives rise to conformation isomers.

→ S-orbitals does head-on overlapping and hence form σ-bonds only

Q Find no. of σ and π-bonds in –

(1.) C_2H_2 (2.) $C_2(CN)_4$ (3.) $C_2(CN)_2$

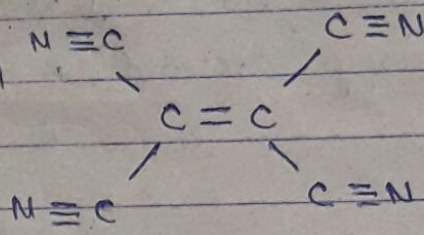
1. C_2H_2 (acetylene)

$H - C \equiv C -$

σ-bonds = 3

π-bonds = 2

2. $C_2(CN)_4$



σ-bonds = 9

π-bonds = 9

3. $C_2(CN)_2$ ⇒

$N \equiv C - C \equiv C - C \equiv N$

σ bonds : 5

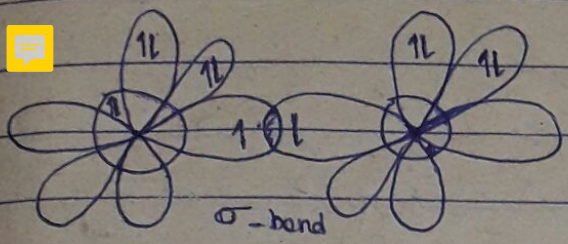
π bonds : 6

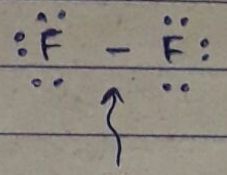
⇒ Delocalisation of π-electrons (due to weak overlapping) give rise to resonance.

Formation of diatomic molecules by VBT

eg. formation of F_2

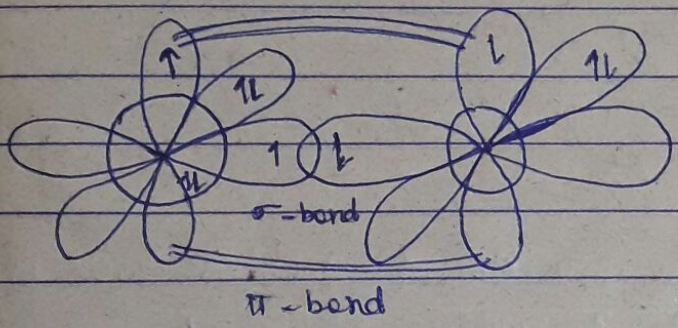
$_9F : 1s^2\ 2s^2\ 2p^5$

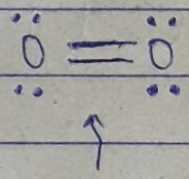




Bond order = 1

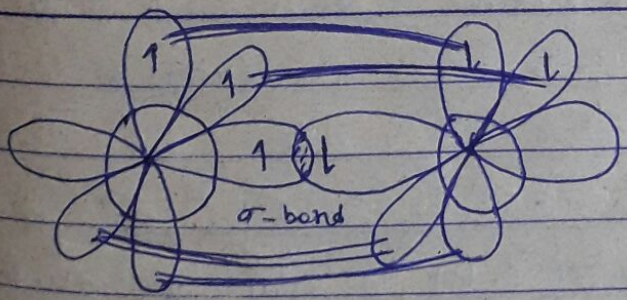
eg. formation of O_2 : $_8O : 1s^2\ 2s^2\ 2p^4 = 1s^2, 2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$ (σ & π bond)



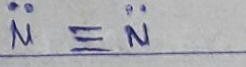


Bond order = 2

eg. formation of N_2 : $_7N : 1s^2\ 2s^2\ 2p^3$



1σ 2π bonds in N_2



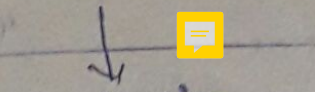
⇒ formation of diatomic molecules are satisfactorily explained (except paramagnetic nature of O_2) by V.B.T

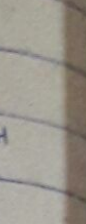
* Formation of polyatomic molecules by V.B.T

eg. formation of CH_4 (methane)

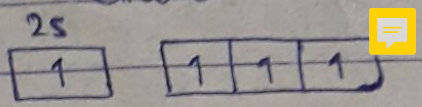
ground state

$_6C$: $1s^2\ 2s^2\ 2p^2$ or $1s^2$,

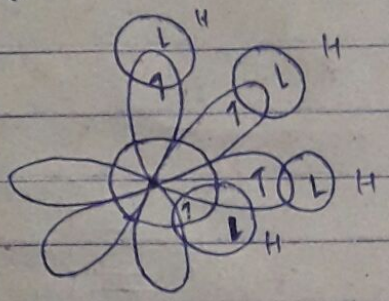




excitation



4 unpaired e^- to form 4 bonds.



formation of CH_4 by VBT

→ According to VBT, 3 orbitals (2p) of 'C' and one '2s' orbital of C overlaps with 1s orbitals of 'H' to form '4' covalent bonds.

⇒ CH_4 as per VBT is expected to have two different types of C-H bonds and three bonds are excepted to be at an angle of 90°.

However,

⇒ Note that all 4 bonds (C-H) in CH_4 are equivalent w.r.t bond energy and bond angle is 109° 28'

Conclusion

VBT fails to explain the bond properties (eg. bond angles and bond length / bond energy) in polyatomic molecules.

⇒ To explain polyatomic molecules, a hypothetical concept called hybridization is adapted which make more complete theory when combined with VBT

OR VBT + Hybridization is more complete theory to explain polyatomic molecules.

Q Select the molecules which are formed in ground state configuration of central atom.

A. BCl_3 C. H_2O

B. CCl_4 D. NH_3

Ans $BCl_3 \Rightarrow {}_5B : 1s^2\ 2s^2\ 2p^1$ (G.S conf^n)

↓ excitation.

$1s^2\ 2s^1\ 2p^2$ or 1s: [↑↓] [↑] [↑|↑|]

(excited state conf^n)

${}_8O : 1s^2\ 2s^2\ 2p^4 \Rightarrow$ 1s: [↑↓] [↑↓] [↑↓|↑|↑] {Ground}

${}_7N : 1s^2\ 2s^2\ 2p^3 \Rightarrow$ [↑↓] [↑↓] [↑|↑|↑] {Ground}

(8) Hybridization:

It is a hypothetical concept according to which appropriate no. of atomic orbitals of central atom undergo intermixing to form equal no. of hybrid orbitals which take part in bond formation with terminal atoms.

(*) Features of Hybridization :-

(1) Atomic orbitals of central atom taking part in hybridization should have proper orientation, same or nearly same energy.

No. of hybrid orbitals formed = No. of atomic orbitals involved in hybdⁿ.

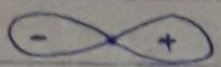
eg. d^2sp^3 / sp^3d^2

$(n-1)d + (n-1)d + ns + np^3$ — Penultimate 'd' orbitals are used in hybdⁿ. and observed in d-block compounds.

$ns + np^3 + nd + nd$ — orbitals of valence shell take part. (observed in compounds of p-block / d-block / s-block)

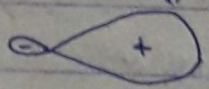
(2) Shape of hybrid orbital consists of two lobes of unequal size, in which the larger lobe takes part in bond formation (σ-bond)

In other words, hybrid orbitals forms stronger bond than the pure 'p' orbital.

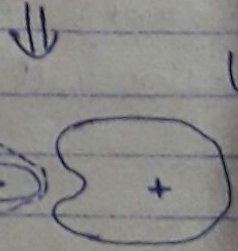


pure 'p' orbital

larger lobe is used for overlapping



Hybrid orbital (HO)



one-nodal surface in HO

(2) Greater is the % s character in hybrid orbital, greater will be the strength of bond (or greater stability of Ho)

50% 33% 25%

$Sp > Sp^2 > Sp^3$

decreasing penetration power of Ho

$Sp-p > Sp^2-p > Sp^3-p > p-p$

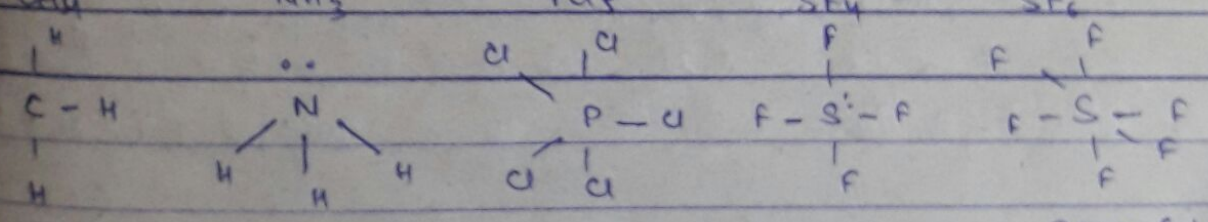
decreasing order of bond strength.

(3) Type of hybridization depends on steric number.

Steric No. of central atom

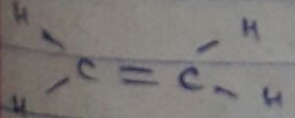
= No. of atoms attached to central atom + No. of lone pair present on central atom.

Q. CH_4 NH_3 PCl_5 SF_4 SF_6



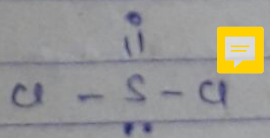
SN = 4+0 SN = 3+1 SN = 5+0 SN = 4+1 SN = 6+0

Q. $CH_2=CH_2$



SN = 3+0 = 3

$SOCl_2$



SN = 3+1 = 4

Note: Atomic orbitals of central atom taking part in hybridization may be fully filled, half filled or vacant.

Steric No.	Hybridization	
2	Sp	$(ns + np)$ → (Linear)
3	sp^2	$(ns + np^2)$ → (Trigonal planar)
4	sp^3	$(ns + npx + npy + npz)$ → (Tetrahedral)
	d^3s	$(n-1)d_{xy} + (n-1)d_{yz} + (n-1)d_{zx} + ns$ → found with d^0 conf^n of d-block. eg. MnO_4^-, CrO_4^{2-}, $TiCl_4$, VO_4^{3-} etc {Tetrahedral}
	dsp^2 (Sq. planar)	$(n-1)\,d_{x^2-y^2} + ns + npx + npy$

5 → Trigonal Bipyramidal (TBP) sp^3d $ns + np^3 + nd_{z^2}$

dsp^3 / sp^3d (Square pyramidal) (observed in some cases of d-block complexes)

$(n-1)d_{x^2-y^2} + ns + np^3$ → $ns + np^3 + n\,d_{x^2-y^2}$

6 → Octahedral (or Square Bipyramidal)

d^2sp^3 / sp^3d^2 → for s & p block

↳ for d-block complexes.

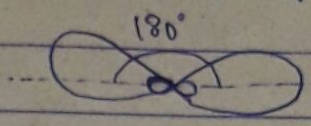
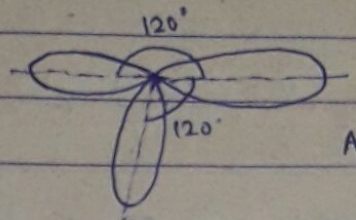
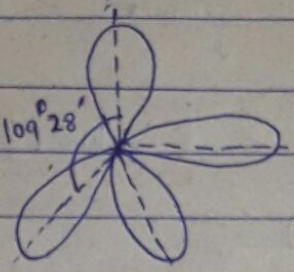
$sp^3d^2 \rightarrow ns + np^3 + nd_{x^2-y^2} + nd_{z^2}$

$d^2sp^3 \rightarrow (n-1)d_{z^2} + (n-1)d_{x^2-y^2} + ns + np^3$

7 → Pentagonal Bipyramidal

$sp^3d^3 \rightarrow ns + np^3 + nd_{z^2} + nd_{xy} + nd_{x^2-y^2}$

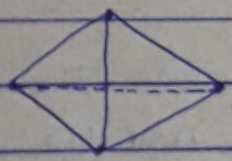
Hybrid Orbitals of Central atom will orient themselves in space such that repulsion between them is minimum.

SN	Hybdⁿ.	Orientation of HO	
2	Sp		Linear
3	Sp^2		Trigonal Planar All three HOs are equivalent.
4	Sp^3		Tetrahedral geometry (Regular tetrahedron)

Each HO faces : 3 repulsions at 109° 28'

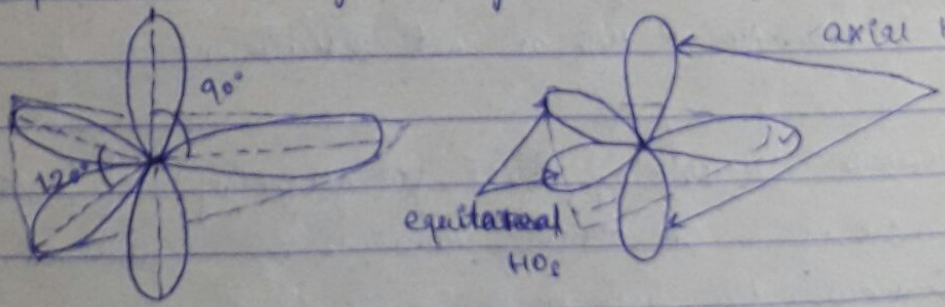
→ All 4 HOs are equivalent

* Tetrahedron : 4 equilateral triangular faces



(4 equilateral triangles)

5 Sp^3d Trigonal Bipyramidal (TBP)

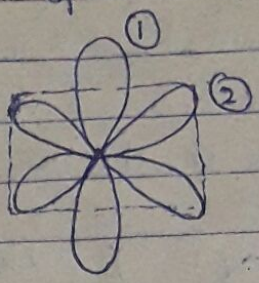


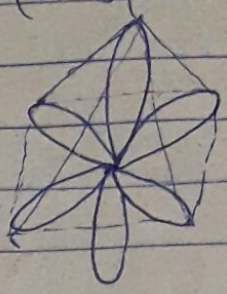
→ Each axial HO faces repulsion = 3 at 90°, 1 at 180°

→ Each equitorial HO faces : 2 at 90° and 2 at 120°

→ All 5 HOs in TBP are not equivalent.

6. sp^3d^2 Octahedral (or Square bipyramidal)

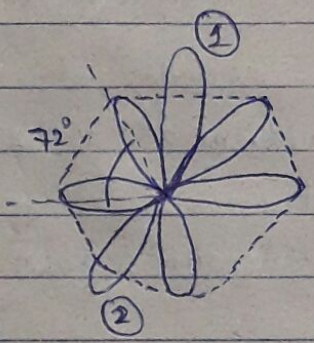




Repulsions faced by HO (1) = 4 at 90°, 1 at 180°
" " " HO(2) = 4 at 90°, 1 at 180°

⇒ In octahedral geometry, all 6 HOs are equivalent.

7. sp^3d^3 (Pentagonal Bipyramidal)



1, 2 → Axial HO

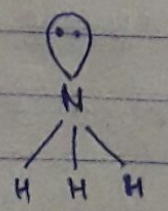
Remaining 5 HOs in Pentagonal plane → Equitorial HOs

Repulsion faced by each axial HO = 5 at 90° and 1 at 180°
" " " " equitorial HO = 2 at 72°, 2 at 90°, 2 at 144°

→ All 7 HOs are not equivalent.

(5) While Visualising shape (or geometry) of a molecule, odd e^- or L.P of e^- on central atom are not visualised. Only bond pairs decide the shape of a species.

eg. $\ddot{N}H_3$
↓
SN = 3+1
= 4 (sp^3)



→ Shape or geometry of NH_3 Pyramidal (Non-planar) (or Trigonal pyramidal)

→ Electron pair geometry of NH_3

→ Tetrahedral.

(*) Valence Shell Electron Pair Repulsion theory :- (VSEPR)

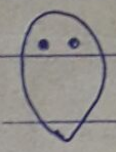
LP - LB > LP - BP > BP - BP

decreasing order of repulsion caused.

Double Bond > Single bond

↑ greater repulsion caused

eg. $\ddot{N}H_3$ (SN = 4) ⇒



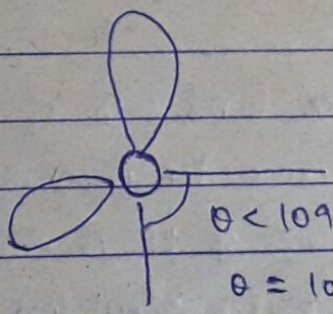
$\theta < 109°28'$

$\Rightarrow \theta \approx 107° < 109°28'$

eg. $H_2\ddot{O}$

SN = 2+2

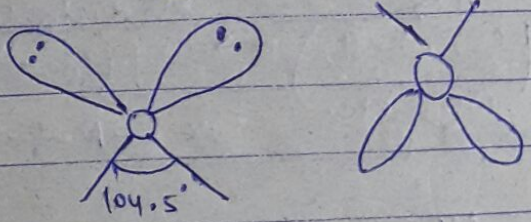
= 4 (sp^3)



$\theta < 109°28'$

$\theta = 104.5°$

(or)



Bent or V-shape or angular Shape (planar)

VSEPR Theory:

LP-LP > LP-BP > BP-BP

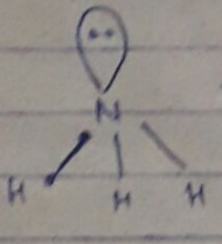
decreasing order of repulsions caused

LP-double bond > double bond-double bond > double bond-single bond > single-single bond

decreasing order of repulsions caused.

NOTE:

(1) Shape of polyatomic compounds get distorted if lone pair or odd e⁻ is present on central atom.

(2) 

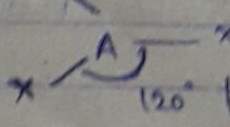
⇒ Since bond pair of e⁻ is controlled by two nuclei whereas L.P is controlled by only one nucleus.
∴ Size of LP is considerably large and hence causes greater repulsion.

SN = 2, AX_2 (CO_2, $BeCl_2$, CS_2 etc.) → No distortion (Linear)

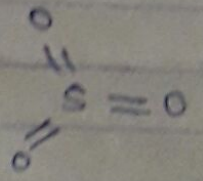
X — A — X

A → Central Atom
X → Terminal Atom
E → lone pair

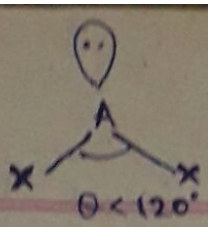
SN = 3, AX_3 (BCl_3, $AlCl_3$, BF_3, SO_3 etc.) → No distortion

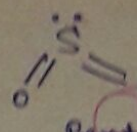


120° (Trigonal Planar)



All the three S-O bond lengths in SO_3 are equivalent.

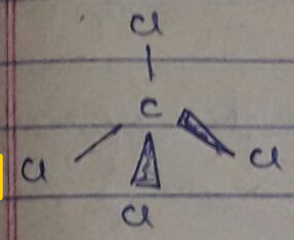
AX_2E (eg SO_2) → 

eg. 
Bent.

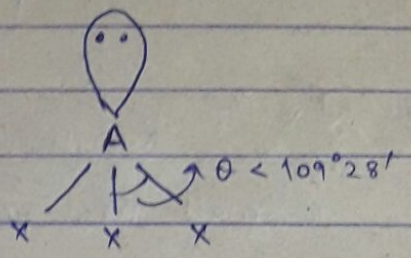
V-shape (or) Angular (or) Bent Molecule.

AX_2 → Hypothetical.

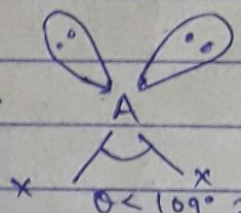
SN = 4 AX_4 (eg. CCl_4, CH_4, $SiCl_4$ etc.) → No distortion.



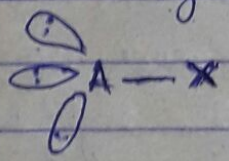
wedge and dash formula of CCl_4

AX_3E ($\ddot{N}H_3$, $\ddot{C}H_3^-$) → 

Pyramidal Structure (Non-planar)

AX_2E_2 (eg. $H_2\ddot{O}$, $:\ddot{N}H_2^-$) → 

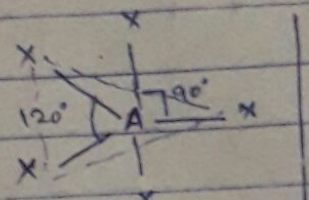
V-shape (or Bent)

AXE_2 ($:\ddot{I}-Cl$) → Always linear 

NOTE: In SN = 2, 3, 4 (as studied above) all the Hos are equivalent and therefore L.P or double bond could be placed at any site or in any hybrid orbital.

Imp $SN = 5$, AX_5 → No distortion (Perfect TBP)

(eg. PCl_5, PF_5 etc)



Trigonal Bipyramidal (TBP) (Non-planar)

→ In TBP, each equitorial HO faces : 2 at 90° & 2 at 120° repulsions

Each axial HO faces : 3 at 90° and 1 at 180°

Imp ** : Repulsions at an angle of 120° and higher are considerably small and not substantial

⇒ Repulsions at 90° and less angle are substantial and affect stability highly.

(*) Conclusion

(1) In TBP, equitorial position is more stable than the axial position.

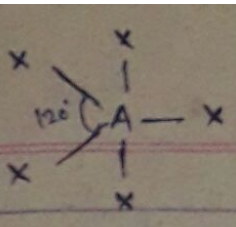
(2) Since L.P and double cause more repulsion and hence should be placed at site / position where angle is more i.e equitorial position in TBP.

In other words, HO having greater 's' character is more stable.

In other words, L.P and double bond need greater 's' character to stabilize.

⇒ Equitorial HO_s are sp^2

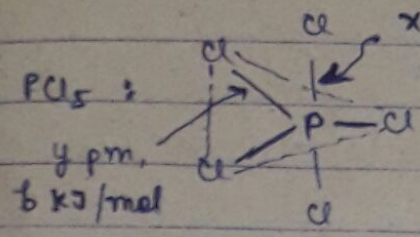
while axial HO_s → p^d



sp^3d $(sp^2 + pd)$ (p_z and d_{z^2})

$ns + np_x + np_y$

→ Equitorial HO in TBP has greater 's' character and hence L.P and double bond in TBP are in equitorial position.



Bond length $\propto \frac{1}{\text{Bond Energy}}$

$\Rightarrow x > y$, $a < b$

→ Equitorial P-Cl bonds are stronger than axial P-Cl bonds.

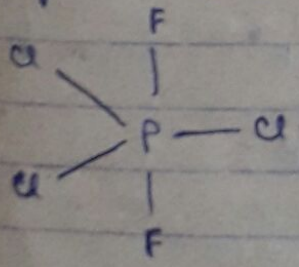
$PCl_5(g) \xrightarrow{\text{heat}} PCl_3(g) + Cl_2(g)$

Bent's Rule :

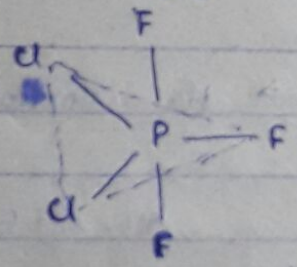
More electronegative substituent is diverted to the hybrid orbital (or site) where 's' character is less (or 'p' character is more)

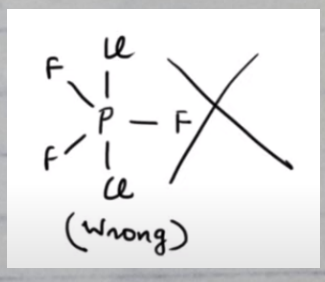
{keep in mind that L.P & ⊕ ⊖B required more 's' character, so will be at equitorial positions at TBP}

eg. PCl_3F_2



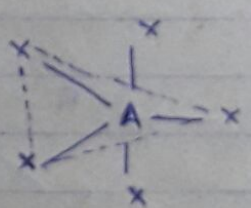
PCl_2F_3



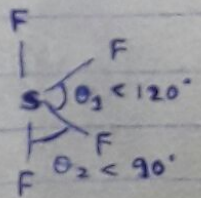


SN = 5

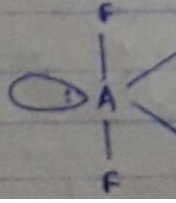
AX_5 [PCl_5, PF_5] → No distortion, TBP



eg.

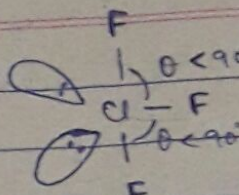


AX_4E (eg $\ddot{S}F_4$) →



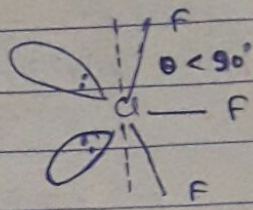
See-saw Structure (Non-planar)

AX_3E_2 (eg. $:\ddot{Cl}F_3$, $:\ddot{Br}F_3$) ⟶

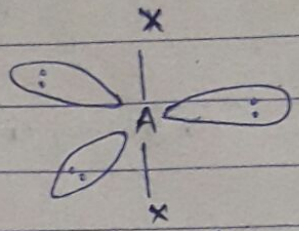


T-shape (Planar)

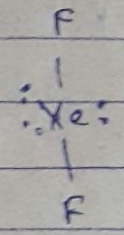
(OR)



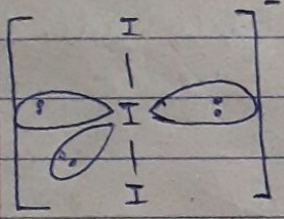
AX_2E_3 (eg. XeF_2, I_3^-, ICl_2^-)



Linear (Planar)



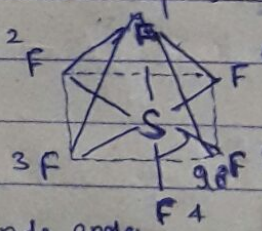
Linear



Linear anion (Triiodide ion)

SN = 6 . sp^3d^2, octahedral geometry (or square dipyramidal)

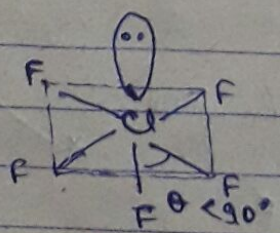
AX_6 (eg SF_6) ⟶ No distortion



(Consists of 8 faces)

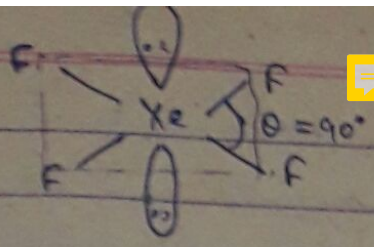
No. of FSF bonds angles.
= 5+4+3+2+1 = 15

AX_5E (eg $\ddot{Cl}F_5$, $\ddot{Br}F_5$) ⟶

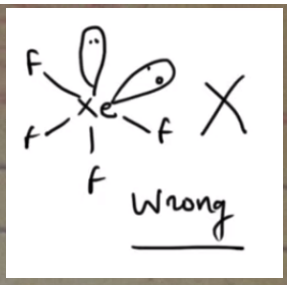


Square pyramidal (Non planar)

AX_4E_2
(e.g. $:\ddot{X}eF_4$, $:\ddot{I}\,Cl_4^-$)

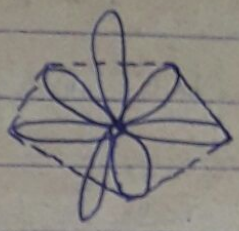


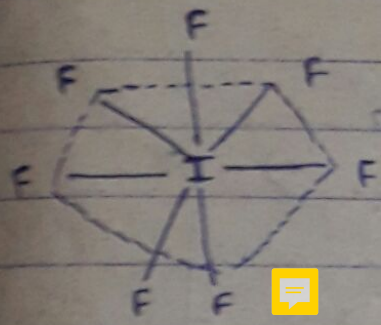
perfectly Square Planar



SN = 7 | sp^3d^3

AX_7 → No distortion
(e.g. IF_7) Pentagonal Bipyramidal

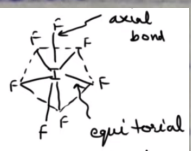




Five I-F equitoral bonds and Two I-F axial bonds.

AX_5 → Pentagonal bipyramidal
$[IF_7]$

→ In IF_7 (or SN = 7), all HOs are not equivalent

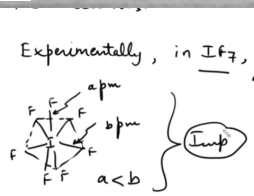


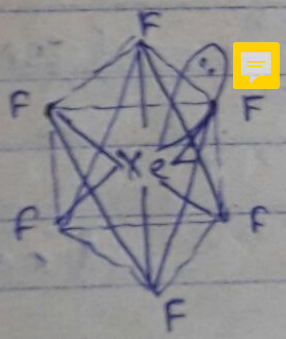
Each axial HO faces repulsions: 5 at 90°, ~~1 at 180°~~
" equitorial HO faces = ~~2 at 72°~~, 2 at 90°, ~~2 at 144°~~

⇒ Repulsions at 72° are intense repulsions which make equitorial HOs being less stable than the axial HOs.

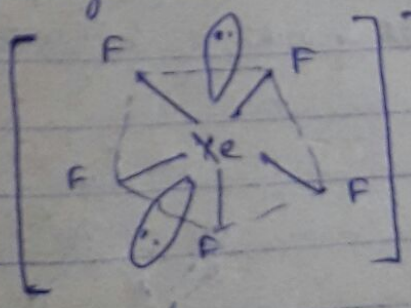
AX_6E → Capped octahedron or distorted octahedron.
(e.g. $\ddot{X}eF_6$)

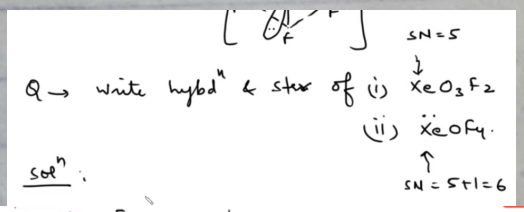
⇒ Experimentally, in IF_7, 2 I-F axial bonds are stronger than the 5 I-F equitorial bonds



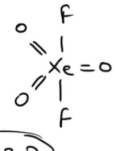


AX_5E_2
(e.g. XeF_5^-) Pentagonal Planar Structure



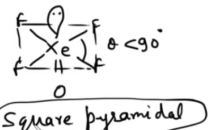


(i) XeO_3F_2



TBP

(ii) $\ddot{X}eOF_4$



Square pyramidal

⊛ Applications of Bent's Rule :-

⇒ Comparison of Bond angle and Bond length.

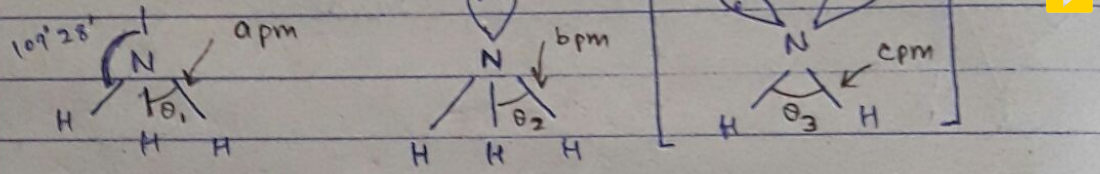
(when comparing in different molecules)

→ When central atom is same and has same hybridization.

→ Higher is the % s character in a bond, greater will be the bond energy.

→ Higher is the % 's' character, higher will be the bond angle

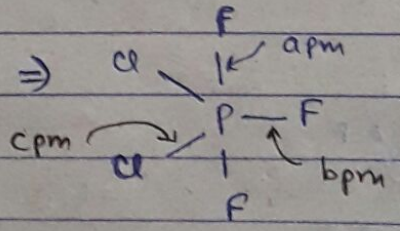
eg. NH_4^+ $\quad$ $\ddot{N}H_3$ $\quad$ $:\ddot{N}H_2^-$



$a < b < c$

$\theta_1 > \theta_2 > \theta_3$

eg. PF_3Cl_2 ⇒

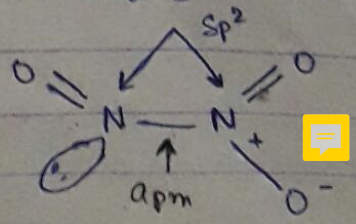


(TBP)

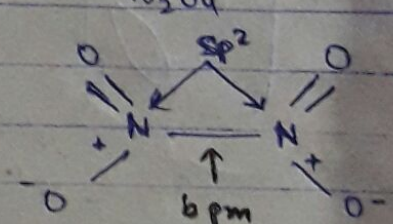
$b < a < c$

Imp → Comparison of bond length / bond strength is done on the basis of Bent's rule (i.e 's' character/ 'p' character) when terminal atoms are same in a molecule.

eg. N_2O_3

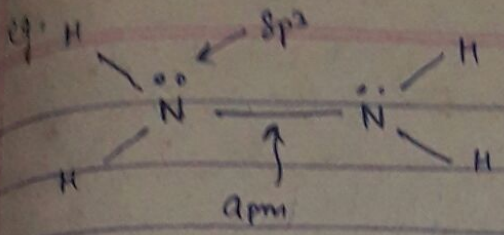


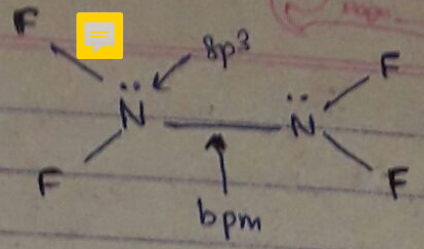
N_2O_4



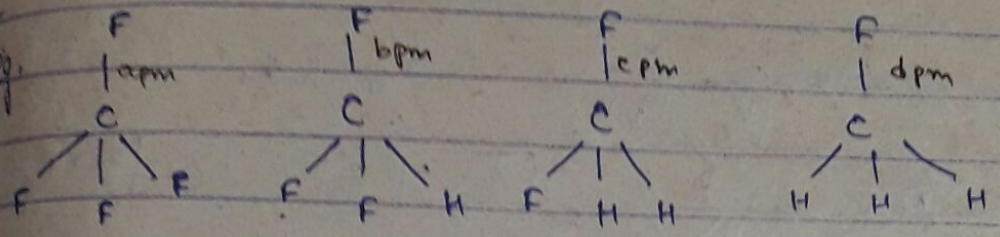
$a > b$

N_2H_4

eg. 



a > b

eg. 

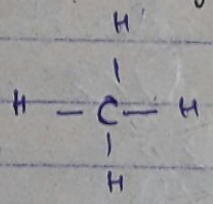
CF_4 CHF_3 CH_2F_2 CH_3F

(sp^3)

a < b < c < d

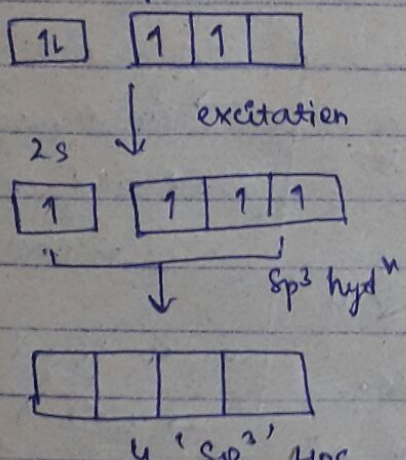
① Formation of polyatomic molecules using VBT + Hyb^n.

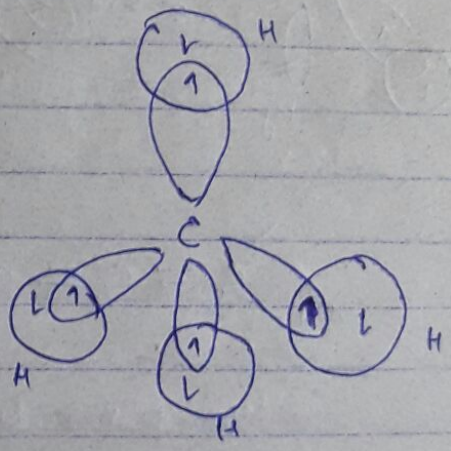
eg. Methane (CH_4)

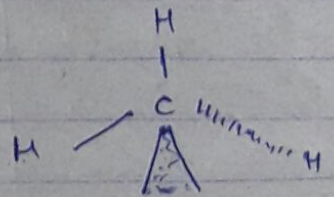


(LDS)

C: $1s^2$ $2s^2$ $2p^2$

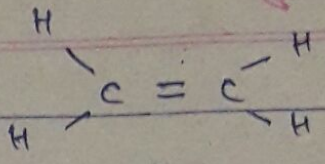




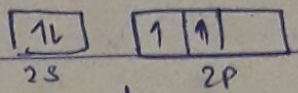


Tetrahedral Structure.

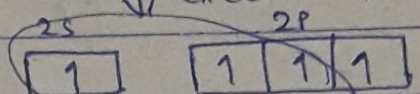
eg. Ethylene (C_2H_4) ⇒



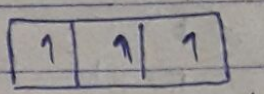
C : $1s^2\ 2s^2\ 2p^2$ or



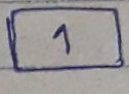
excitation



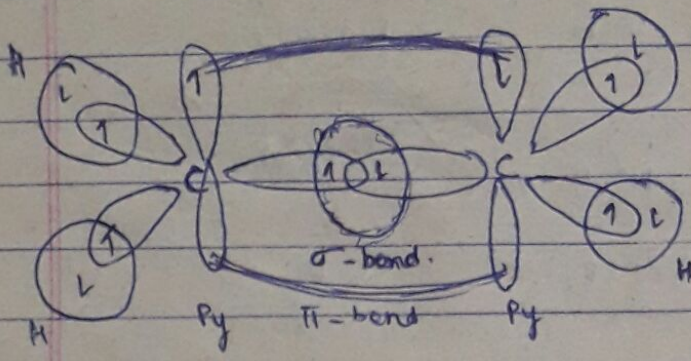
sp^2 hybridn.



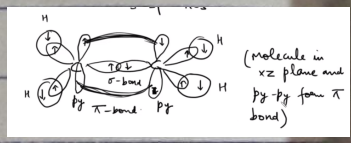
3 sp^2 HOs



→ pure 'p' orbital

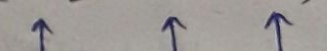


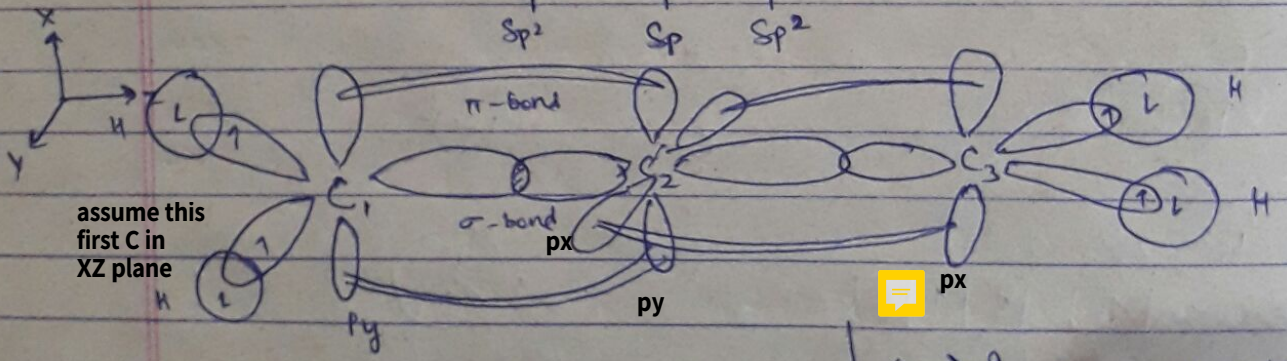
molecule in XZ plane
py orbital used to form pi bond



(molecule in xz plane and py-py form π bond)

eg. Propadiene ($CH_2 = C = CH_2$) (Cumulated diene)



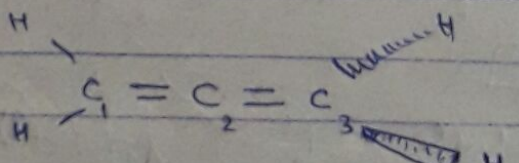


assume this first C in XZ plane

px

py

px



Non-planar Molecule.

→ In cumulated dienes,

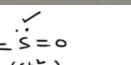
⇒ No. of double bonds

→ even (Non-planar)

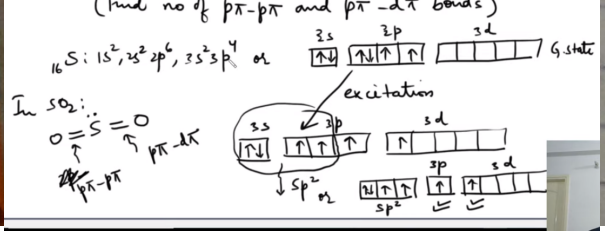
⇒ No. of double bonds

→ odd (eg 1, 3, 5 etc)

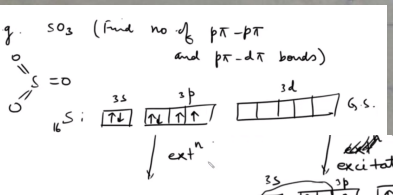
Planar.

e.g SO_2 ⇒ 

(find no of pπ-pπ and pπ-dπ bonds)



e.g SO_3 (find no of pπ-pπ and pπ-dπ bonds)

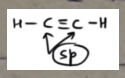


In SO_3, one pπ-pπ, Two pπ-dπ } Ans

Additional examples of different kinds of hybridisation :

⇒ Examples of Sp hybdⁿ :-

eg. CO_2, CS_2, C_2H_2, C_3O_2, $BeCl_2(g)$



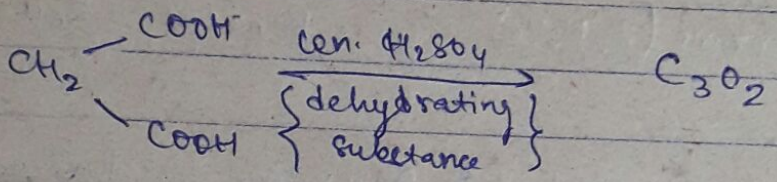
carbon suboxide

O = C = O (sp)

S = C = S

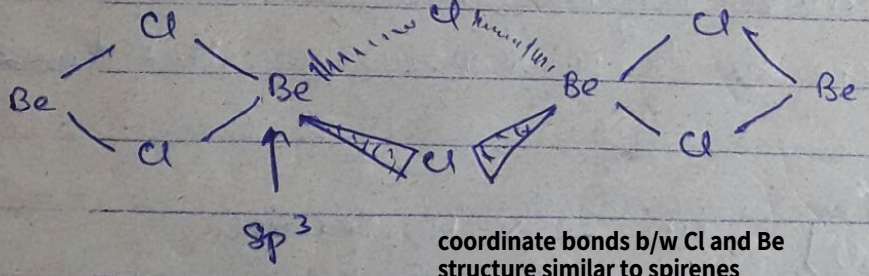
CS_2 is a well known non-polar solvent.

⇒ C_3O_2 O = C = C = C = O (sp)



⇒ Cl – Be – Cl in 'g'

⇒ At room temperature $BeCl_2(s)$ is a polymeric chain like structure (Sp^3)

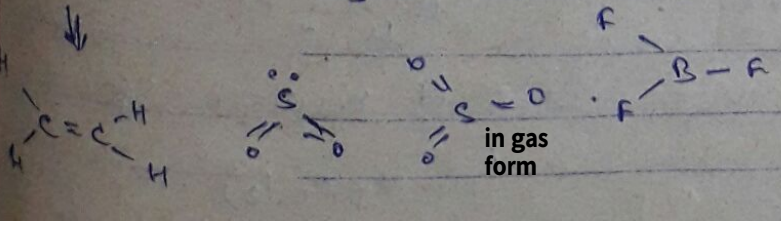


coordinate bonds b/w Cl and Be
structure similar to spirenes

Non-planar structure.

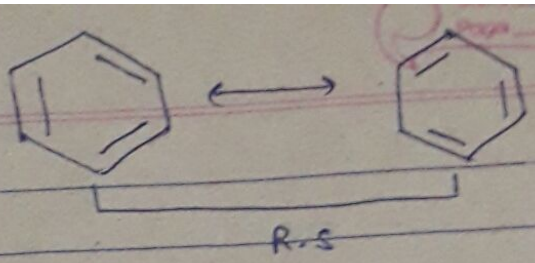
* Examples of sp^2 hybdⁿ :

C_2H_4 SO_2 SO_3 BF_3 BCl_3, $AlCl_3$ etc

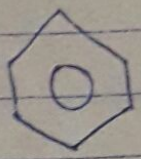


in gas form

eg. benzene, C_6H_6

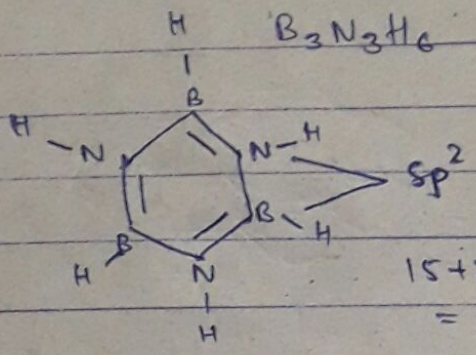


R.S



Resonance hybrid.

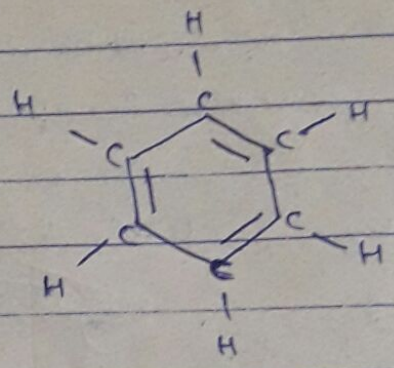
(V.Imp) Inorganic Benzene (or) Borazine (or) Borazole



$B_3N_3H_6$

sp^2

15+21+6 = 42 e⁻



42 e⁻ total

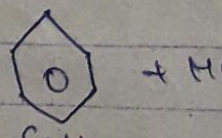
Inorganic Benzene ($B_3N_3H_6$) Benzene (C_6H_6)

→ Both are planar and sp^2, resonance stabilized

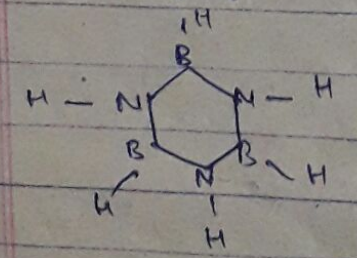
→ Both are isoelectronic and non-polar.

⇒ Benzene is very less reactive due to non-polar nature of C-C bonds

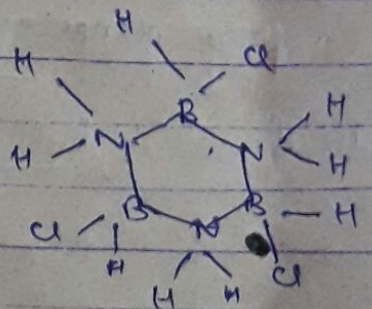
⇒ However, inorganic benzene is much more reactive than benzene due to polar nature of N-B bonds.



C_6H_6 + HCl ⟶ No Reaction



+ 3HCl ⟶



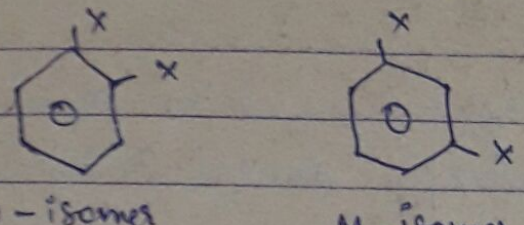
B & N now are sp^3.

($B_3N_3H_9Cl_3$)

⇒ Benzene + $H_2O(l)$ ⟶ No rex, no mixing.

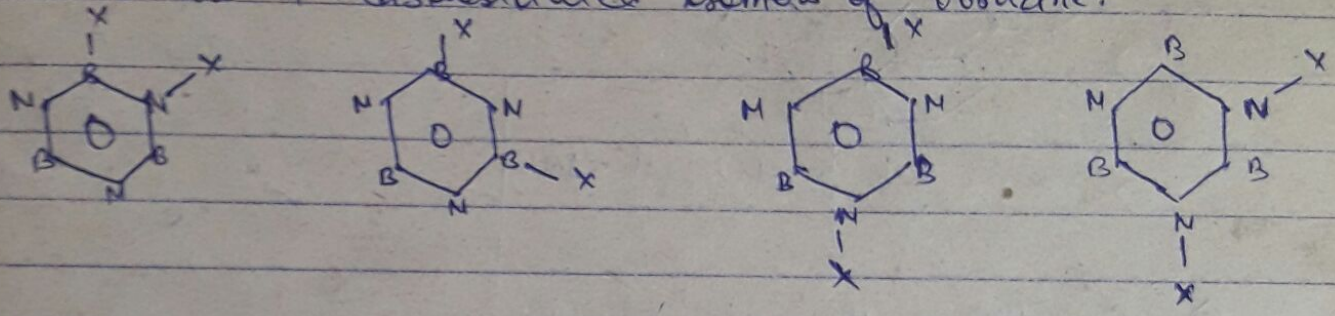
⇒ $B_3N_3H_6 + H_2O(l) \xrightarrow[\text{Hydrolysis}]{\text{Slowly}} NH_3 + H_3BO_3 + H_2$

⇒ Benzene has three disubstituted isomers

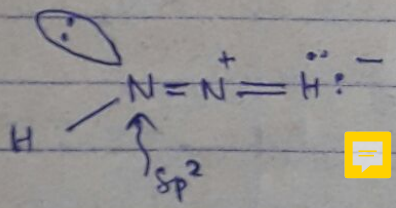


o-isomer M-isomer P-isomer

⇒ There are 4 distubstituted isomers of borazine.

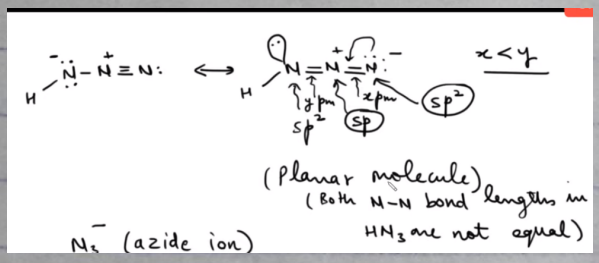


⇒ Hydrazoic acid, $HN_3 \xrightarrow[\text{heat}]{} H_2 + N_2$
(a weak acid)

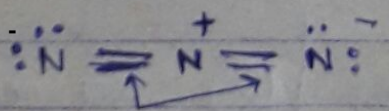


(Planar Molecule)

both N-N bond lengths are not equal



N_3^- (azide ion)

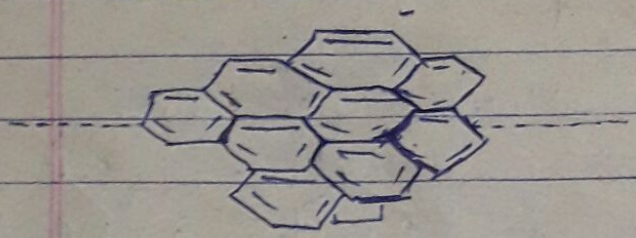


Both N-N bond lengths in N_3^- are equal.

(*) Allotropes of Carbon : (Three crystalline forms) → Graphite, Diamond, Fullerene.

Graphite

(1) Each carbon is sp^2, forming a sheet like covalently bonded network structure.



C-C bond length = 1.41Å

⇒ Graphite is thermodynamically more stable than diamond due to delocalisation of π-electrons in the sheet ($\Delta H^\circ_f = 0$) at 25°C

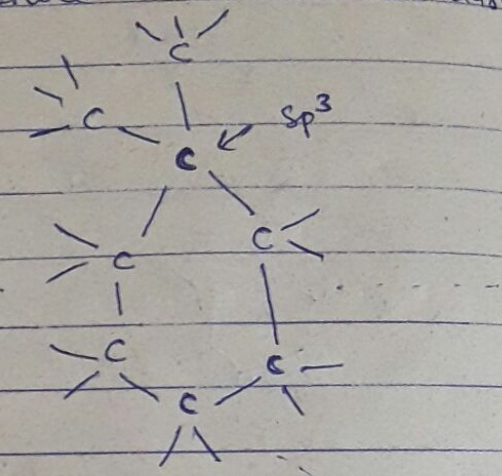
(2) Graphite acts as solid lubricant as one sheet can slide over the other.

(3) Good electrical conductor along the sheet due to delocalisation of $\pi - e^-$ along the sheet.

(4) Graphite is metallic conductor along the sheet and semi-conductor across the sheet.

Diamond

(1) Each carbon is sp^3 forms covalently bonded 3D-network structure



(Similar covalently bonded 3D structure are found in SiO_2, SiC, Si, Ge etc)

(2) No lubricating property

(3) Not good conductor of electricity.

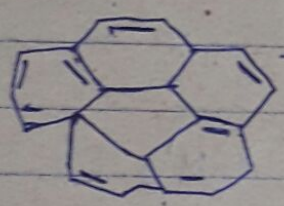
T↑, E.C ↑ across the sheet so Semiconductor.

⇒ Graphite is anisotropic wrt electrical properties

④ Hardest known substance (natural), denser than graphite. A very good conductor of heat and used in making tools to cut glass stones. Also higher MP than graphite.

* Buckminster fullerene (C_{60})

→ A football or bucking ball structure have 20 hexagonal and 12 pentagonal rings of carbon atoms which are sp^2 hybridised.



→ There are no dangling bond in buckminsterfullerene

↳ It is an unsatisfied valence or immobilised free radical.

* Graphite and diamond may have dangling bonds.

→ There are two different C-C bonds lengths in buckminsterfullerence.

→ C_{60} < graphite < diamond

increasing density, M.P

ΔH°_f (graph) = 0 , ΔH°_f (dia) = +x kJ/mol , ΔH°_f (C_{60}) = +y kJ/mol

y > x

$B_3N_3H_6 \xrightarrow{\Delta} (BN)_x$

(Borazine)

(Graphite like 'sheet' structure)

(High melting solid similar to graphite)

$(BN)_x$

Graphite like structure

→ hexagonal crystal structure

→ Acts as good insulator

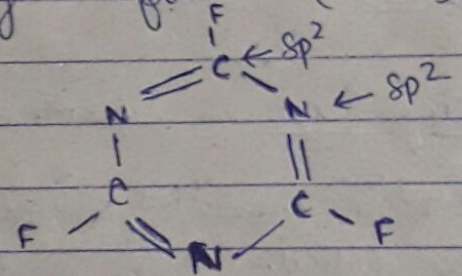
Diamond like structure

→ Cubic crystal structure

→ hard substance like diamond

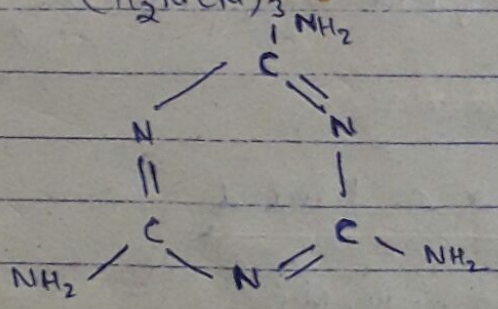
→ known as 'Borazon'

⇒ Cyanic fluoride $(FCN)_3$



⇒ Cyanic amide (Melamine)

$(H_2NCN)_3$



Examples of sp^3 :-

Alkanes (CH_4, C_2H_6), SiH_4, $SiCl_4$, $\ddot{N}H_3$, $H_2\ddot{O}$, H_2O_2, CCl_4 etc

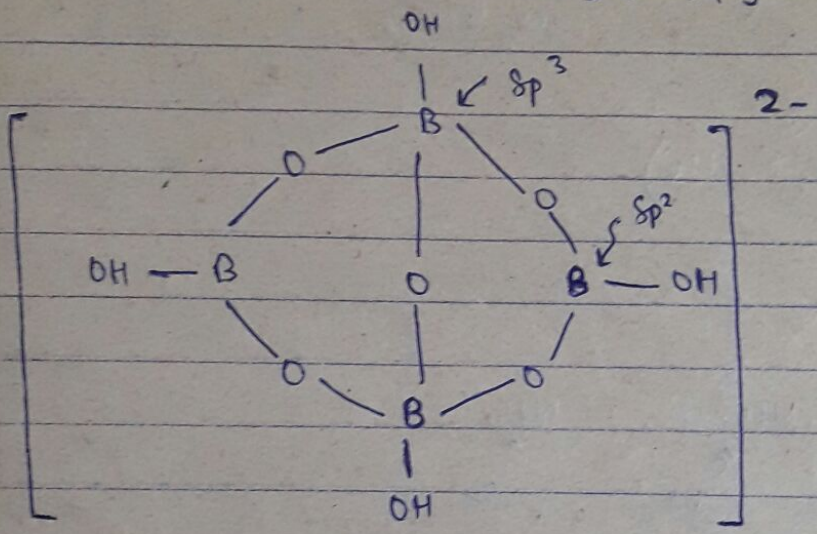
eg: $C_2H_6 \Rightarrow CH_3-CH_3 \xrightarrow[\text{fission of C-C bond}]{\text{heterolytic}} \overset{+}{C}H_3 + \ddot{C}H_2^-$

(CH_3^+: sp^2; $\ddot{C}H_2^-$: sp^3)

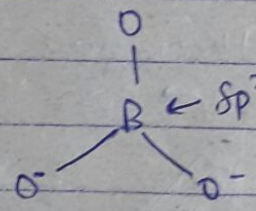
eg: Borax (white powder) , $Na_2B_4O_7 \cdot 10H_2O$

$\Downarrow$

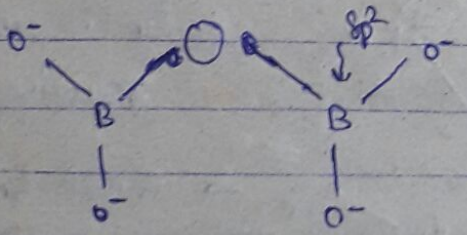
$2Na^+[B_4O_5(OH)_4]^{2-}$ and $8H_2O$



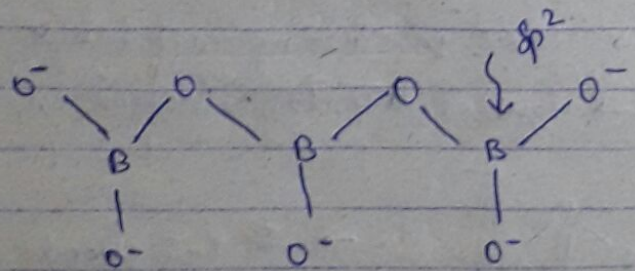
→ Orthoborate ion, BO_3^{3-}



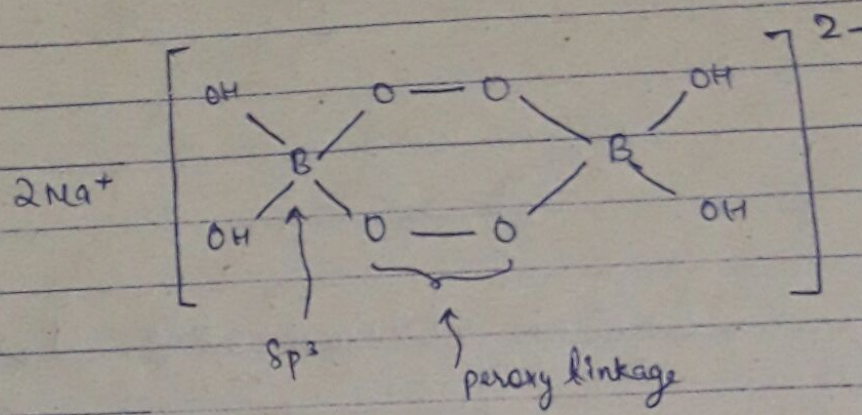
→ diborate ion, ($B_2O_5^{4-}$)



→ Triborate ion, ($B_3O_7^{5-}$)



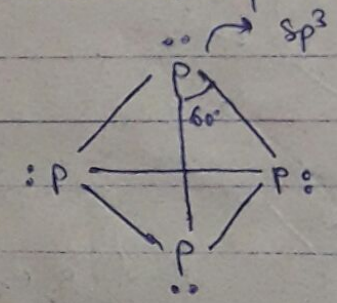
eg. Sodium perborate or sodium peroxoborate
(used as brightener in washing powder)



($Na_2B_2O_8H_4$)

eg. $\ddot{N}H_3$, NH_2OH , NH_4^+
(sp^3) (sp^3) (sp^3)

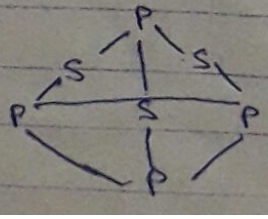
eg. white Phosphorous (P_4)



→ due to high angle strain, P_4 is unstable and highly reactive.

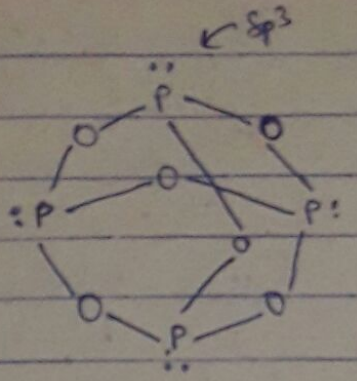
→ 6 P-P bonds in P_4 molecule.

eg. P_4S_3 → used in making match box
→ ignites due to friction on surface

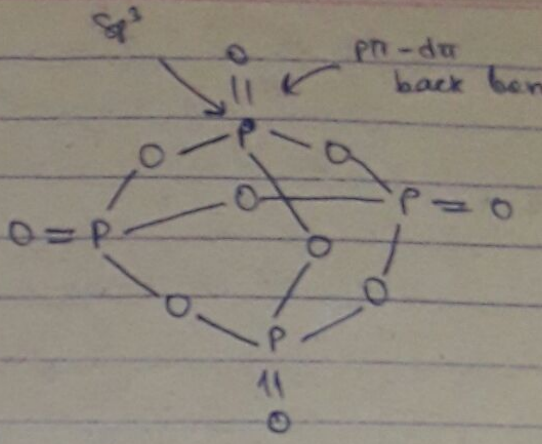


→ 6 P-S bonds
and 3 P-P bonds.

eg. P_4O_6 (+3)



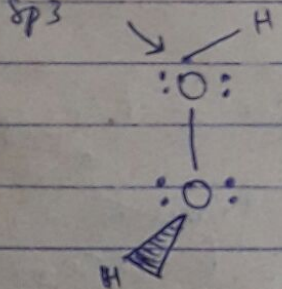
P_4O_{10} (+5)



eg. $H_2\ddot{O}$ ↑ sp3 $\quad H_3O^+$ ↑ sp3 { Hydronium ion }

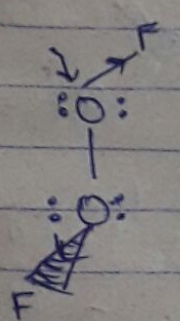
eg. H_2O_2 (hydrogen peroxide)

↳ open book like structure (Non-planar)



This O-O bond is longer bond {and weaker} than expected. This is due to long repulsion between L.P of small sized 'O' atoms.

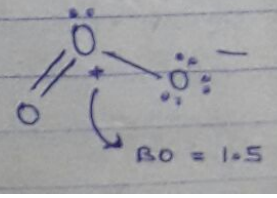
eg. O_2F_2



O-O bond length in O_2F_2 is considerably smaller than that in H_2O_2. It is due to high EN of F atom which diverts L.P electron cloud to itself and hence repulsion between L.Ps of 'O' reduced greatly.

$H_2O_2 > O_3 > O_2F_2$

decreasing O-O bond length.

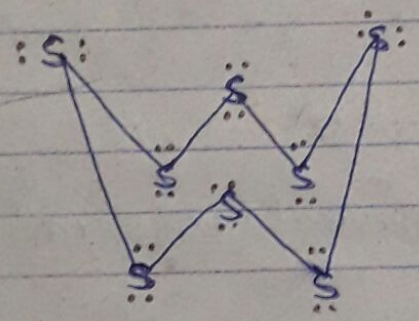


⇒ The same effect of heavy repulsion in L.P on second period small elements is observed in H_2O_2, NH_2OH, F_2 etc.
{ F-F bond is weaker than Cl-Cl bond. }

{ N-O bond is longer and weaker than expected }

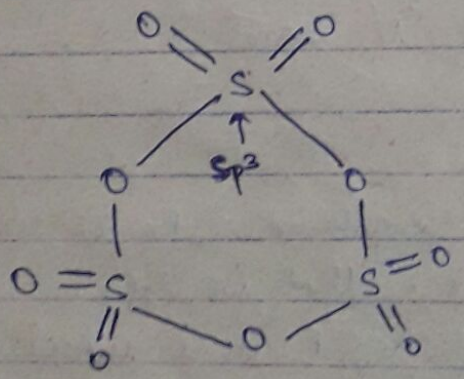
eg. S_8 (rhombic or monoclinic) sulphur :

S_8 → Crown like structure.

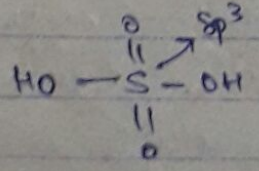


eg. At room temperature SO_3 exist as $(SO_3)_3$ (s)

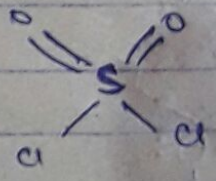
Cyclic Trimer of SO_3
($\gamma - SO_3$)



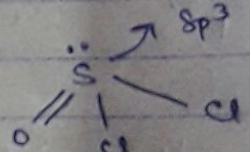
eg. H_2SO_4,



eg. SO_2Cl_2 { Sulphuryl Chloride }

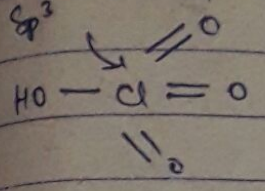


eg $SOCl_2$ (Thionyl chloride)

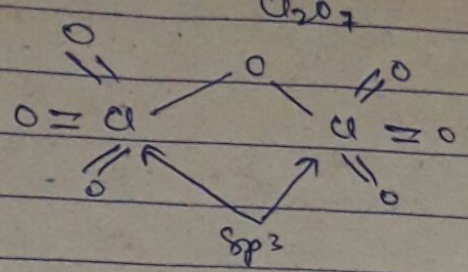


Pyramidal

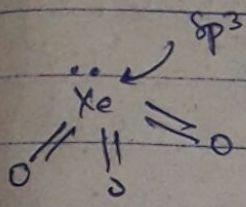
eg. $HClO_4$



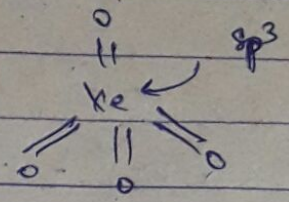
Cl_2O_7



eg. XeO_3



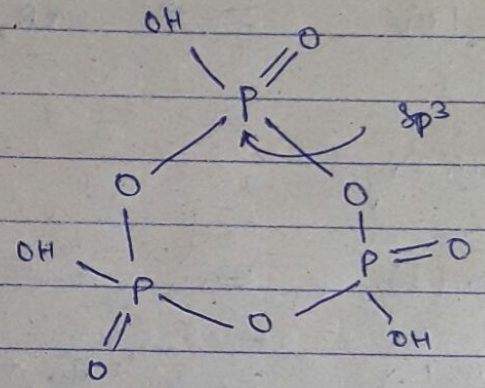
XeO_4



eg. $(HPO_3)_3$

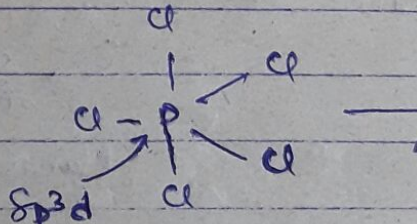
cyclopolymeta phosphoric acid

$\Rightarrow$



Q eg. of sp^3d / sp^3d^2 / sp^3d^3 etc

eg. $PCl_5(g) \Rightarrow$



$\xrightarrow{\Delta} PCl_3(g) + Cl_2$

→ Some compounds in solid or liquid state become ionic (ie dissociate into ions partially or fully)

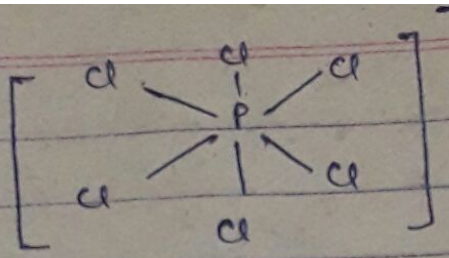
eg. $2PCl_5(s) \longrightarrow PCl_4^+ + PCl_6^-$

or

$PCl_4^+ \quad PCl_6^-$

$\uparrow \quad \uparrow$

$sp^3 \quad sp^3d^2$

* PCl_6^- ⇒ 

Covalency of 'P' = 6

eg. $PBr_5(s) \longrightarrow PBr_4^+ Br^-$

PBr_6^- is unstable due to steric crowding by 6 large Br^-

$PI_5(s) \longrightarrow PI_4^+ I^-$

eg. $2BrF_3(l) \rightleftharpoons BrF_2^+ + BrF_4^-$

↑ electrically conducting.

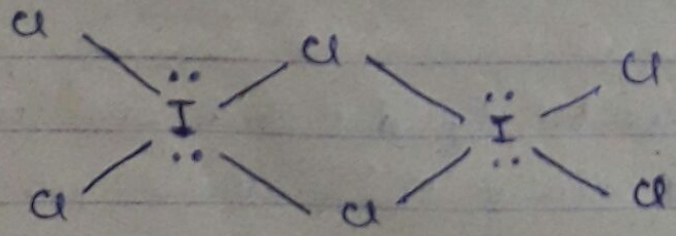
eg. $N_2O_4(l) \rightleftharpoons NO^+ + NO_3^-$

eg. $N_2O_5(s) \rightleftharpoons NO_2^+ NO_3^-$

eg. dimer of ICl_3 i.e I_2Cl_6 (s)

↳ solid at room condition

(Planar, sp^3d^2, 'I' cations



Planar molecule

⇒ $I_2Cl_6(l) \rightleftharpoons :\ddot{I}Cl_2^+ + \ddot{I}Cl_4^-$

$:\ddot{I}Cl_2^+$: sp^3 (Bent molecule)

$\ddot{I}Cl_4^-$: sp^3d^2 (~~Square planar~~)

⇒ $Cl_2O_6 \longrightarrow ClO_2^+ ClO_4^-$

Note: PF_5 is not ionic in solid state due to very strong P-F bonds.

OR

PF_5 remains covalent in solid state also.

Eg. $XeF_6(s) \longrightarrow \ddot{X}eF_5^+ F^-$

$\ddot{X}eF_5^+$: sp^3d^2

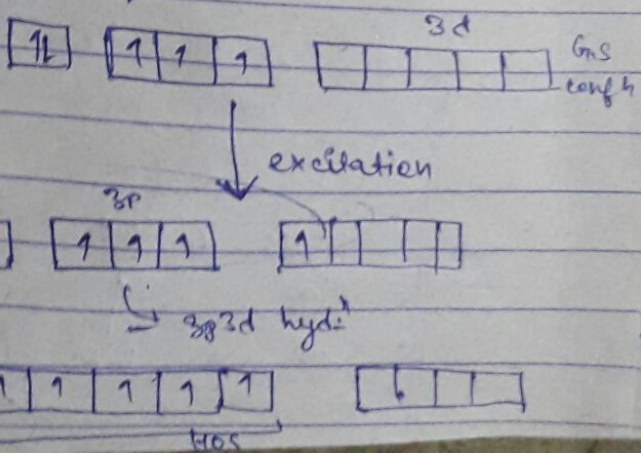
Q which can conduct electricity.

A. Graphite
B. $BrF_3(l)$
C. Sodium Metal
D. Diamond.

Q Explain why PCl_5 exist but not PH_5 ?

Soln: PCl_5 : P : $3s^2\ 3p^3\ 3d^0$

1s

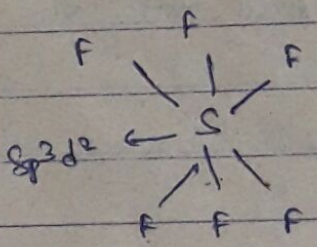


PCl_5 exist (or forms) because 'Cl' being more EN causes a partial +ve charge on 'P' which decreases size and energy of outer orbitals of 'P' due to which hybⁿ become easier.

* 'H' has nearly same EN as 'P' and hence does not decrease energy of outer orbitals of 'P' and hence hybd^{n} does not occurs so, PH_5 does not exist.

* Similarly SF_6 exist but not SH_6

Q) SF_6 exist but not SCl_6 (or SBr_6 or SI_6) ?

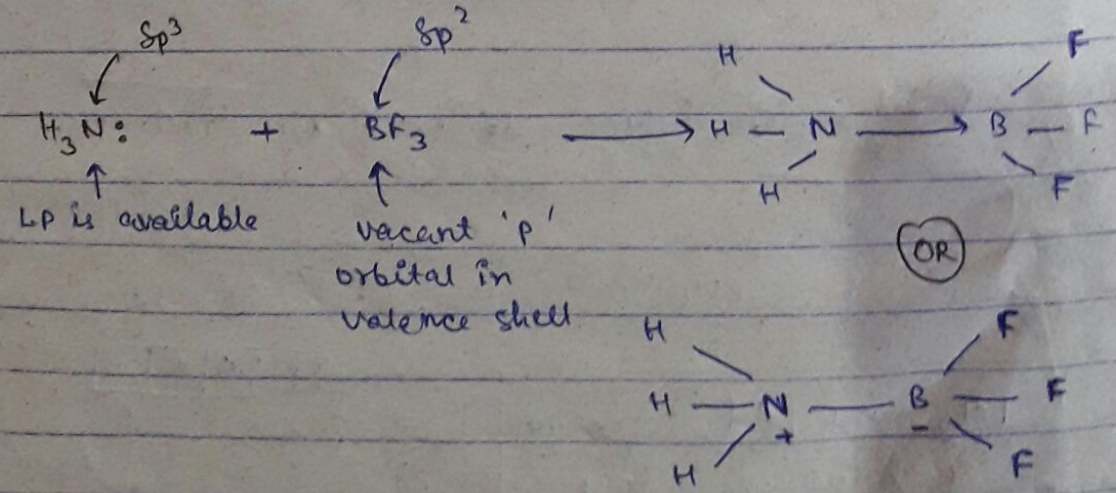
Soln 

→ 6 'Cl' being large in size, cause steric hindrance around 'S' and SCl_6 does not exist.

(*) Lewis Acid - Base Theory :-

→ lone pair donor is called lewis Base (L·B) and L.P acceptor is called lewis acid (L·A)

→ The bond formed between L·A and L·B in coordinate bond.

eg. 

eg. $H_3N:$ + H^+ ⟶ NH_4^+

↑ L·B ↑ L·A

eg. of lewis base : $\ddot{N}H_3$, $\ddot{P}H_3$, X^- (F^-, Cl^-, Br^-, I^-), $H_2\ddot{O}$, $:\ddot{O}H^-$, O^{2-} etc.

eg. of lewis acids → There may be a variety in types of lewis acids.

* Types of lewis acids :-

(1) Species having incomplete octet in valence shell.

eg. BF_3, BCl_3, BBr_3, BI_3, $AlCl_3$, BeF_2 etc.

$$\underset{\text{L.A}}{BeF_2} + \underset{\text{LB}}{2F^-} \longrightarrow BeF_4^{2-}$$

$$\underset{\text{LA}}{BF_3} + \underset{\text{LB}}{F^-} \longrightarrow BF_4^-$$

(2) Compounds having vacant 'd' orbitals in their valence shell.

eg. PCl_3, PCl_5, SiF_4, $SiCl_4$ etc.
BrF_3, ClF_3 etc.

$$\underset{\text{LA}}{PCl_5} + \underset{\text{LB}}{Cl^-} \longrightarrow PCl_6^-$$

$$\underset{\text{LA}}{SiF_4} + \underset{\text{LB}}{2F^-} \longrightarrow SiF_6^{2-}$$

$$\underset{\text{LA}}{:\ddot{B}rF_3} + \underset{\text{LB}}{F^-} \longrightarrow BrF_4^-$$

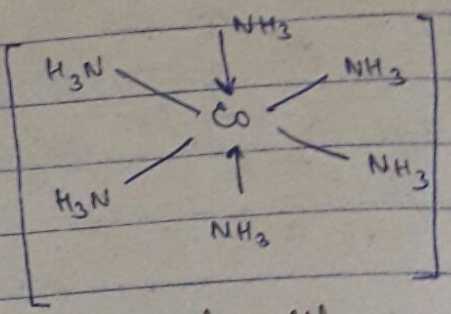
(3) Metal cations of small size :-

eg. LA $\{ H^+, Be^{2+}, Ag^+, Fe^{2+}, Fe^{3+}, Co^{2+}, Co^{3+}, Au^{3+}, Mg^{2+}, Zn^{2+}, Al^{3+}$ etc.

eg. $H_3N: + H^+ \longrightarrow NH_4^+$

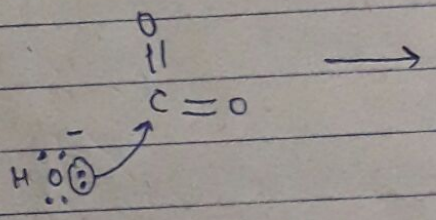
↑ L.A

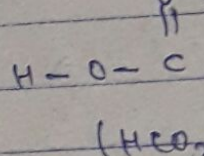
$Co^{3+} + 6NH_3 \rightarrow [Co(NH_3)_6]^{3+}$



(4) Molecules in which central atom is attached with EN atom with multiple bonds :

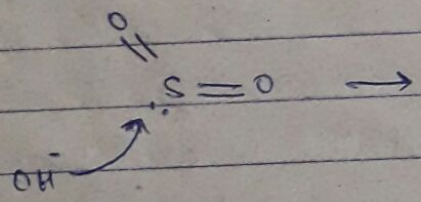
eg. CO_2, SO_2

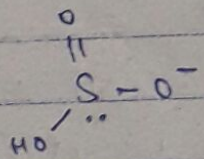




(HCO_3^-)

(Bicarbonate ion)





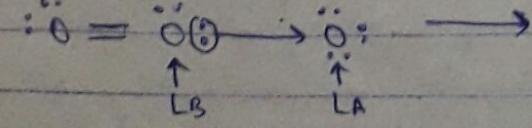
$[HSO_3^-]$

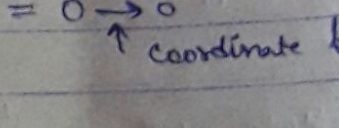
(Bisulphite ion)

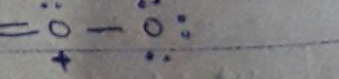
(5) When combining atoms has 6 e^- in valence shell -

eg. O_3

$O_2 + O \rightarrow O_3$







(Resonating Structure)

Q. Which cannot act as lewis acid ?

A. SiF_4
B. $AlCl_3$
C. $ZnCl_2$
✓ CCl_4

↳ No vacant 'p' or 'd' orbital in valence shell.

⇒ $Co^{3+} > Co^{2+}$

↑ Strongest LA

$H^+ > Li^+ > Na^+ > K^+ > Rb^+ > Cs^+$

decreasing lewis acid strength
{size factor}

Size ↑ lewis acid strength ↑

Eg. $BeCl_2 > AlCl_3 > ZnCl_2$

decreasing lewis acid strength.

* Comparing the lewis base strength :-

(a) Along the period from left to Right. EN ↑
tendency to donate L.P decreases and hence lewis Base strength decreases.

$\ddot{N}H_3 > H_2\ddot{O} > H\ddot{F}:$

decreasing basic nature

$\ddot{P}H_3 > H_2\ddot{S} > H\ddot{C}l:$

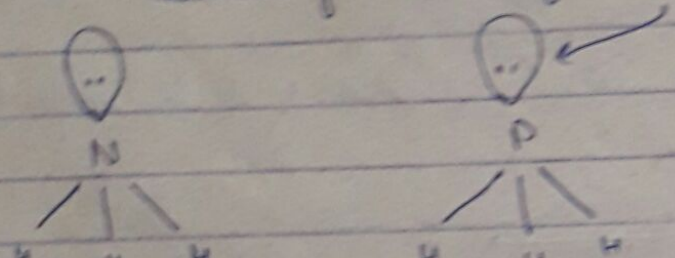
decreasing basic nature.

(b) Draw the order :
Size of L.P donating atom is deciding factor.

→ From T to B L.B strength decreases.

$NH_3 > PH_3 > AsH_3 > SbH_3$
decreasing L.B strength

$H_2O > H_2S > H_2Se$
decreasing L.B strength



diffused lone pair as it is present in pure 's' orbital as per hybd". in PH_3
(Bragg's Rule)

Better L.B than PH_3

(*) Comparison of Bond angle :-

→ According to VSEPR theory, bond angle in polyatomic molecules depends on basis.
- State of hybd"
- No. of L.P on Central atom
- EN of central atom.
- Size of terminal atoms/groups.

→ VSEPR theory says :

LP - LP > LP - BP > BP - BP
decreasing order of repulsions caused.

Case I : State of hybd. hybridization is different

→ % s character ↑ , Bond angle ↑

eg. $BeCl_2$ > BCl_3 > CCl_4

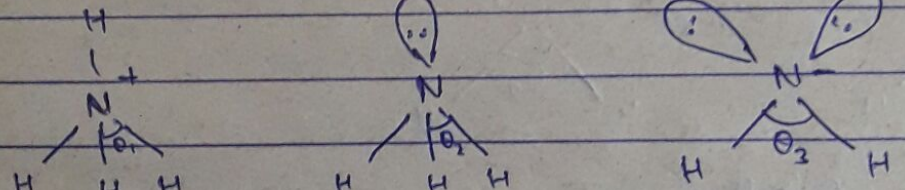
(120° 120° 109°28')

	$BeCl_2$	BCl_3	CCl_4
	sp	sp^2	sp^3
% s →	50%	33.3%	25%

→ decreasing Bond Angle

Case II - Hybd^n is same, terminal atom is same, No. of lone pairs are different.

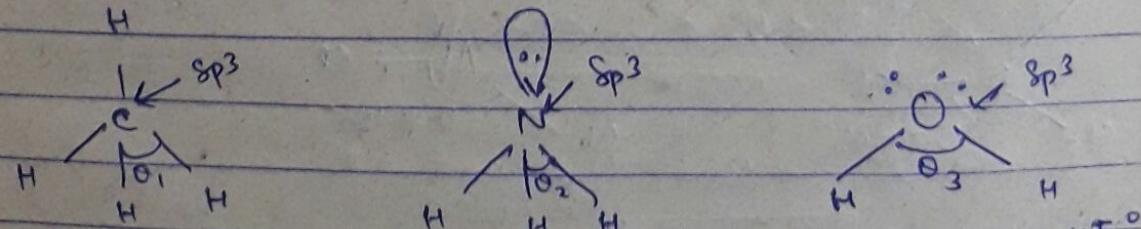
eg. NH_4^+ $\ddot{N}H_3$ $:\ddot{N}H_2^-$



$= 109°28'$ $= 107°$

$\theta_1 > \theta_2 > \theta_3$

eg. CH_4 NH_3 H_2O



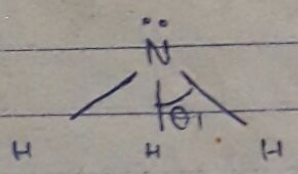
$109°28'$ $\theta_1 > \theta_2 > \theta_3$ $107'$ $= 104.5°$

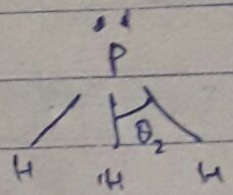
No. of L.P ↑, Bond angle ↓

Case - II Terminal atoms ~~of~~ Same, No. of L.P are same. Central atom is different. ($\neq 0$)

EN of Central atom ↑, B.A ↑

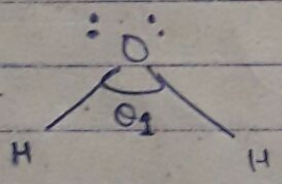
Eg. NH_3 PH_3

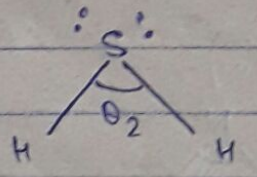




$\theta_1 > \theta_2$

Eg. H_2O H_2S

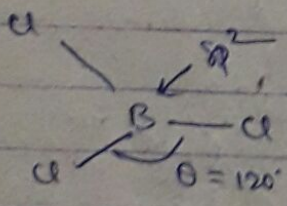


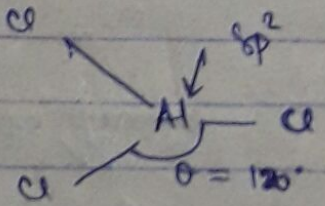


$\theta_1 > \theta_2$

Eg. BCl_3 $AlCl_3$

θ_1 θ_2





$\boxed{\theta_1 = \theta_2}$

Alternate Approach —

(Drago's Rule) –

NH_3 H_2O

Bond Angle ≃ 90° ← PH_3 H_2S → In these cases, Drago's Rule applies.

No Hybridisation ↙ AsH_3 H_2Se

SbH_3 H_2Te

3rd & highes period elements when attached to terminal atoms of low EN (≤ 2.5) eg H, C, then there is no hybd^n of central atom if central atom has one or more L.P in polyatomic species. (Drago's Rule) →

In such cases pure 'p' orbitals of Central atom, form bond with terminal atoms (like H or C)

Logic : Same explanation why SF_6 exist exist but not SH_6 [or PF_5 exist but not PH_5]

eg. 107° NH_3 | 93.5° PH_3 | 92° AsH_3 | 91° SbH_3

104.5° H_2O | 92° H_2S | 91 H_2Se | 90° H_2Te

→ Experimental evidence of these bond (≈90°) indicate no hybd^n and pure 'p' orbitals take part in bond formation.

Q. Which has/have bond angle close to 90°

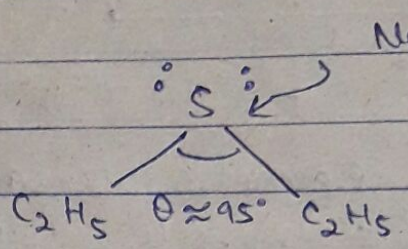
A. PF_3

B. PH_3 ✓

C. H_2O

D. ✓ $(C_2H_5)_2S$

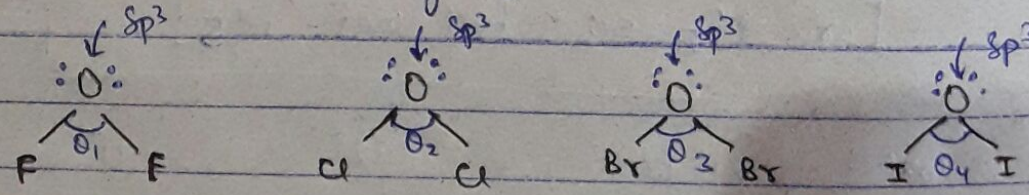
$(C_2H_5)_2S$ ⇒ Thioether



Case - IV Hybd^n same, No. of L.P same, central atom same, terminal atom is different.

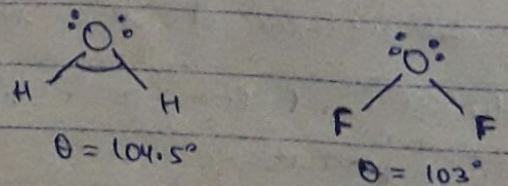
⇒ In general, larger is the size of terminal atom, greater will be the bond angle. (Except in case of 'F' terminal atom where EN is the deciding factor.)

eg.

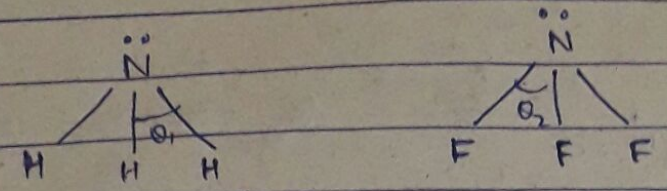


$$\theta_1 < \theta_2 < \theta_3 < \theta_4$$

* To compare between H_2O and OF_2

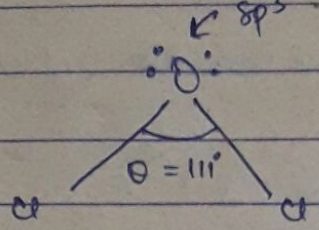


Eg. NH_3 and NF_3 (Same logic of EN will be deciding factor)



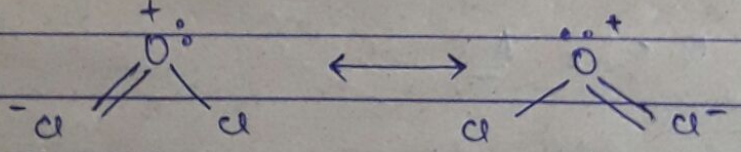
$\theta_1 > \theta_2$ Ans.

Imp Note : OCl_2 or Cl_2O



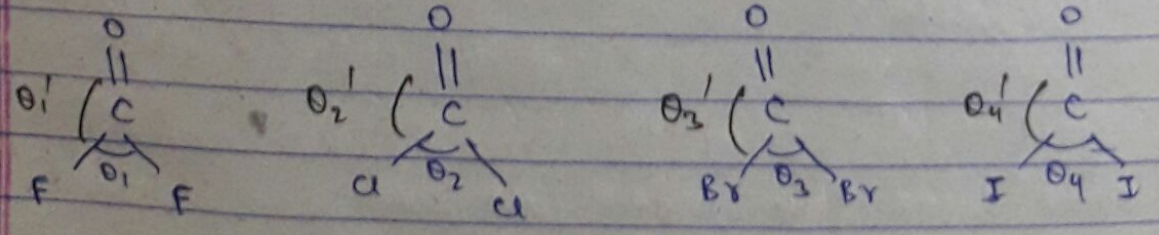
Q Why bond angle in Cl_2O is greater than the tetrahedral angle of 109° 28'.

Ans: Back Bonding between L.P of 'O' and vacant 'd' orbital of Cl atoms.
($p\pi$-$d\pi$ back bond)



⇒ Double bond character in O-Cl bond causes greater repulsion or L.P repulsions on bonds is reduced, as a result bond angle increases and is found to be 111°

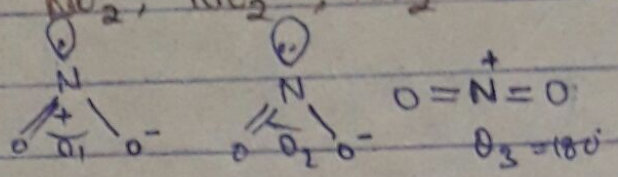
Eg. COF_2 $COCl_2$ $COBr_2$ COI_2



$\theta_1 < \theta_2 < \theta_3 < \theta_4$
$\theta_1' > \theta_2' > \theta_2' > \theta_4'$ } order of Bond angle }

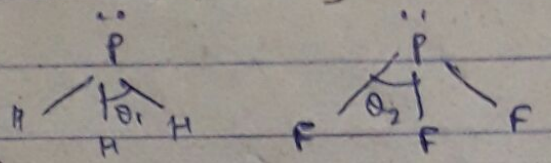
Q. Compare bond angles among the following –

(i) NO_2, NO_2^-, NO_2^+



$\theta_2 < \theta_1 < \theta_3$ Ans

(ii) PH_3, PF_3

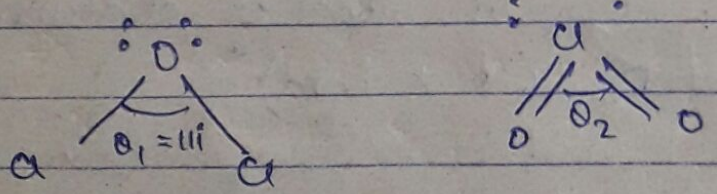


$\theta_1 \not< \theta_2$ $\theta_1 < \theta_2$

(iii) BF_3, BCl_3, BBr_3, BI_3

All bond are equal as there is no lone pair on central atom

(iv) Cl_2O, Cl_2O_2



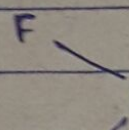
$\theta_1 < \theta_2$ Ans

(*) Back Bonding :-

Any π-coordinate bond is back bond.

↑ Bond formation between a vacant and fully filled orbitals.

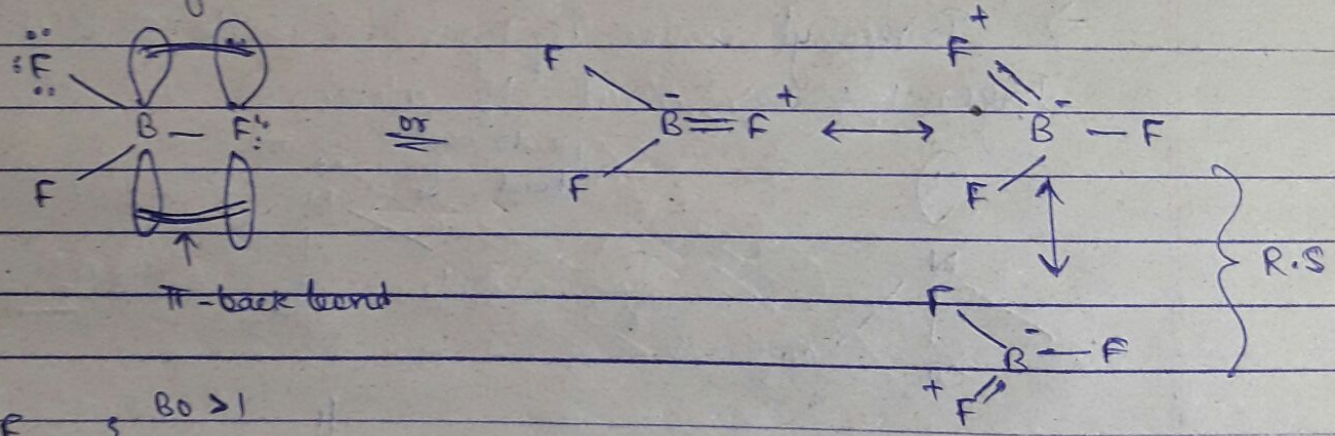
eg. BF_3 (sp^2)

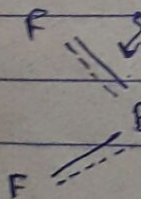


{ a good lewis acid }

↳ B-F bond in BF_3 is found to be Stronger and shorter than expected.

→ To explain this concept of backbonding is used in which vacant p-orbital of boron overlaps with fully filled p-orbital of 'F' forming $p\pi - p\pi$ back bond.

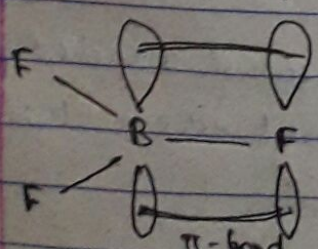


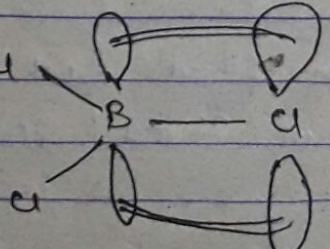


→ Experimentally all B-F bonds are equivalent in BF_3 molecule.

Imp → $BF_3 > BCl_3 > BBr_3 > BI_3$

decreasing order of extent of back bonding.





weak ($p\pi - p\pi$) back bond due to large size of Cl atom

V. Imp

$BF_3 < BCl_3 < BBr_3 < BI_3$

increasing lewis acid strength.

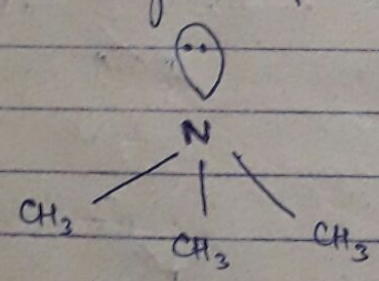
eg. Trimethyl amine $(CH_3)_3N$: Trisilyl amine $(SiH_3)_3N$

$CH_4 \xrightarrow[-H]{} CH_3$ Methyl

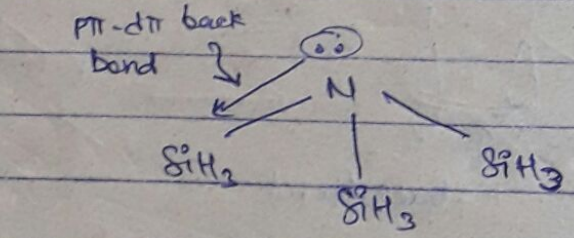
$SiH_4 \xrightarrow[-H]{} SiH_3$

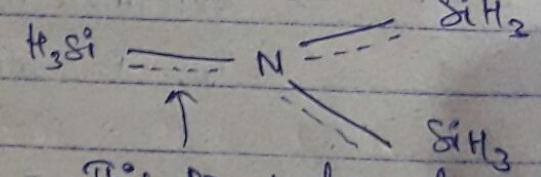
Silane Silyl group

⇒ Trimethyl amine is pyramidal while Trisilyl amine is trigonal planar about 'N' atom.



Pyramidal





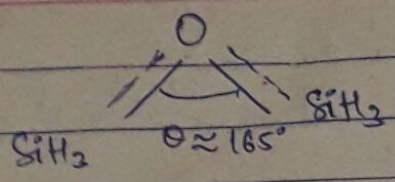
Trigonal planar

This pπ-dπ back bond is delocalised in entire molecule.

Imp

* Silicon always form pπ-dπ back bond when attached to 'N', 'O' atoms as in silica (SiO_2), silyl ethers ($H_3Si-O-SiH_3$)

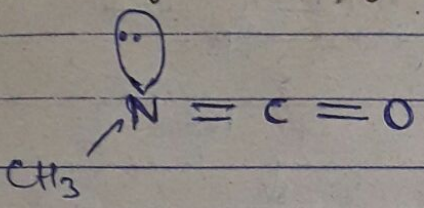
Silyl Ethers



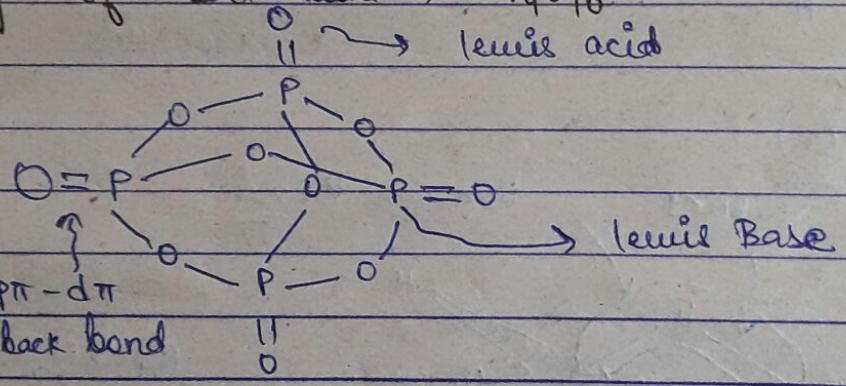
Silyl isothiocyanate $SiH_3 = N = C = O$ (Linear around 'N' atom)

↑ $p\pi - d\pi$ back bond.

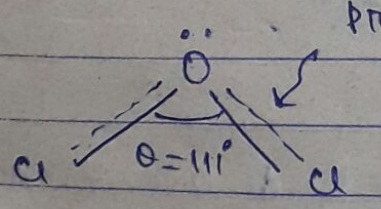
eg. Methyl isothiocyanate,



eg. of back bond, P_4O_{10}

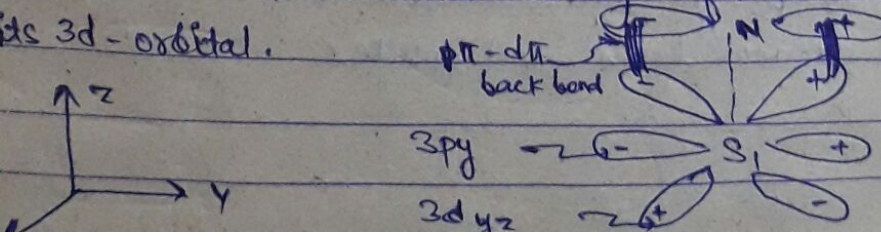


eg. OCl_2



gmp

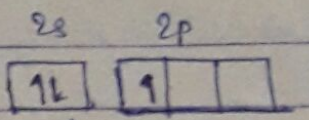
Note: Silicon is not comfortable in making π-bonds with its 3p-orbital, rather it easily forms π-back bond with its 3d-orbital.



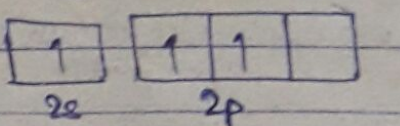
⊛ 3c - 2e⁻ bonds (or Banana Bonds)
{ Electron deficient bonds }

eg. Diborane, B_2H_6 { A powerful Reducing Agent, A good rocket fuel }

$_5B$: $1s^2\ 2s^2\ 2p^1$

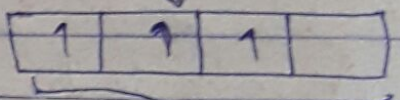


↓ Excitation

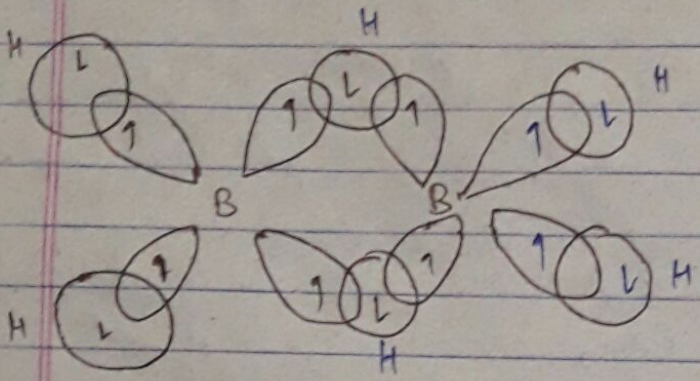


sp^3 hybd^n

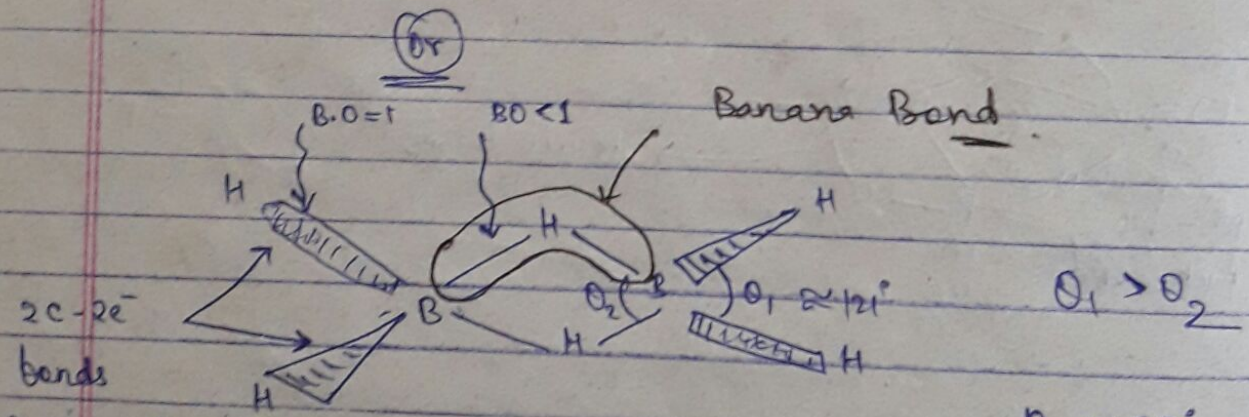
↓



$4\ sp^3$ HOs



(or)



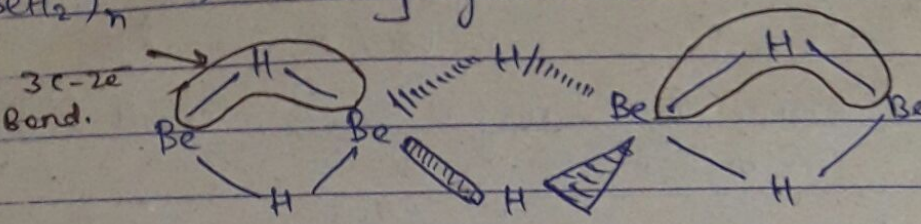
→ In diborane, terminal B-H bonds are stronger than the bridged B-H bonds.

→ Since B_2H_6 is an electron deficient molecule hence it invites attack from nucleophiles.

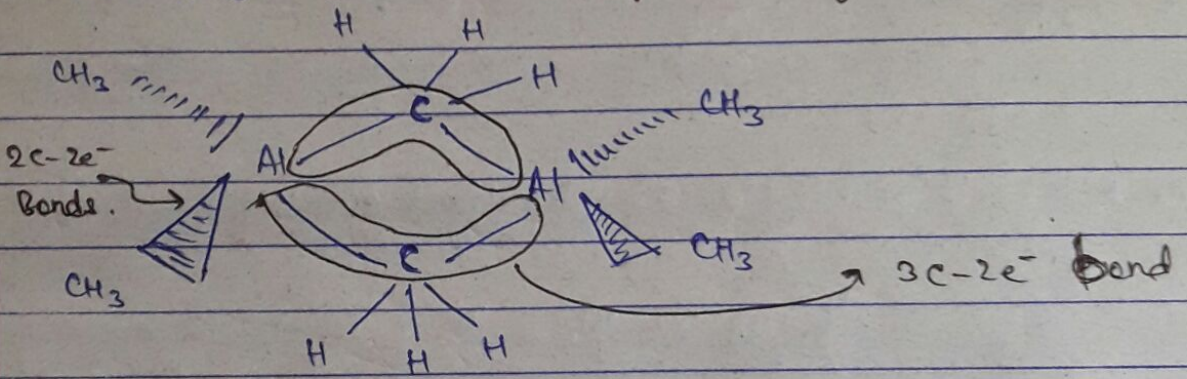
Other examples of $3c-2e^-$ bonds are :-

$(BeH_2)_n$, $(AlH_3)_n$, $(GaH_3)_n$, $Al_2(CH_3)_6$, $Ga_2(CH_3)_6$ etc.

$(BeH_2)_n$ → It is polymeric chain like solid at room temp.



$Al_2(CH_3)_6$ (It is dimer of trimethyl aluminium)

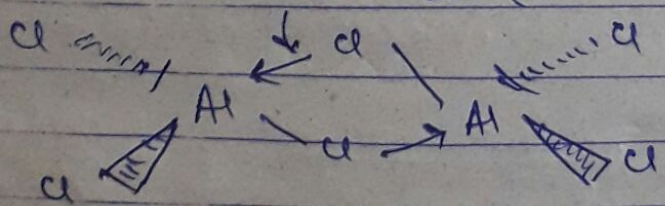


Q. Which has / have $3c-2e^-$ bonds.

(A) $(AlH_3)_n$ (B) B_2H_6

(C) Al_2Cl_6 (D) Be_2Cl_4

Sol^n : In Al_2Cl_6 and Be_2Cl_4 have $2c-2e^-$ bonds



Non-planer Molecule

Q. BH_3 monomer does not exist and exist as dimer i.e B_2H_6 whereas BCl_3 exist as ~~dimer~~ monomer Why?

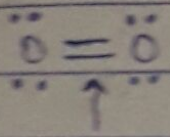
Sol^n BH_3 tries to completes its valency by forming dimer

(H atom easily gets bridged due to small size)

→ BCl_3 does not form dimer as 'Cl' is large atom and 4 'Cl' atoms around small sized 'B' cause heavy steric hindrance.

(*) Molecular Orbital Theory - (MOT)

→ According to VBT,
O_2 molecule is expected to be diamagnetic



no unpaired e^- as per V.B.T

⇒ But the fact is that liquid O_2 sticks between the poles of magnet indicating paramagnetic behaviour.

⇒ Paramagnetic behaviour can be successfully explained using M.O.T.

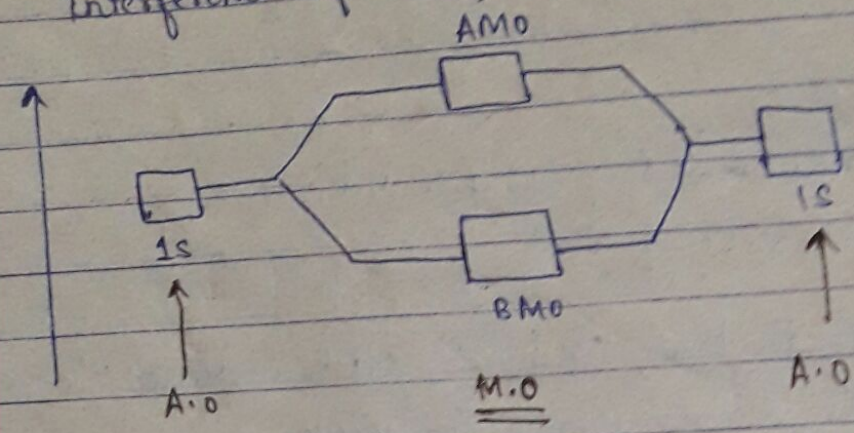
⇒ formation of Molecules like C_2, B_2 etc can be easily understood using MOT

* Features of MOT

① Appropriate atomic orbitals of combining atoms combine through LCAO (linear combination of Atomic Orbitals) and form equal number of molecule orbitals.
After the formation of molecule orbitals, there remain no existance of atomic orbitals.

② In Molecular Orbitals (MO), all electrons remains under the influence of all nuclei and hence MO is called polycentric. However, atomic orbitals in VBT are called monocentric.

(3) When two appropriate AO of bonding atoms do combine through LCAO, two MOs are formed (Viz BMO is formed by constructive interference and AMO is formed by destructive interference of AOs)



Due to in-phase combination of wave functions of AOs, BMO is formed.

$$\psi_{BMO} = \psi_1 + \psi_2$$ [when same sign wave functions ψ of AOs combine]

$$\psi^2_{BMO} = \underbrace{\psi_1^2} + \underbrace{\psi_2^2} + \underbrace{2\psi_1\psi_2}$$

↑ Electron density in BMO increases by a factor of $2\psi_1\psi_2$, so BMO is more stable than AO.

When "out of phase" combination occurs (i.e destructive interference)

$$\psi_{AMO} = \psi_1 - \psi_2$$ [when opposite sign wave functions do combine]

$$\psi^2_{AMO} = \psi_1^2 + \psi_2^2 - \underbrace{2\psi_1\psi_2}$$

↑ Electron density decreases by a factor of $2\psi_1\psi_2$ so AMO is less stable than AOs.

(4) Atomic orbitals taking part in LCAO, should have equal or nearly equal energy.

⇒ In homonuclear diatomic molecules (O_2, N_2, F_2 etc) only following combination shall occur.

1s - 1s
2s - 2s
2p - 2p

⇒ However in heteronuclear molecules & different types of AOs may combine depending upon their energies.

1s - 1s
1s - 2s
1s - 2p
} In heteronuclear molecules. (CN, NO etc) HF

(5) Molecular orbitals are classified as σ or π depending upon cylindrical symmetry.

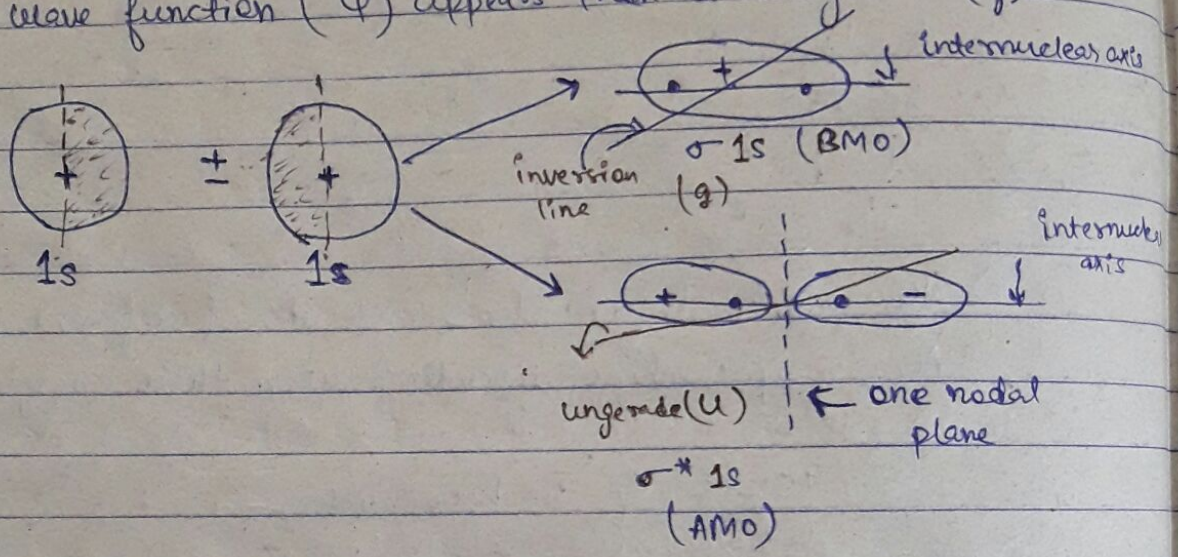
→ If MO has cylindrical symmetry, then it is σ, otherwise it is π MO.

→ Cylindrical Symmetry:

Rotate MO about its internuclear axis through 180°, if same sign of wave function appears, then MO has cylindrical symmetry and hence it will be designated as σ (sigma) MO.

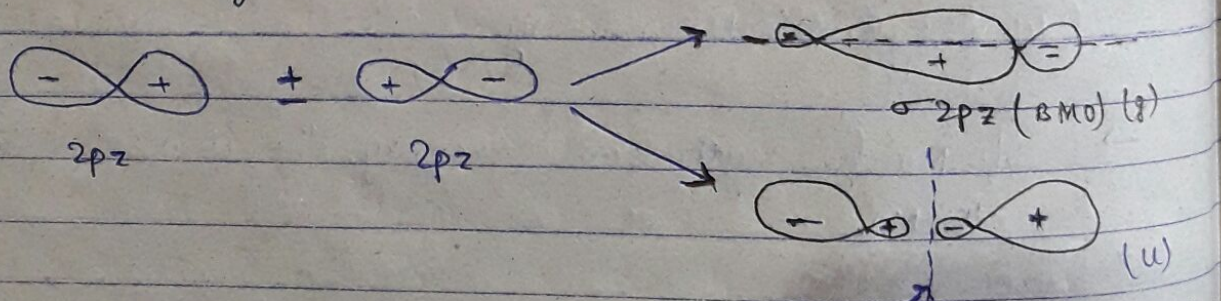
→ MO can also be designated as gerade (g) (even in german language) or ungerade (u) {odd}. ~~depending upon its nature to~~

* Move equidistant from centre of MO on inversion line, if same sign wave function (ψ) appears then it is gerade (g).

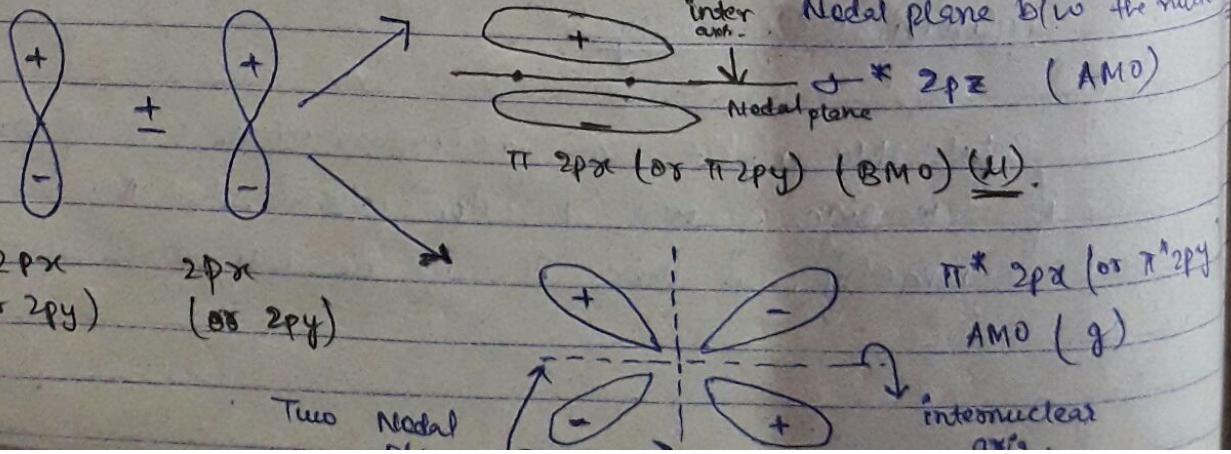


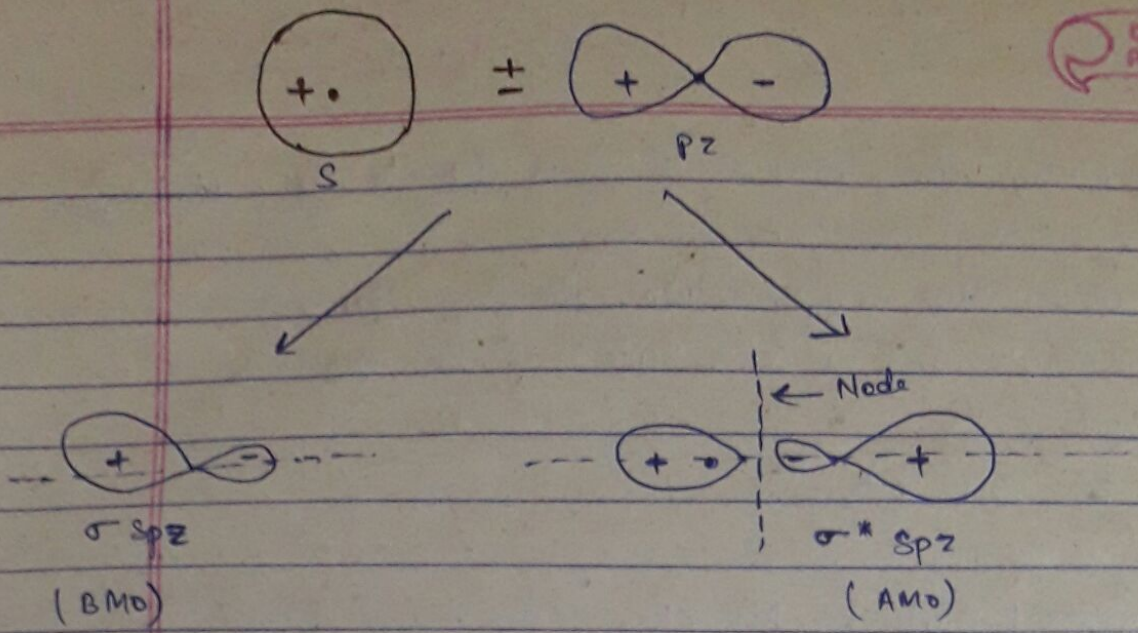
* AMO has one extra node (or nodal plane) relative BMO when two AOs combine

⇒ In our analysis, z axis is taken to be the internuclear axis.



⇒





'g' & 'u' are not defined in such unsymmetrical cases.

(6) Energies of various MOs are experimentally measured for homonuclear diatomic molecules as under :-

(a) for total e^- > 14 :
(F_2, O_2 etc)

z → internuclear axis

(Normal energy order)

$\sigma 1s$, $\sigma^* 1s$, $\sigma 2s$, $\sigma^* 2s$, $\sigma 2pz$, $\underbrace{\pi 2px = \pi 2py}_{\text{degenerate}}$,
$\pi^* 2px = \pi^* 2py$, $\sigma^* 2pz$

→ increasing energy order

(b) for total $e^- \leq 14$.
(eg. B_2, C_2, N_2 etc)

(exceptional energy order)

$\sigma 1s$, $\sigma^* 1s$, $\sigma 2s$, $\sigma^* 2s$, $\pi 2px = \pi 2py$, $\sigma 2pz$
$\pi^* 2px = \pi^* 2py$, $\sigma^* 2pz$

→ increasing energy order.

* This exception (for total $e^- \leq 14$) is due to s & p mixing.

* while write MO configuration, the three rules Pauli's, Afbau's and Hund's rule will be applied.

eg. H_2 (2) : $\sigma 1s^2, \sigma^* 1s^0$ .....

Bond order (B.O) = $\frac{N_b - N_a}{2}$

where N_b = No. of e^- filled in BMO
N_a = No. of e^- filled in AMO

B.O = $\frac{2-0}{2}$ = 1 (equivalent to single bond)

H_2 ⟶ Diamagnetic

Note :

(1) Greater is the bond order, greater will be the thermodynamic stability. Zero Bond order indicates that species does not exist.

V. Imp (2) while comparing among the derivatives of a species, if BO is equal, then greater are the no. of anti-bonding electrons, lesser will be the thermodynamic stability.

(3) A fractional B.O indicates paramagnetic nature of species.

V. Imp (4) N_2^- (Total $e^- = 15$) : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2px^2 = \pi 2py^2, \sigma 2pz^2, \pi^* 2px^1 = \pi^* 2py^0, \sigma^* 2pz^0$.

V.V. Imp ★ While writing MO confin. of an ion, the energy order corresponding to its neutral molecule shall be followed.

$$B.O = \frac{10-5}{2} = 2.5$$

(Paramagnetic)

No. of unpaired $e^- = 1$

Spin mag. moment $= \sqrt{x(x+2)}$ B.M

$= \sqrt{3}$ BM

eg. O_2^{2+} : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2pz^2, \pi 2px^2 = \pi 2py^2, \pi^* 2px^0 = \pi^* 2py^0, \sigma^* 2pz^0$

(total $e^- = 14$)

B.O = 3, (Diamagnetic)

eg. ~~He2~~ : He_2 : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^0 \ldots$

$BO = \frac{2-2}{2} = 0$

(He_2 does not exist)

Q Compare stability of H_2^+ and H_2^-

(1) $H_2^+ = \sigma 1s^1, \sigma^* 1s^0$, $BO = 1/2$

(3) $H_2^- = \sigma 1s^2, \sigma^* 1s^1 = BO = 1/2$

H_2^+ is more stable than H_2^- } Refer to point (2)

or Bond E H_2^+ > B.E H_2^- }

eg. B_2 ($10e^-$) : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^1 = \pi 2p_y^1, \sigma 2p_z^0$ ....

HOMO — Highest occupied MO

LUMO — Lowest unoccupied MO

Bond order $= \frac{6-4}{2} = 1$

Alternatively. $B.O = \frac{2-0}{2} = 1$ (π-bond)

$B_2 \rightarrow$ Paramagnetic

eg. C_2 ($12e^-$) : $\sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^0$ ....

HOMO, LUMO

$B.O = \frac{4-0}{2} = 2$ (Both bonds in C_2 are π-bonds)

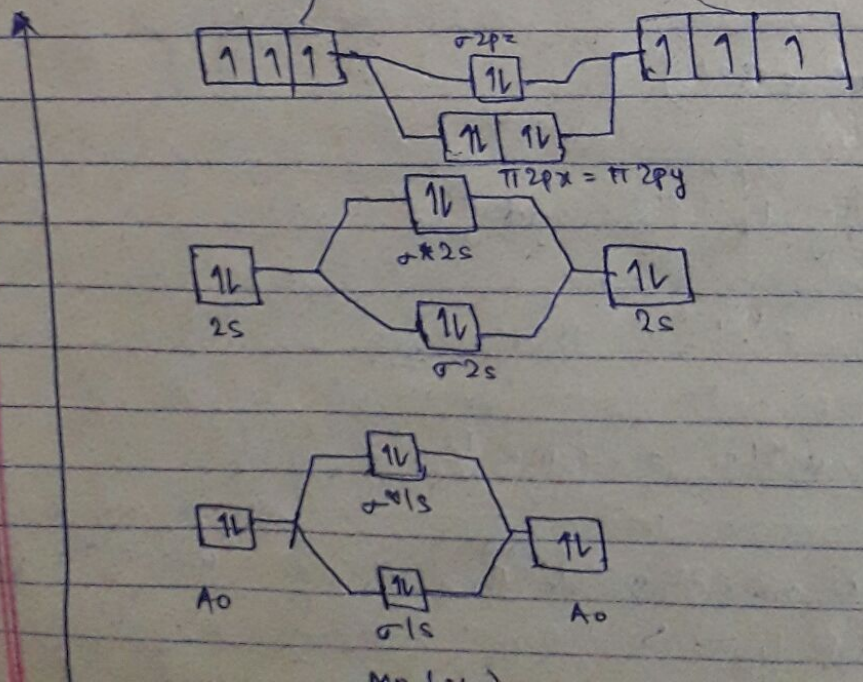
$C_2 \rightarrow$ Diamagnetic

eg. N_2 ($14e^-$) : KK $\sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^2, \pi^* 2p_x^0 = \pi^* 2p_y^0$

HOMO, LUMO

$BO = \frac{6-0}{2} = 3$ ($2\pi, 1\sigma$) bond

$N_2 \rightarrow$ Diamagnetic



O_2 (16e⁻) : KK, $\sigma 2s^2$, $\sigma^* 2s^2$, $\sigma 2pz^2$, $\pi 2px^2 = \pi 2py^2$, $\pi^* 2px^1 = \pi^* 2py^1$, $\sigma^* 2pz^0$

($\pi^* 2px^1 = \pi^* 2py^1$: HOMO ; $\sigma^* 2pz^0$: LUMO)

Increasing order of energy of MOs →

$BO = \frac{6-2}{2} = 2$ (one σ and one π)

O_2 → Paramagnetic.

IIT-JEE ADVANCED

Eg. (*) S and P- mixing :-

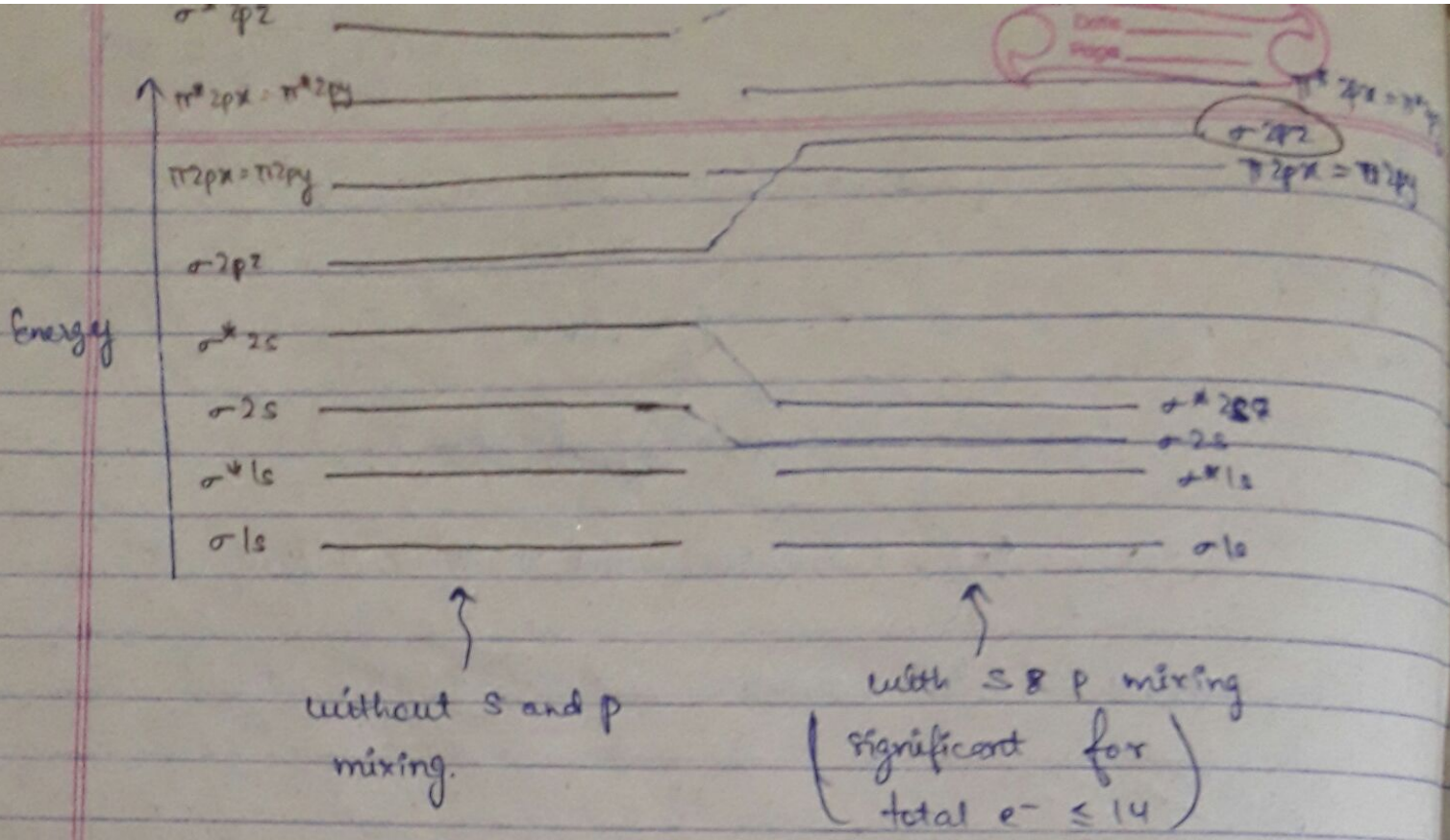
(Modification in Energies of MOs due to S & p mixing)

(or known as Symmetry Interaction)

→ It is experimentally observed that MOs of similar symmetry and energies do interact and repel each other, due to which energy of high energy MO is further increased and energy of low energy MO is further reduced.

→ In cases of total $e^- \leq 14$, in homonuclear diatomic molecules, $\sigma 2s$ and $\sigma 2pz$ have similar symmetry and hence interact to form modified MOs in which energy of $\sigma 2s$ is reduced and that of $\sigma 2pz$ is increased.

* Also interaction between $\sigma^* 2s$ and $\sigma^* 2pz$ occurs in similar manner.



Note: Alternate theory of s & p mixing is that appropriate atomic orbitals of similar symmetry and energy first undergo hybridization and these HOs under go mixing through LCAO to form modified MOs.

2s + 2pz → Two sp hybrid orbitals → Two HOs → LCAO → σ2s and σ2pz → Modified HOs

2s + 2pz: atomic orbitals undergoing hybridization

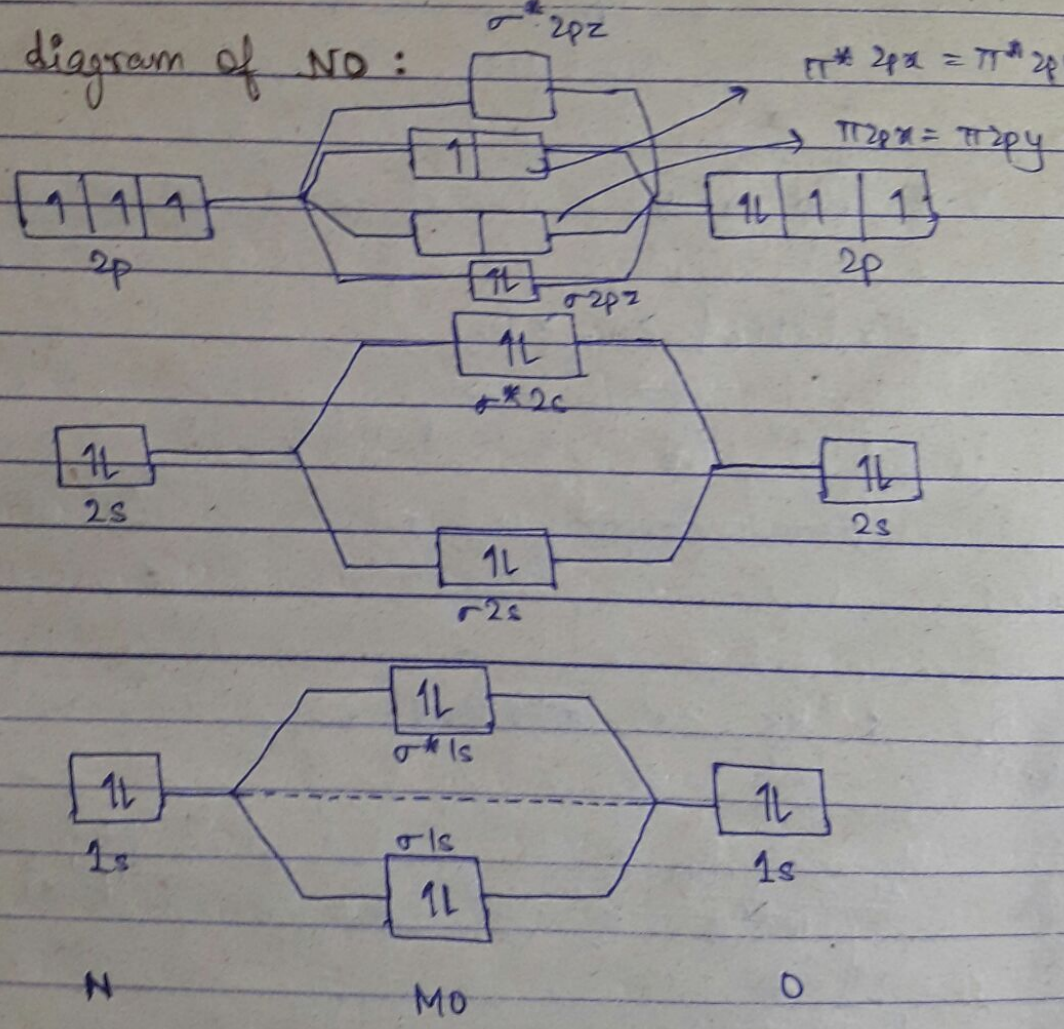
* (Heteronuclear Molecule)

eg. NO : (15e⁻) KK $\sigma 2s^2, \sigma^* 2s^2 \{ \sigma 2pz^2, \pi 2px^2 = \pi 2py^2$
$\pi^* 2px^1 = \pi^* 2py^0, \sigma^* 2pz^0$

B.O = $\frac{6-1}{2}$ = 2.5, one unpaired e⁻,
spin magn. moment = $\sqrt{3}$ BM

{Paramagnetic}

MO diagram of NO :



Q. Compare B.O among Na_2O_2, KO_2, O_2

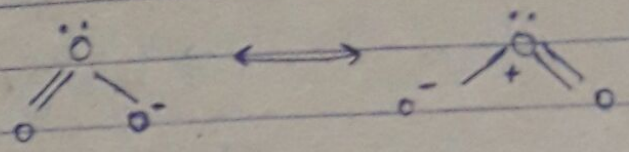
$Na_2O_2 \Rightarrow 2Na^+ \quad O_2^{2-}$
$KO_2 \Rightarrow K^+ \quad O_2^-$

$O_2 \rightarrow$ B.O = 2
$O_2^- \rightarrow$ B.O = 1.5
$O_2^{2-} \rightarrow$ B.O = 1

$Na_2O_2 < KO_2 < O_2$
increasing B.O

Q. Compare B.O among O_3, O_2, $O_2[PtF_6]$

Sol^n: O_3 ⇒



B.O in O_3 = 1.5

B.O in O_2 = 2

⇒ $O_2[PtF_6]$ ⇒ $O_2^+[PtF_6]^-$

↑

B.O = 2.5

⇒ O-O bond order,

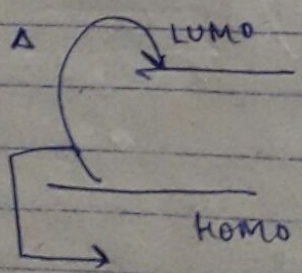
$O_2[PtF_6] > O_2 > O_3$

decreasing O-O bond order.

Note:

O_2 (16 e⁻) : KK KK', $\sigma 2p_z^2$, $\pi 2p_x^2 = \pi 2p_y^2$, $\pi^* 2p_x^1 = \pi^* 2p_y^1$ HOMO

$\sigma^* 2p_z^0$ LUMO



$\Delta = \frac{hc}{\lambda}$ → if this λ is in between 4000 A° - 7500° then colour will appear.

⇒ Despite having two unpaired e^-, O_2 is colourless because HUMO and LUMO energy difference does not correspond to any wavelength in visible region (4000 A° to 7500 A°)

⇒ F_2 is light yellow gas even though it has zero unpaired e^- because HOMO and LUMO energy difference in F_2 corresponds to one of the wavelength in visible region (4000 to 7500 A°)

* Exceptional Case of CO: ← Carbon Monoxide.

wrong M.O confⁿ.

CO (14e⁻) : $KK, \sigma 2s^2, \sigma^* 2s^2, \pi 2p_x^2 = \pi 2p_y^2, \sigma 2p_z^2, \pi^* 2p_x^0 = \pi^* 2p_y^0$

$$B.O = \frac{6-0}{2} = 3$$

CO^+ ⇒ Expected B.O in CO^+ = 2.5

* It was expected that C-O bond length in CO^+ to be longer than that in CO.

Imp * However, note the data

C-O bond length in CO^+ = 1.115 A°

C-O bond length in CO = 1.128 A°

OR

$$B.O_{(CO^+)} > B.O_{(CO)}$$

Dipole Moment { A measure of ionic character in a polar covalent bond. }

→ A dipole is the two charges separated by a certain distance (called charge separation).

→ Consider a polar diatomic molecules

eg. $H^{\delta+}$ —— $Cl^{\delta-}$

←——— d ———→

→ Dipole moment (or bond moment) of a dipole is the product of magnitude of charge and charge separation (or bond length). It is a vector quantity and represented by an arrow pointing towards more EN atom.

$\delta+$ ——→ $\delta-$

A ——— B

←— d ———→ ↑ More EN

Dipole moment (μ) = $\delta \times d$

eg. $H^{\delta+}$ —— $F^{\delta-}$

μ ——→

$\mu = \delta \times d$

Unit of 'μ' = Coulomb-meter (in SI)
= esu-cm (in CGS)

→ More commonly used unit of dipole moment is Debye

1 Debye = 10^{-18} esu-cm

$1e^-$ charge = 4.8×10^{-10} esu

= 1.6×10^{-19} Coulomb

$\Rightarrow$ 1 esu = $\frac{1.6 \times 10^{-19}}{4.8 \times 10^{-10}} = \frac{1}{3} \times 10^{-9}$ Coulomb.

$1D = 10^{-18} \times \frac{1}{3} \times 10^{-19} C \times 10^{-2} m$

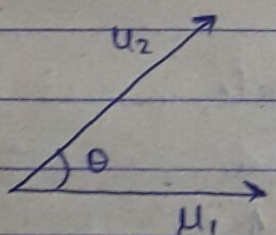
$1D = \frac{1}{3} \times 10^{-29}$ C-m

$1D = 3.33 \times 10^{-30}$ C-m → In SI

$1D = 10^{-18}$ esu-cm → In CGS

→ Net dipole moment of a polyatomic molecule is the vector sum of all dipoles

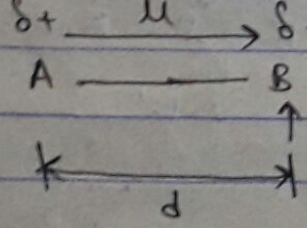
eg.



$\mu_{net} = \sqrt{\mu_1^2 + \mu_2^2 + 2\mu_1\mu_2\cos\theta}$

$\theta \downarrow$, $\mu_{net} \uparrow$

Note (1)



$\mu = \delta \times d$

δ → depends on Δ(EN)

→ out of 's' and 'd', 's' is the dominant factor i.e $\delta\uparrow$, i.e Δ(EN)$\uparrow$, $\mu\uparrow$

eg. $\underset{\text{decreasing dipole moment}}{\underrightarrow{H-F > H-Cl > H-Br > H-I}}$

Important Exception :—

C-Cl > C-F

↑ 1.5 D ↑ 1.4 D

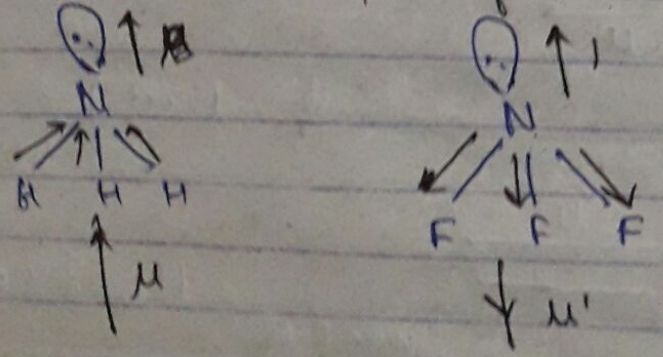
In this case Bond length is the dominant factor.

Imp $\mu_{CH_3Cl} > \mu_{CH_3F}$

↑ greater dipole moment

(2) Lone pair also contributes in dipole moment but it's contribution can't be quantified.

eg.



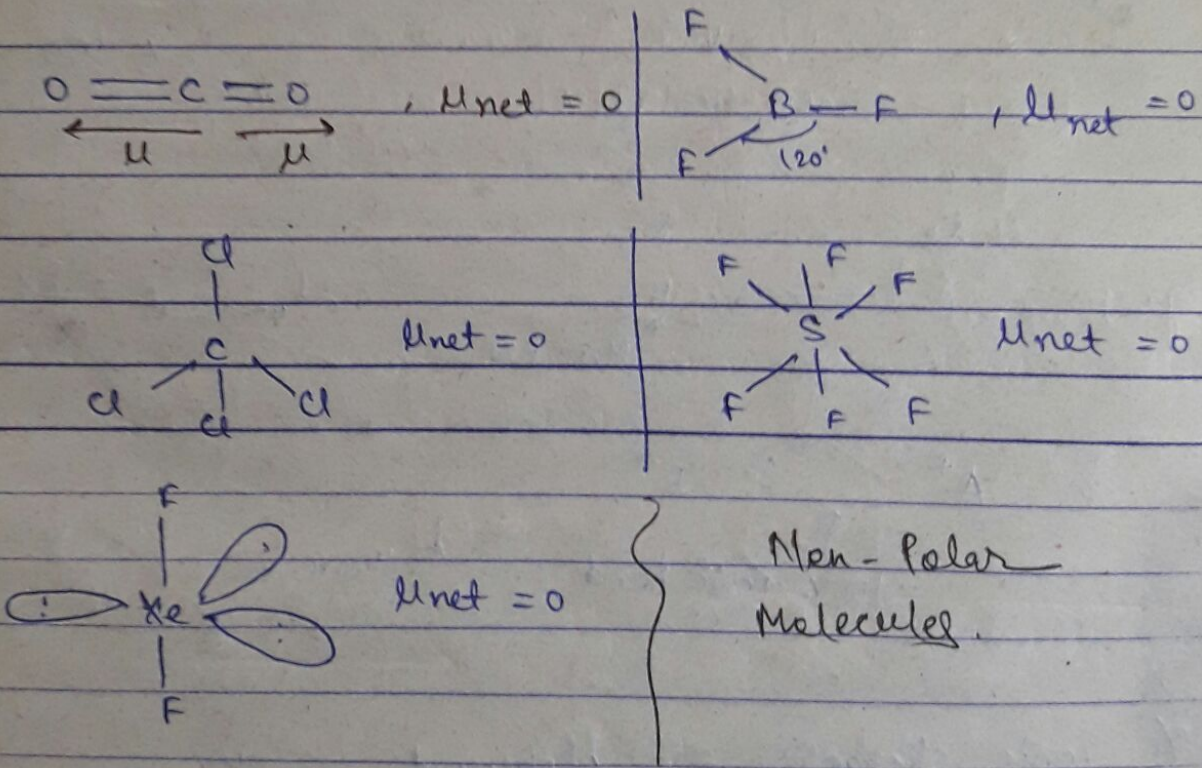
$\mu_{NH_3} >> \mu_{NF_3}$

* NH_3 is a highly polar molecule of high net dipole moment whereas NF_3 has a little dipole moment.

→ Dipole moment of L.P is taken in the direction away from atom bearing L.P

(3) If net dipole moment of a polyatomic molecule is zero, then we can say that it is a symmetrical molecule and such molecules with $\mu_{net} = 0$ are called Non-polar Molecules.

eg. CO_2, CS_2, BF_3, BCl_3, CH_4, CCl_4, $SiCl_4$, SF_6, XeF_2, XeF_4, PCl_5 etc.



→ Polar polyatomic molecules have $\mu_{net} \neq 0$

eg. of polar molecules

NH_3, H_2O, PCl_3, SF_4, SF_2, ClF_3 etc

Polar molecules

Q. Which is/are polar molecules?

A. PF_2Cl_3
B. $H.XeF_4$
C. IF_5
D. D_2O

Q. Which is/are planar as well as polar?

A. Borazine
B. BrF_3
C. XeF_6
D. H_2O

(4) % Ionic character in Polar Covalent Bond

$$= \frac{\text{Observed dipole moment}}{\text{theoretical D.M (considering 100\% ionic Bond)}} \times 100$$

eg. $\overset{\delta+}{A} \text{———} \overset{\delta-}{B}$ (B: More EN)

$\mu_{observed} = \delta \times B.L$

$A^+ \quad B^-$ ↑ 100% Ionic bond

$\mu_{theoretical} = 1e^- \times B.L$

$$\% \text{ Ionic character} = \frac{\mu_{observed}}{\mu_{theoretical}} \times 100$$

$$= \frac{\delta \times B.L}{1e^- \times B.L} \times 100$$

$$\% \text{ Ionic character} = \frac{\delta}{1e^-} \times 100$$

→ for H-Cl, H-F, LiCl

of molecule is like ZnS, MgS etc

$$\% \text{ ionic character} = \frac{\mu_{obs}}{\mu_{theo}} \times 100$$

$$= \frac{\delta \times B.L}{2e^- \times B.L} \times 100$$

V. Imp Some important Dipole Moment orders :-

→ H-F > H-Cl > H-Br > H-I
→ H_2O > H_2S
→ NH_3 > NF_3

} decreasing μ_{net}

Vimp

→ CH_3Cl > CH_2Cl_2 > $CHCl_3$ > CCl_4

decreasing μ_{net} ; $\mu_{net} = 0$

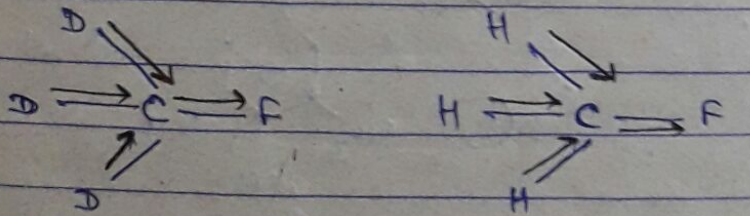
→ CH_3Cl > CH_3F

↑ greater D.M

Imp

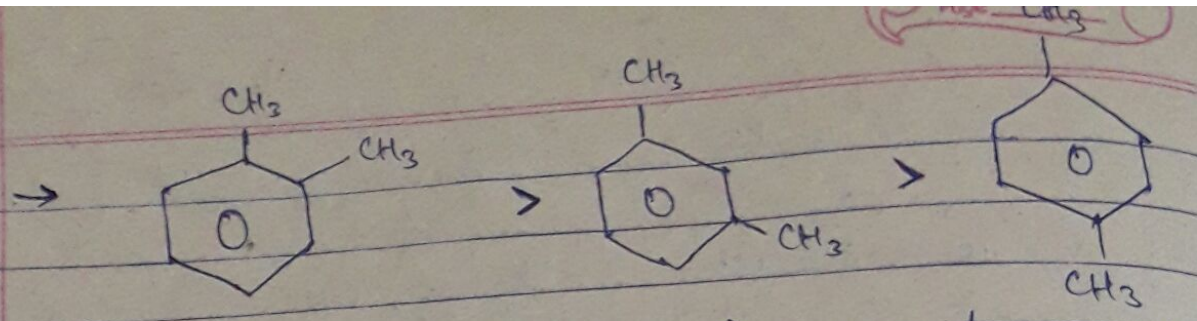
→ CD_3F > CH_3F

↑ greater D.M (indicates that $EN_D < EN_H$) (slightly)



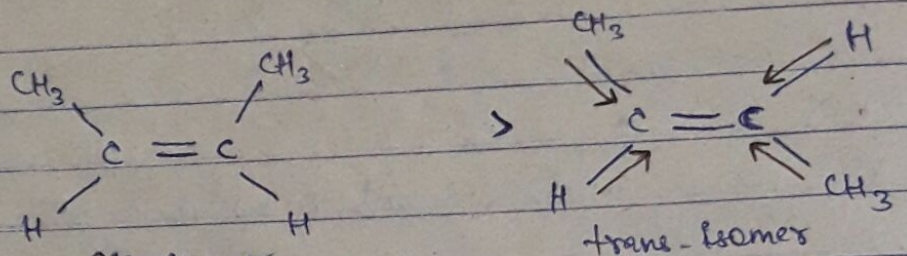
$\Delta(EN)$ in C-D > $\Delta(EN)$ in C-H

$\therefore$ $EN_D < EN_H$ (Imp)



decreasing dipole moment

($\mu_{net} = 0$)



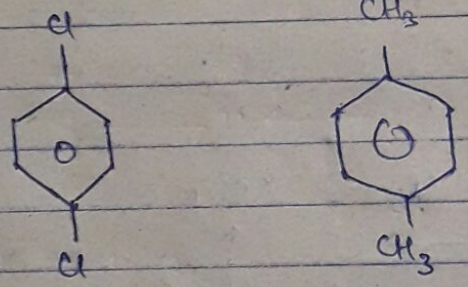
cis-isomer

($\mu_{net} \neq 0$)

trans-isomer

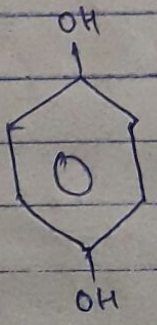
($\mu_{net} = 0$)

V. Imp

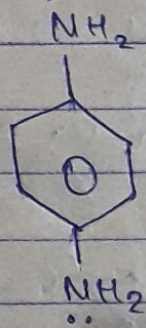


$\mu_{net} = 0$ $\mu_{net} = 0$

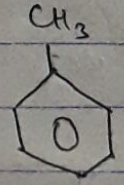
Very Imp But



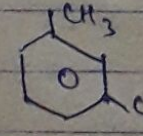
, $\mu_{net} \neq 0$

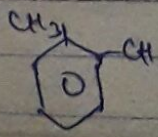


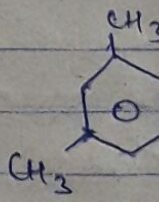
, $\mu_{net} \neq 0$

Q. Net dipole moment of  is x 'D' then in which

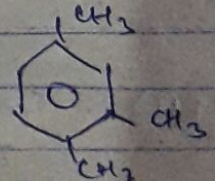
Options, $\mu_{net} \geq x$ 'D'

(A) 

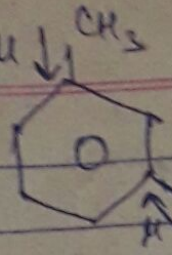
(B) 

(C) 

, $\mu_{net} = 0$

(D) 

Sol^n (A)



$\mu_{net} = \sqrt{\mu^2 + \mu^2 + 2\mu^2 \cos 120^\circ}$

$= \mu$

Q Find % ionic character in H-Cl molecule

$\mu_{observed}(HCl) = 1.00\,D$

H-Cl Bond length = 1.3 A°

Sol^n % ionic character $= \dfrac{1 \times 10^{-18}\ esu - cm}{4.8 \times 10^{-10} \times 1.3 \times 10^{-8}} \times 100$

$= \dfrac{100}{1.3 \times 4.8} \simeq 17\%$ Ans

(*) Hydrogen Bonding

→ It is a special type of dipole-dipole attraction in which one of the pole of is H-atom which is covalently bonded with small sized highly EN atom like F, O, N.

eg. H—F ····· H—F ····· H—F

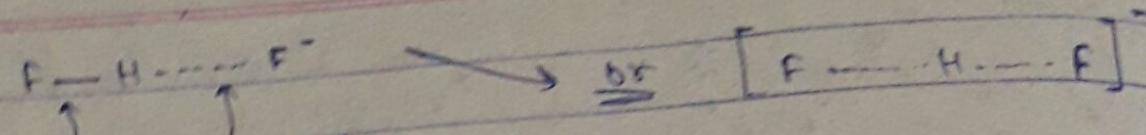
H-bonds

→ In general H-bond (BE in range 8 to 42 kJ/mol) is much weaker than a covalent bond (200 to 450 kJ/mol)

→ In KHF_2 [or $NaHF_2$ etc], H-bond is exceptionally strong.

$KF + HF(l) \longrightarrow K^+[HF_2]^-$

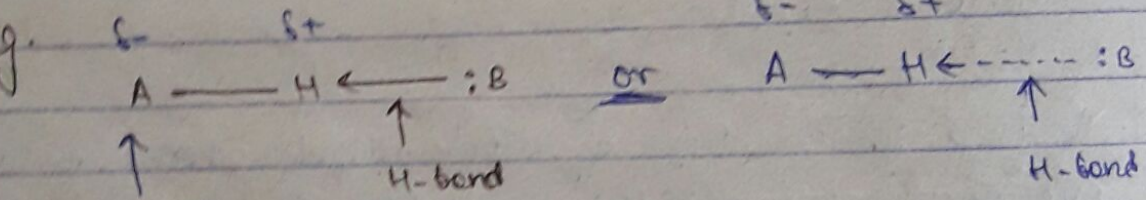
difluoride ion
(or Bifluoride ion)



$x = y = 161.5$ kJ/mol
(found experimentally)

* unlike HF_2^-, HCl_2^-, HBr_2^-, HI_2^- do not exist

⇒ Infact H. bond is the interaction between partially positively charged H atom and a L.P present on an EN atom



V. Imp Strength of H-bond depends on the following two factors

(a) Higher is EN difference, greater will the partial +ve charge on H-atom, greater will be the strength of H-bonding
(DOMINANT FACTOR)

(b) Greater is the ease of donation of L.P to H-atom, greater will be the strength of H-bond

N > O > F
→
decreasing tendency to donate L.P

Q. I. F —— H ····· F (δ+, apm)

II. F —— H ····· O (δ+, bpm)

III. O —— H ····· N (cpm)

IV O —— H ····· O (dpm)

V N —— H ····· O (epm)

Compare a, b, c, d and e (i.e bond length of H-bonds)

Soln

beaceedee

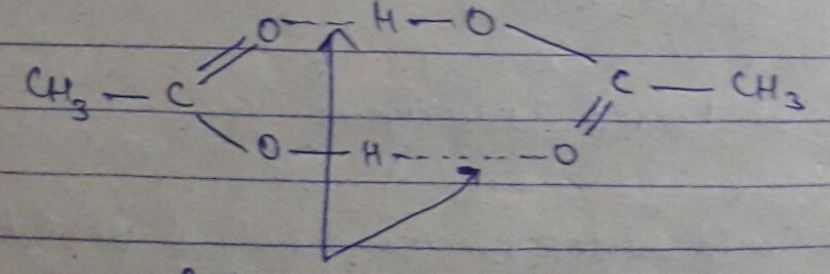
Increasing H-bond length

(*) Types of Hydrogen Bonding.
- Intermolecular H-Bond
- Intramolecular H-Bond

(a) Intermolecular H-Bonding
When two or more molecules are attached through H-bonds.

Eg. H_2O, D_2O, CH_3OH, C_2H_5OH, RNH_2, R_2NH, RCOOH etc
HF, NH_3 etc.

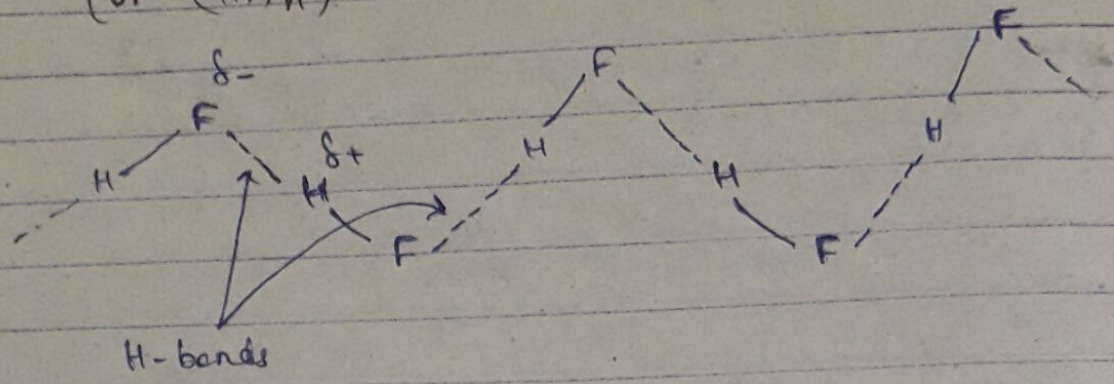
Eg. dimer formation of CH_3COOH or C_6H_5COOH in non-polar solvent through H-bonding



Intermolecular H-bonds

Fig. Dimer of CH_3COOH

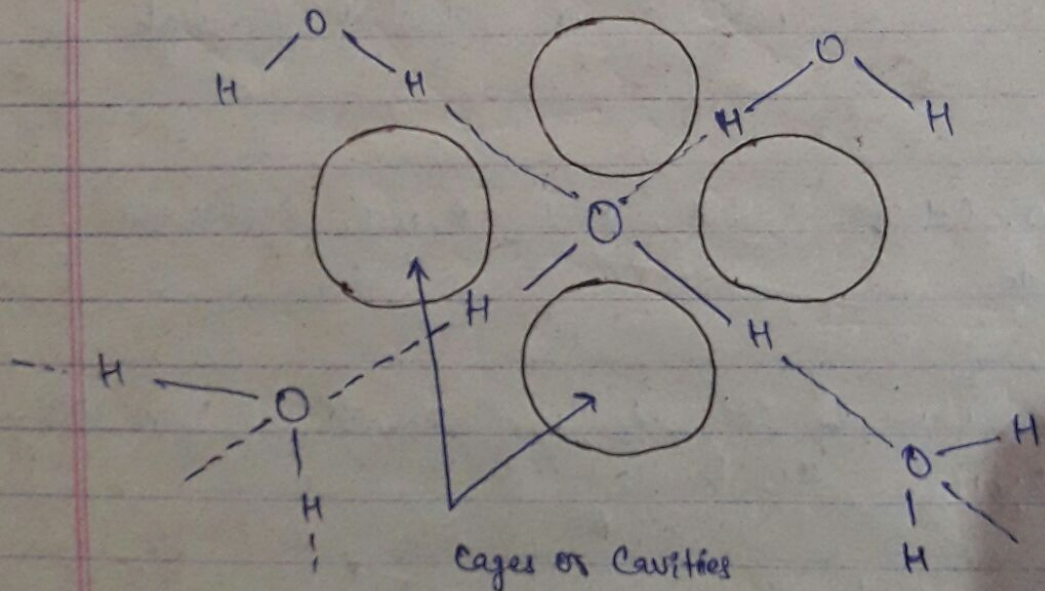
eg. zig-zag chain like H-bonded structure of solid HF
(or $(HF)_n$)



⇒ Solid > Liquid > Gas

decreasing extent of H-Bonding in a Substance

V. Imp → Cage like H-bonded structure of ice
(3D- Tetrahedral structure of ice)



3D network H-bonded structure of ice.

⇒ Ice floats over liquid water because density of ice < density of water which is due to cages / cavities in extensively H-bonded ice.

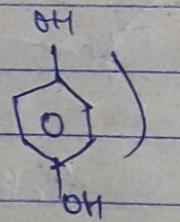
⇒ When ice is melted by giving heat, no. of cages/cavities decrease due to breakage of H-bonds and density increases upto 4°c water. Thereafter density of water ↓ as thermal expansion occurs.

⇒ Explain why $D_2O(s)$ floats over $D_2O(l)$ but $D_2O(s)$ sinks in $H_2O(l)$.

Reason: $EN_D < EN_H$, therefore $\Delta(EN)$ between D and O in D_2O is greater and hence stronger H-bonds (or infact D-bond in D_2O than in H_2O

∴ $D_2O(s)$ has greater density than that of $H_2O(l)$ and hence $D_2O(s)$ sinks in $H_2O(l)$

→ Reason why $D_2O(s)$ floats over $D_2O(l)$ is same as why $H_2O(s)$ floats over $H_2O(l)$

→ Ice floats and solid quinols  form clathrate with gases like Xe, Kr, Br_2, Cl_2 etc.

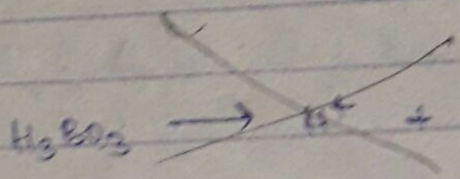
↳ Clathrate is the entrapment of appropriately size gas molecu into the cavities/voids in solid ice and solid quinol.

→ There is a weak dispersion force (or london forces) between trapped gases and ice/quinol.

→ He/Ne gas atoms are not trapped in cages of ice/quinol due to small size of gas.

eg. Boric acid is solid white powder under room condition due to H-bonded layered structured.

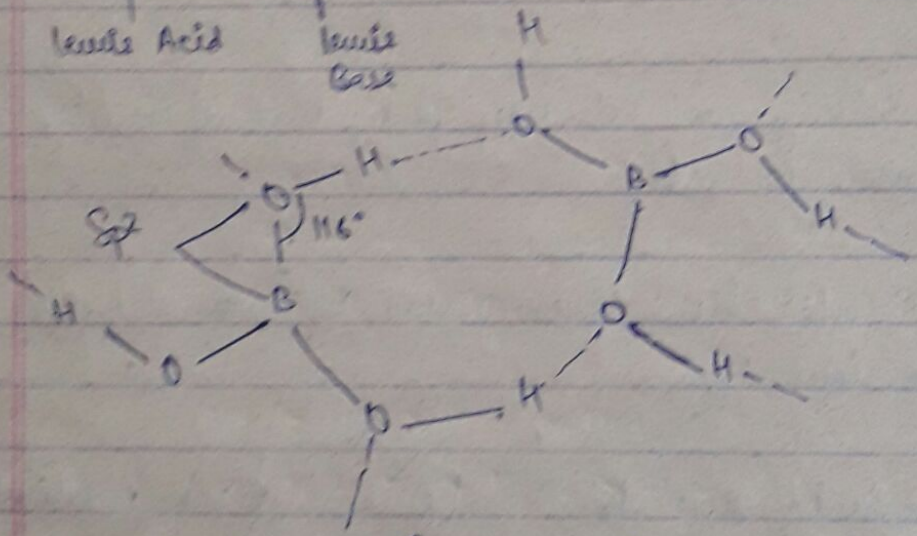
$B(OH)_3$ or H_3BO_3 → acts as a lewis acid (Monobasic acid) and it is not a protonic acid.



$$H_3BO_3 \longrightarrow H^+ + H_2BO_3^-$$

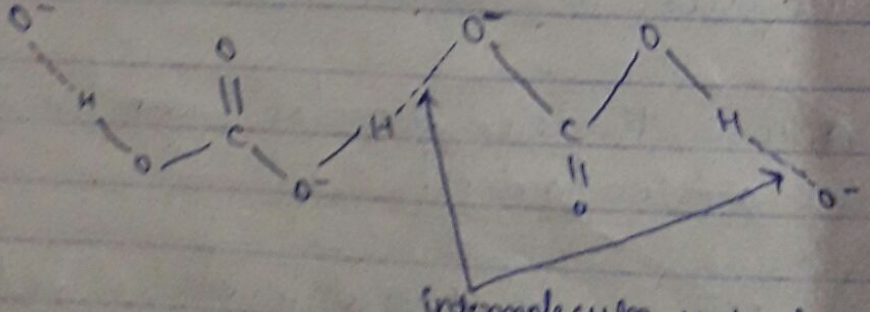
$$B(OH)_3 + OH^- \longrightarrow B(OH)_4^-$$

↑ Lewis Acid ↑ Lewis Base



Intermolecular H-Bond

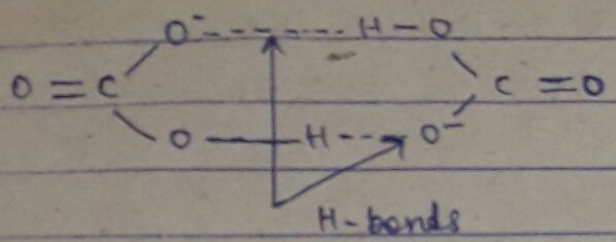
eg. $NaHCO_3$ → HCO_3^- units are H-bonded to form a chain like structure.



Intermolecular H-bonds

eg. $KHCO_3$

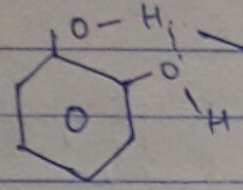
↑ HCO_3^- units form dimer (H-bonded)



(b) Intramolecular H-Bonding :-

→ It is the H-bonding within the same molecule

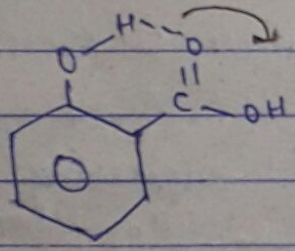
eg.



→ intramolecular H-Bond.

Catechol.

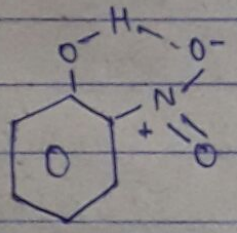
eg.



→ intramolecular H-Bond

Salicyclic Acid

eg.

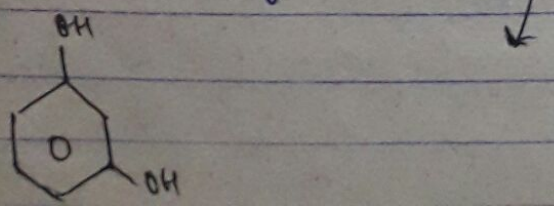


Nitrophenol

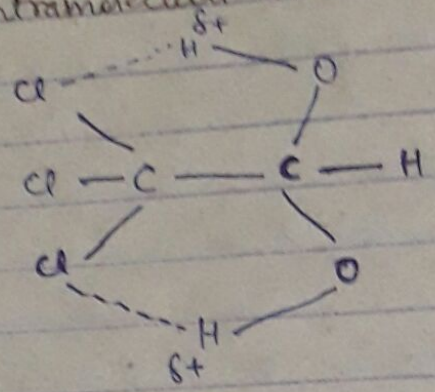
⇒ Resorcinol and quinol do not show intramolecular H-Bond.

(m-isomer) (p-isomer)

Rather they may show intermolecular H-bonding.



eg. Intramolecular H-Bonding in Chloral hydrate ($CCl_3 \cdot CH(OH)_2$)



Two intramolecular H-bonds per molecule.

(*) Applications of H-bond :

H-bond affects physical/chemical properties mainly B.P, viscosity, ΔH_{vap} etc.

⇒ Intermolecular H-Bonding increases

→ B.P
→ ΔH_{vap}
→ Viscosity.

→ Solubility of compound form H-bond with solvent.
eg. CH_3OH, CH_3COOH, C_2H_5OH etc.

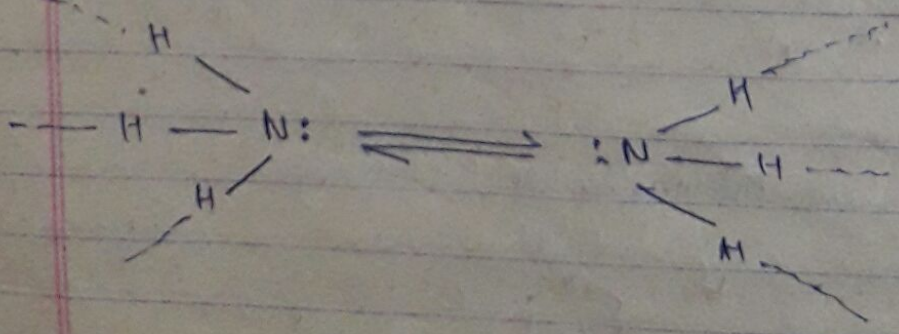
V. Imp

eg. $HF > HI > HBr > HCl$
$H_2O > H_2Te > H_2Se > H_2S$
$SbH_3 > NH_3 > AsH_3 > PH_3$

} decreasing Boiling Point or ΔH_{vap}.

Stibene — Arsene — Phosphene

B.P of NH_3 is not highest among hydrides of pnictogens due to flipping.



Eg CH_3OH, C_2H_5OH, CH_3COOH, $HCOOH$ etc. are highly soluble in water as they all form H-bond with water.

$CH_3COOH > C_2H_5OH > CH_3CHO$ $\Rightarrow$ $CH_3-C(=O)-H$

decreasing B.P

greater extent of H-Bonding.

no H-bonding in act acetyldehyde

→ Wooden Sticks are used (& not the plastic sticks) for icecream bars / kulfi

Reason: wood contain cellulose fwhich has extensive -OH linkages which form H-bond with water Concent/ice of icecream and holds the icecream longer.

→ $H_2SO_4(l)$, $H_3PO_4(l)$, $H_2O_2(l)$, $HNO_3(l)$ etc are more ~~viscous than water~~. why?

Reason: Extensive H-Bonding in H_2SO_4, H_3PO_4, H_2O_2, $HNO_3(l)$ etc (greater than that in water)

→ Glycerol is highly viscous. why?

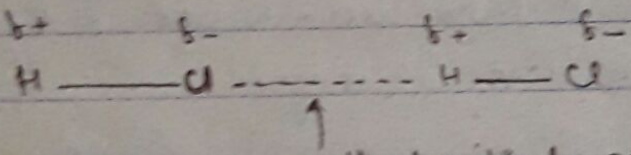
CH_2-OH
$CH-OH$ } Extensive H-Bonded.
CH_2-OH

* Various types of intermolecules forces :
(weak forces of a interaction, also known as Vanderwaal's forces)

(1.) Dipole - Dipole attraction :-

→ This is the interaction between polar molecules
→ Interaction energy (or Potential Energy $\propto \frac{1}{r^3}$

→ operate between permanent dipoles (or polar molecules like HCl, HBr, H_2S, PH_3 etc)

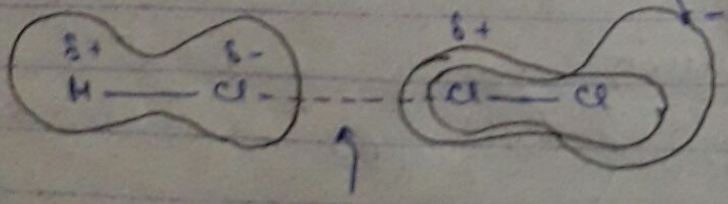


→ H-Bond is the Special (stronger) type of dipole - dipole attraction.
eg. HF, NH_3, H_2O, ROH, RNH_2 etc.

(2) Dipole - induced dipole attraction :

→ operates between a polar and non-polar molecules.

eg. $Cl_2(g) + HCl(g)$

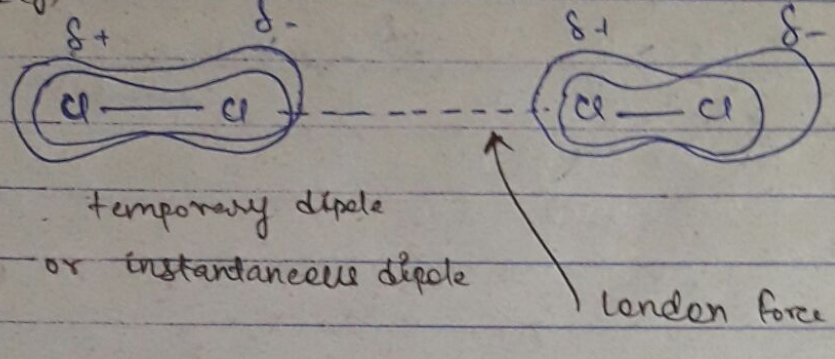


→ P.E $\propto \frac{1}{r^6}$ { dipole - induced dipole attⁿ. very weak forces of }

imp (3) Instantaneous dipole - induced dipole att^n :
(also known as london forces or dispersion forces)

→ operate between non-polar molecules / atoms
eg. Cl_2, Br_2, I_2, F_2, He, Ne, Ar, etc.

eg. $Cl_2(g)$



→ P.E $\propto \frac{1}{r^6}$ (very weak)

imp → Dispersion forces always accompany other types of bonds (covalent bond / ionic bond / metallic bond) and other intermolecular forces.

→ Dispersion forces are due to polarisation of e^- cloud (omni present)

→ Extent of polarisation ↑, Strength of london forces ↑

* In general, higher is the molar mass, greater will be the extent of polarisation of e^- cloud and hence stronger are the london forces.

$$He < Ne < Ar < Kr < Xe < Rn \longrightarrow$$

increasing london forces.

$\Rightarrow$ $F_2 < Cl_2 < Br_2 < I_2$ → solid (at room temp.)

gas ← increasing London forces.

↓ liquid

* NOTE : (Exception)

$H_2 > He$ } Remember !

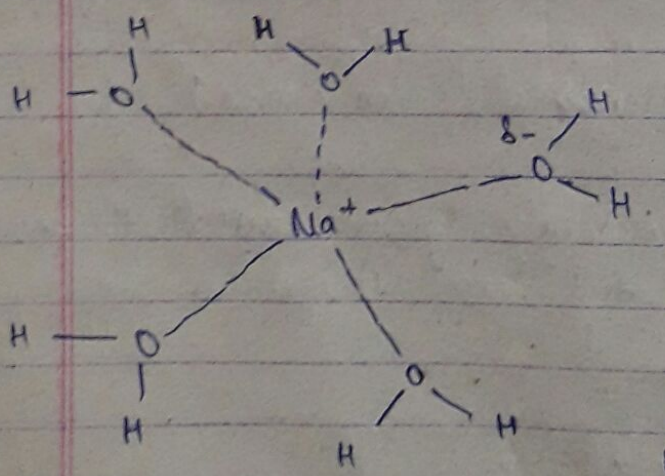
↑ greater extent of dispersion forces

* Interaction between ionic compounds and covalent compounds:
(Not to call them vander waal's forces)

(1) Ion-dipole Attraction :

→ It is the interaction between ion and a polar molecule (or permanent dipole)

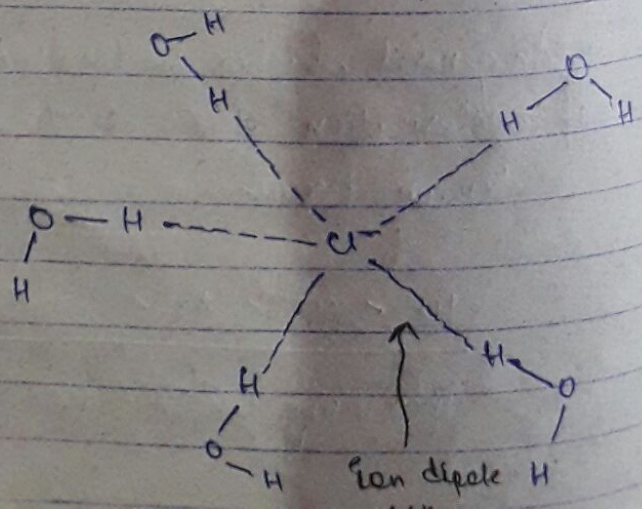
eg. $NaCl(s) + H_2O(l) \longrightarrow [Na(H_2O)_x]^+_{(aq)} + [Cl(H_2O)_y]^-_{(aq)}$



Hydration of ion

OR

$Na^+(aq)$ $Cl^-(aq)$



→ P.E $\propto \frac{1}{r^2}$

imp (2) Ion-induced dipole attⁿ :-

→ It is the interaction between ion and a non-polar molecule

eg. $KI + I_2 \longrightarrow KI_3$

or

$I^- + I_2 \longrightarrow I_3^-$

ion — non-polar molecule

$I_3^- > Br_3^- > Cl_3^-$

decreasing (Stability order)

* I_2 undergoes greater extent of polarisation due to larger size by I^-

→ P.E $\propto \frac{1}{r^4}$

(*) METALLIC BOND —
(Observed in Metals and in alloys)

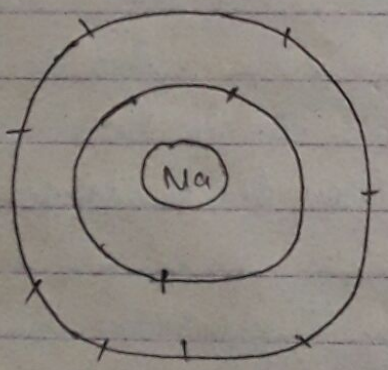
(a) Electron Sea Model (oversimplified approach)

→ According to this approach, metal kernels occupy lattice position in crystal structure of metal and are surrounded by electron gas (or sea of electrons)

Metallic bond is the electrostatic force of attraction between metal kernels and valence electrons (or free electron or mobile electrons)

eg. Sodium Metal, Na : $1s^2, 2s^2, 2p^6, 3s^1$ ← valence e^-

↑ experiences low Z_{eff}. and hence easily delocalised.



Metal kernal (Na^+) (or atomic core)

eg. Magnesium Metal.
↳ Two valence e^- per metal atom.

eg. Al metal
↳ 3 valence e^- per metal atom.

→ Strength of Metallic Bond

$$\propto \text{no. of valence } e^- \text{ per metal atom}$$

$$\propto \frac{1}{\text{size of metal}}$$

→ Particularly in case of d-block metals, unpaired 'd' electrons also take part in metallic bond formation.

→ In general, greater are the no. of unpaired 'd' e^- in 'd' block metals, greater will be the strength of metallic bonds.

→ Many mechanical properties of Metal can be related with Strength of metallic bond eg. Melting point, Hardness.

Melting point of Metal & Hardness $\propto$ Strength of Metallic Bond.

→ $Li > Na > K > Rb > Cs$

$Be > Mg > Ca > Sr > Ba$

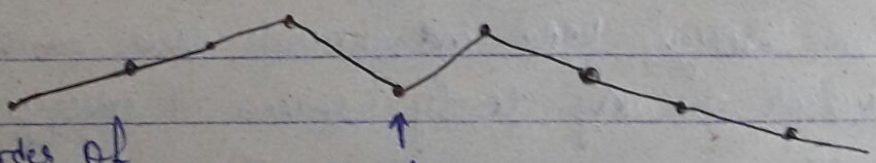
$Al > Mg > Na$

} decreasing Strength of metallic Bond, and therefore decreasing order of Hardness and M.P.

→ Alkaline earth metals are harder and have higher M.P than the corresponding alkali metals.

→ In 3d Series (Sc to Zn), Melting point first increase, reach max in middle and then decreases.

Sc, Ti V Cr Mn Fe Co Ni Cu Zn



Order of Melting Point.

Mn is Exception

→ Zn, Cd, Hg have very low M.P (relative to other d-block metals) as they do not have any unpaired 'd' e^-. That is the reason why Zn, Cd, Hg are called Volatile metals

(concept is used in Metallurgy)

Imp → Cr has highest M.P in 3d series (and not Mn)
lower M.P of Mn (than Cr and Fe) is probable due to stable electronic conf^{n} ($3d^5, 4s^2$) which resist in delocalisation

→ Metals show good electrical conductivity due to the present of valence (or free e^-) which move (delocalise) under applied electric field. Ag (among metals) is the best conductor of electricity.

→ Metals show metallic lusture (silvery white surface when freshly cut) due to absorption of visible light by valence electrons on surface.
lusture is due to de-excitation of e^- and emit light

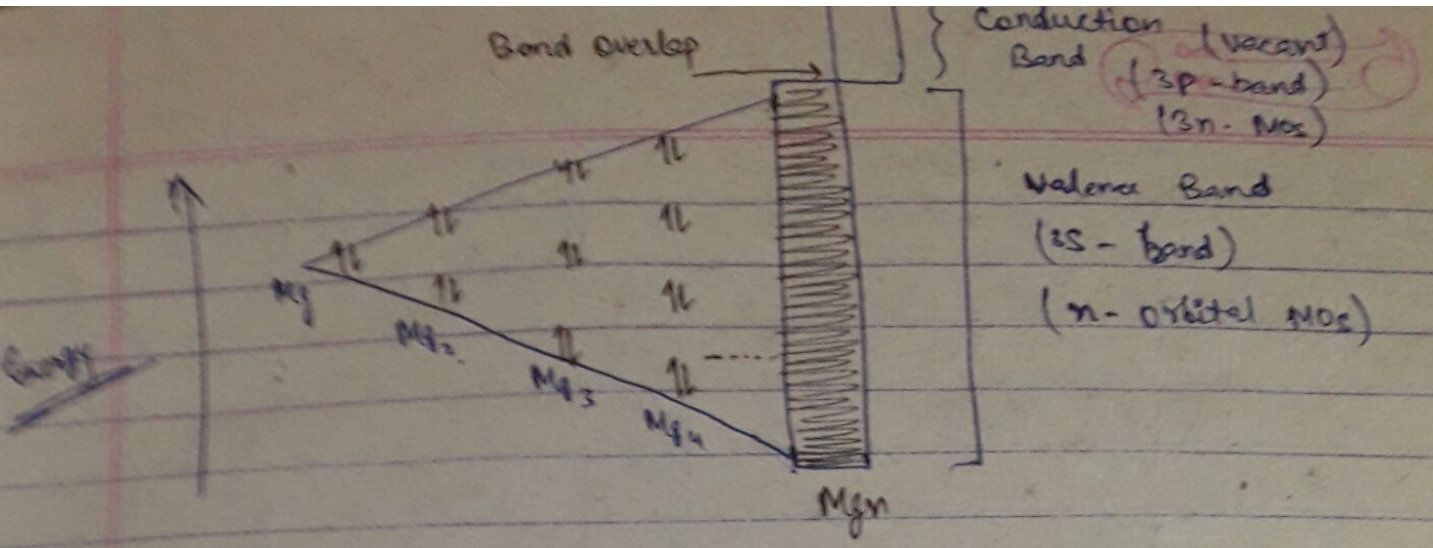
→ Metals show ductility and malleability due to the presence of 'slip planes'.

② Band Model (based on MOT)

→ In a metal, a large number of atoms of metals undergo combination through LCAO and generate equal no. of molecular orbitals which are very closely spaced in energy.

eg. Mg metal, $_{12}Mg : 1s^2, 2s^2, 2p^6, \underbrace{3s^2, 3p^0}_{\text{valence shell}}$

eg. Mg_n where $n \rightarrow$ very large

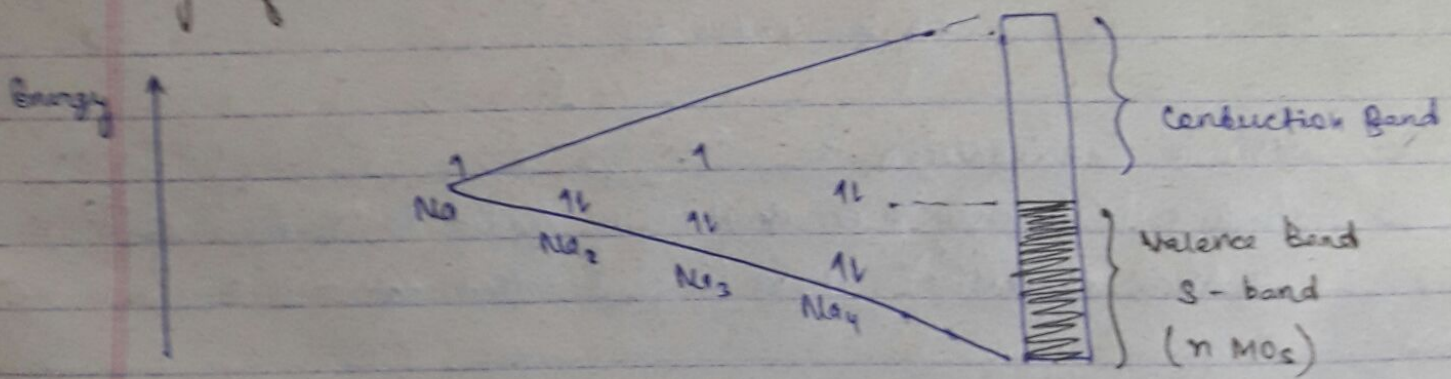


→ In metals (conductors), band either overlap or there is infinite simally small energy difference in valence band and conduction band (or infinite small band gap)

→ Band gap (or known as forbidden energy difference) is the energy difference between Valence band, and conduction Band.

Eg. of Sodium Metal, Na_n

$Na: [Ne]\ 3s^1\ 3p^0$



→ Based on band gap, we define, conductors, insulators, Semi Conductors.

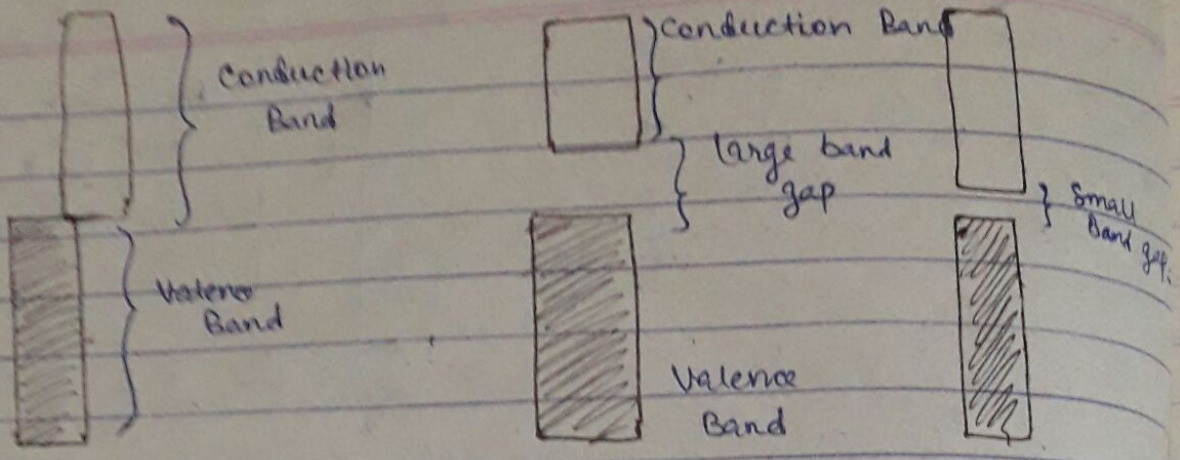
Small band gap (< 1.1 ev)

Si : Gap = 1.1 evolt

Ge : Gap = 0.7 evolt

Infinitesimally small band gap or Bonds overlap

large Band gap (> 3ev)
eg. diamond, phosphorus etc.



→ At low temperature, conductivity of intrinsic Semiconductors (eg. Pure Si, Ge) is low but on heating e^- easily jump from valency band to conduction band, and hence

T↑, Electrical Conductivity ↑
for semiconductors.

→ Extrinsic Semiconductor :
Impure (impurity is deliberately added to improve their electrical properties) Si or Ge are called extrinsic Semiconductors.

eg n-type
p-type

⊛ Hydrolysis :

⇒ Hydration : It is cluster formation of water molecules around an ion through ion-dipole attractions.

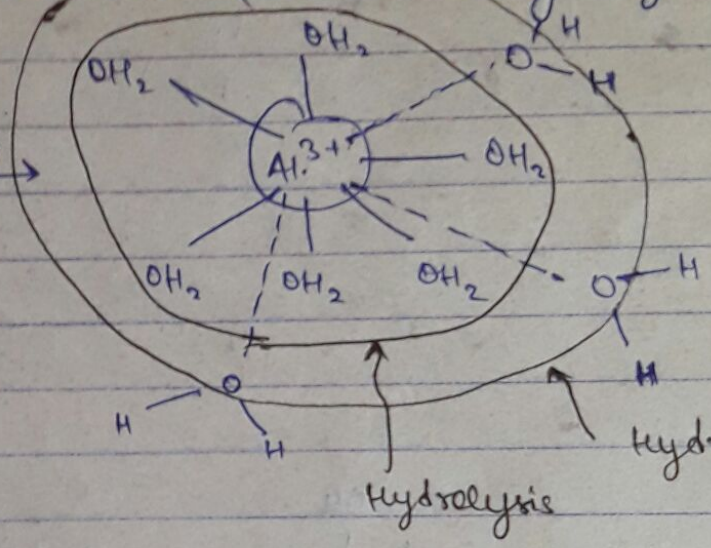
→ In general, ionic compounds undergo mainly hydration (though they may undergo hydrolysis too)

⇒ **Hydrolysis:** Chemical Rkn. with water to form covalent / coordinate Bond.

→ Covalent compounds usually undergo hydrolysis along with a little hydration.

eg. LiCl, NaCl, $AlCl_3$ → They mainly under hydration in water. remain as hydrated ions (But under hydrolysis too to a contain extent)

eg. $AlCl_3$ $\xrightarrow[\text{(excess)}]{+H_2O(l)}$



These cations exist as

$[Li(H_2O)_4]^+_{(aq)}$, $[Be(H_2O)_4]^{2+}_{(aq)}$, $[Al(H_2O)_6]^{3+}_{(aq)}$

→ Al^{3+} makes acidic solution.

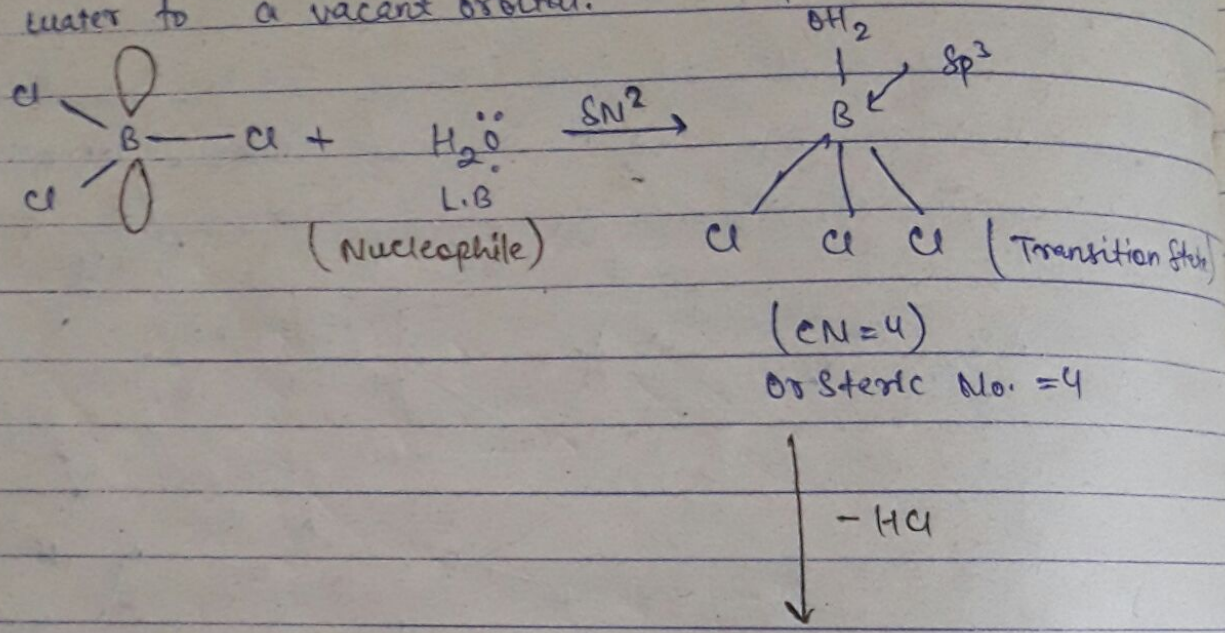
$$[Al(H_2O)_6]^{3+} + H_2O(l) \rightleftharpoons [Al(H_2O)_5(OH^-)]^{2+} + H_3O^+$$

Solution is (acidic)

(x) Hydrolysis of Covalent Halides :

eg. Hydrolysis of BCl_3 :

↳ Hydrolysis occurs as a nucleophile (Lewis Base) attack of water to a vacant orbital.



$$BCl_3 + 3H_2O(l) \longrightarrow B(OH)_3 + 3HCl(aq)$$

(Solution becomes strongly acidic)

Boric acid

$$BX_3 + H_2O(l) \longrightarrow H_3BO_3 + HX$$

[X → Cl/Br/I]

* Note : BF_3 undergoes partial hydrolysis.

$BF_3 + 3H_2O(l) \rightleftharpoons B(O$

$BF_3 + 3H_2O(l) \longrightarrow B(OH)_3 + 3HF$

$3BF_3 + 3HF \longrightarrow 3HBF_4$ (Not a hydrolysis rxn.)

fluoroboric acid

$4BF_3 + 3H_2O(l) \longrightarrow B(OH)_3 + 3HBF_4$

1 : 3

Molar Ratio

* A small amount of $B(OH)_4^-$ is also present

$B(OH)_3 + H_2O(l) \longrightarrow H^+ B(OH)_4^-$

⇒ $BCl_3 + HCl \longrightarrow HBCl_4$ is unstable as size of Cl atom is large and 'B' (small size) can't accomodate 4 Cl atoms { Steric Hindrance }

→ CCl_4 does not undergo hydrolysis at room temperature.

$CCl_4 + H_2O(l) \xrightarrow{R.T} \text{No. Rxn.}$

↑

No vacant orbital on 'C' in valence shell

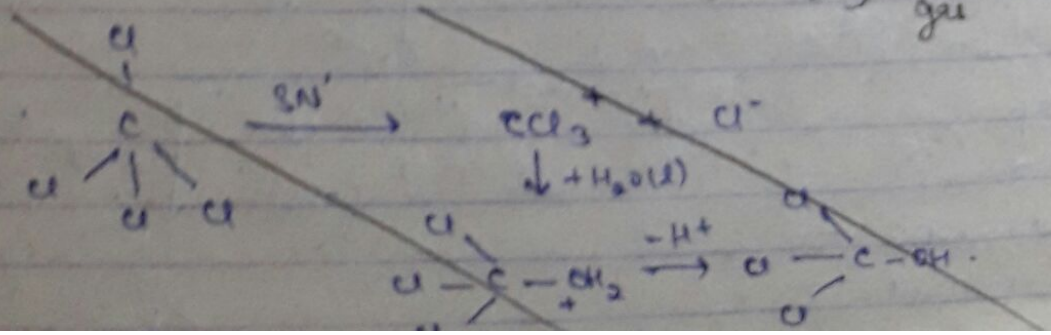
⇒ However, $CCl_4 + H_2O(g) \xrightarrow{\approx 500^\circ C} COCl_2 + 2HCl$

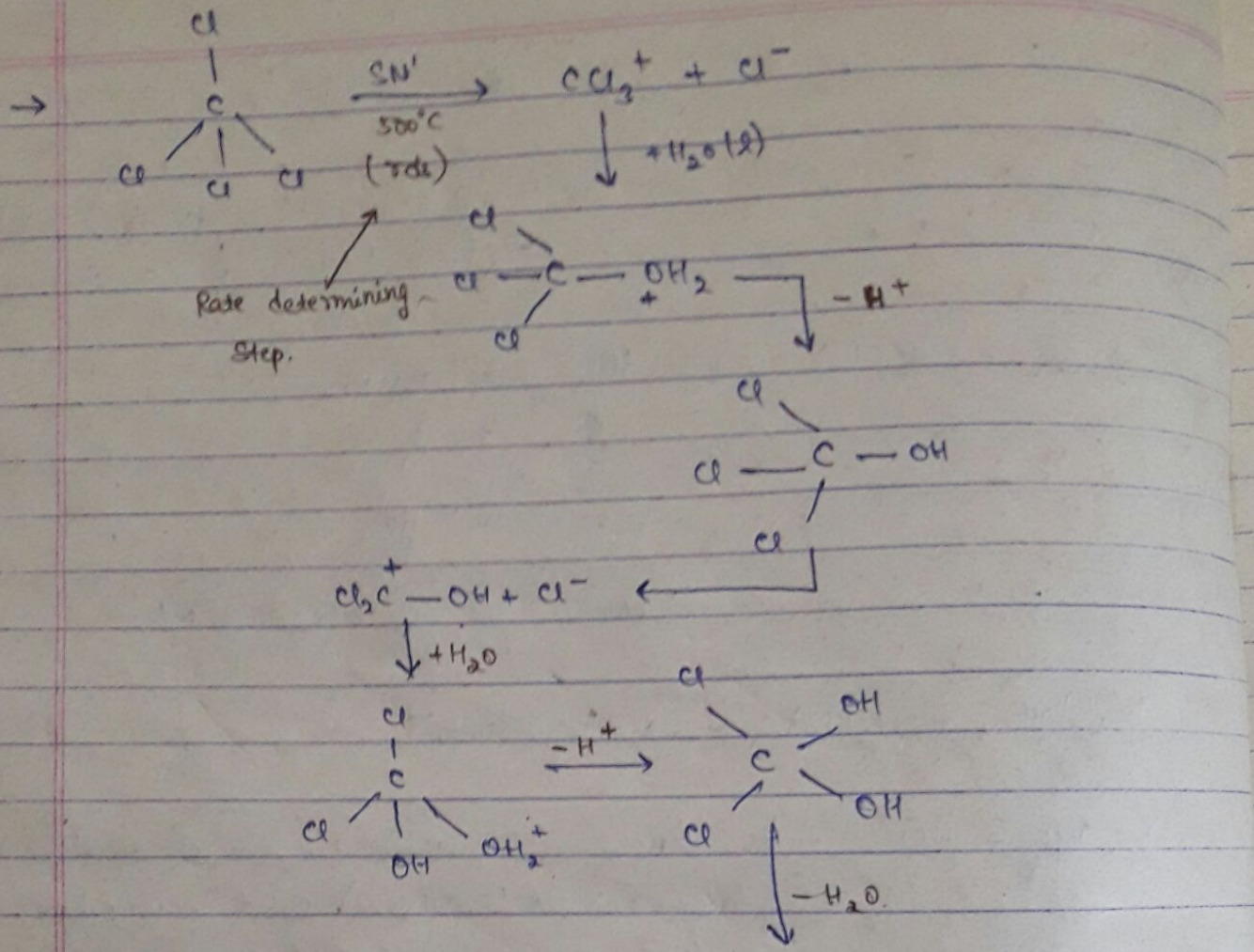
(Steam)

↑

Phosgene (a highly poisonous gas)

→



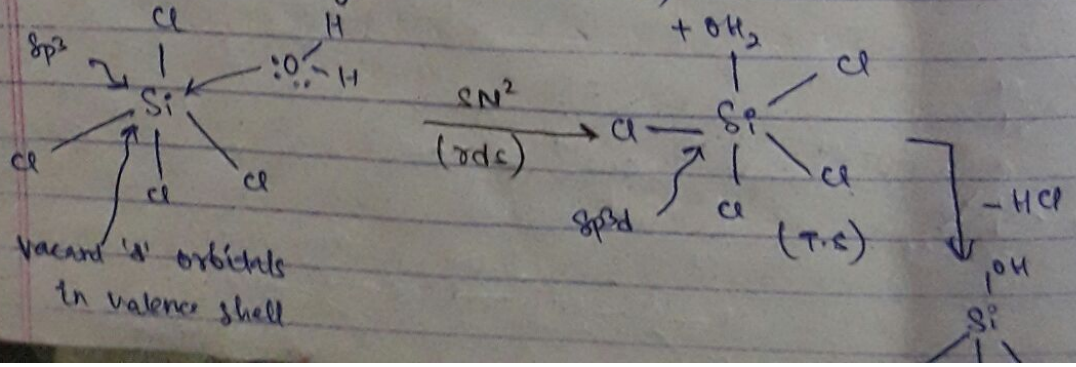


→ $SiCl_4 + 4H_2O(l) \longrightarrow Si(OH)_4 + 4HCl$

$\downarrow -H_2O(l)$

H_2SiO_3 (Metasilicic acid)
(exist in polymeric form)

Mechanism (Through SN^2 mechanism)



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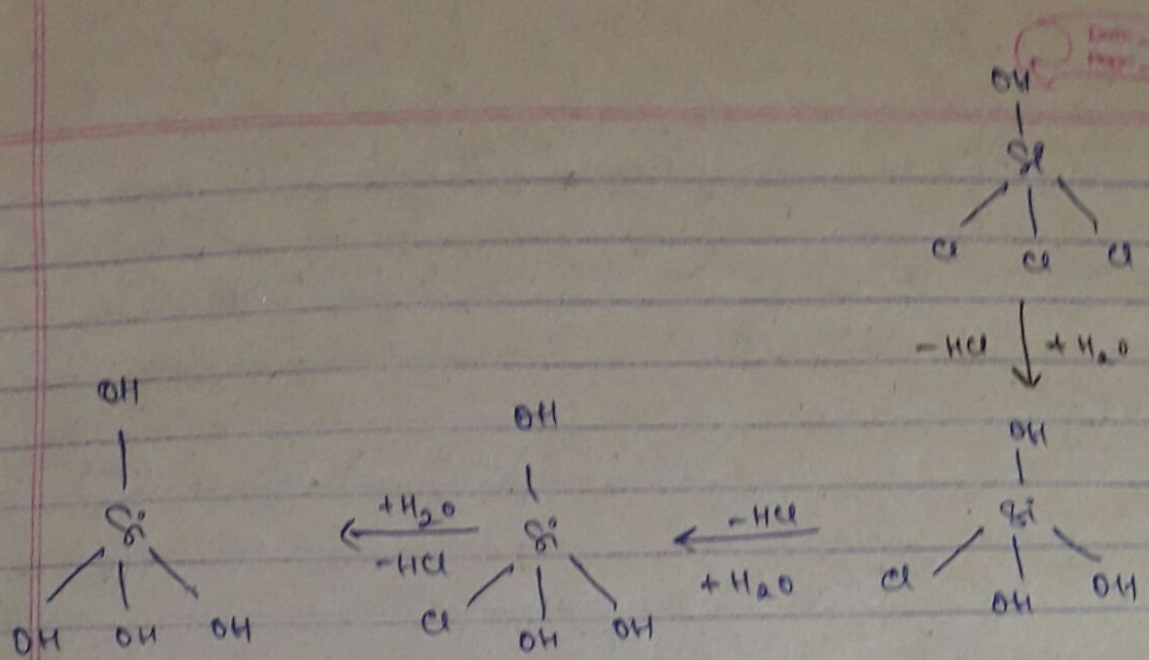


Silicic Acid

→ SiF_4 undergoes partial hydrolysis.

$$SiF_4 + 4H_2O(l) \longrightarrow Si(OH)_4 + 4HF$$
$$SiF_4 + 2HF \longrightarrow H_2SiF_6 \quad \times (2)$$
(Fluorosilicic Acid)

$$3SiF_4 + 4H_2O \longrightarrow Si(OH)_4 + 2H_2SiF_6$$
$$Si(OH)_4 \xrightarrow{-H_2O} H_2SiO_3$$

H_2SiF_6 : H_2SiO_3 = 1 : 2

only 33% of SiF_4 undergoes hydrolysis.

(+4)

⇒ $SnCl_4 + H_2O(l) \longrightarrow H_2SnO_3 + HCl$
Metastannic Acid

V. Imp ⇒ $SnCl_2 + H_2O(l) \longrightarrow Sn(OH)Cl + HCl$
Cloudy Solution
(Partial Hydrolysis)

⇒ $Sn(OH)Cl$ is partially soluble in water.

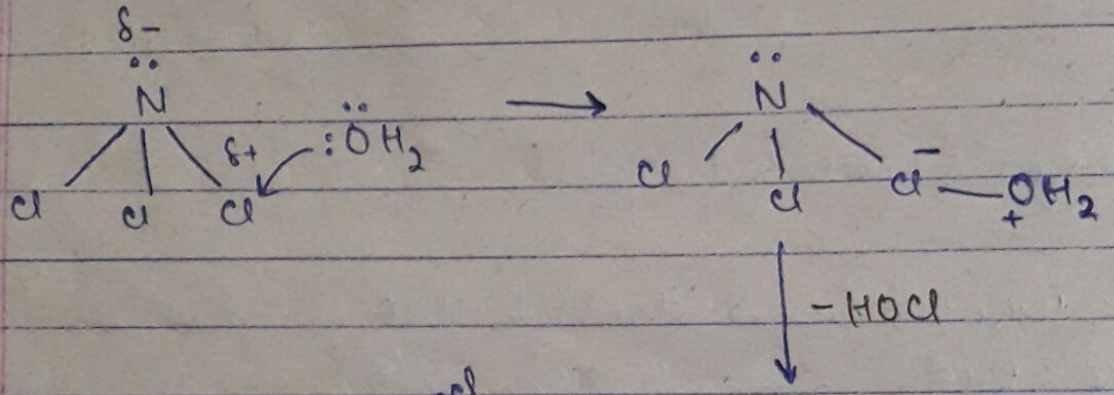
⇒ $PbCl_4 + H_2O(l) \longrightarrow PbO_2 \downarrow + HCl\,(aq)$

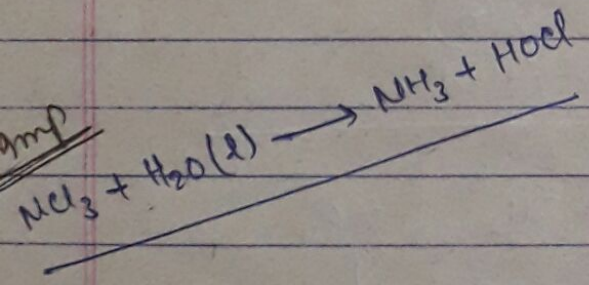
Hydrolysis $\ddot{N}Cl_3$:

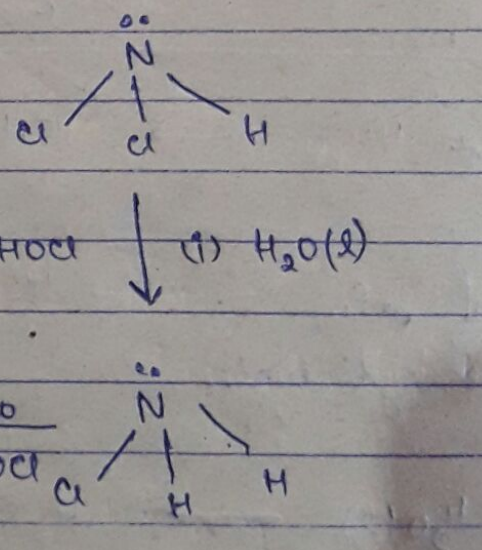
$$\ddot{N}Cl_3 + H_2O(l) \longrightarrow NH_3 + HOCl$$

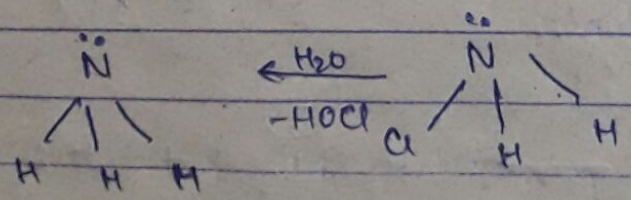
↑ 'N' has no vacant orbital in valence shell.

↑ Hypochlorous acid









→ $\ddot{N}F_3$ does not hydrolyse at room temperature.

→ $\overset{(+3)}{NF_3} + H_2O(g)$ (Steam) $\xrightarrow{300^\circ C}$ $\overset{(+3)}{N_2O_3} + HF$

$\downarrow$

$NO + NO_2$

→ $PCl_5 + H_2O(l) \xrightarrow{SN_1}$ $\overset{(+3)}{H_3PO_3}$ + HCl
Phosphorous acid

→ $PCl_5 + H_2O(l) \xrightarrow{SN_2}$ $\overset{(+5)}{H_3PO_4}$ + HCl
orthophosphoric acid

* PCl_5 + moisture ($H_2O(l)$) (limited) → $POCl_3$ + HCl
(partial hydrolysis)
$+ H_2O(l)$ ↓
H_3PO_4 + HCl

→ $AsCl_3 + H_2O(l) \longrightarrow \overset{(+3)}{H_3AsO_3} + HCl$

3mf → $SbCl_3 + H_2O(l) \longrightarrow Sb^+Cl^-\downarrow + HCl$
white turbidity

3mf → $BiCl_3 + H_2O(l) \longrightarrow Bi^+Cl^-\downarrow + HCl$
white turbidity

→ $\overset{(+2)}{SF_2} + H_2O(l) \longrightarrow S\downarrow + SO_2 + HF$

→ $\overset{(+2)}{SCl_2} + H_2O(l) \longrightarrow S\downarrow + SO_2 + HF$

→ $\overset{(+1)}{S_2Cl_2} + H_2O(l) \longrightarrow S\downarrow + SO_2 + HF$

→ $\overset{(+4)}{SF_4} + H_2O(l) \longrightarrow H_2SO_3 + HF$
unstable
↓
$SO_2\uparrow + H_2O$

V. Imp → SF_6 is kinetically inert and does not react with water.

$SF_6 + H_2O(l) \longrightarrow$ No rxn.

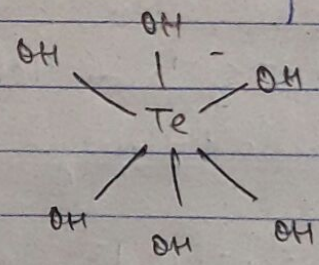
'S' is sterically crowded by 6F atoms and H_2O is not able to approach it.

→ $SeF_4 + H_2O(l) \longrightarrow \overset{(+4)}{H_2SeO_3} + HF$

→ $SeF_6 + H_2O(l) \longrightarrow H_2SeO_4 + HF$

→ $TeF_6 + H_2O(l) \longrightarrow Te(OH)_6 + HF$

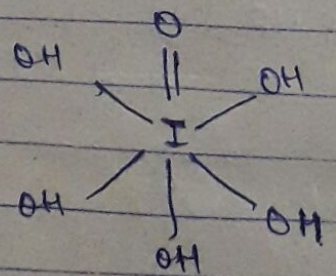
↑ large size



→ $\overset{(+3)}{ClF_3} + H_2O(l) \longrightarrow HF + \overset{(+3)}{HClO_2}$

↑ oxyacid for large halogen.

→ $IF_7 + H_2O(l) \longrightarrow HF + \overset{(+7)}{HIO_4}$ $\xrightarrow{+2H_2O}$ $HIO_4 \cdot 2H_2O$ or H_5IO_6



Octahedral.

→ $XeF_2 + H_2O(l) \longrightarrow Xe(g) + O_2(g) + HF$

→ $XeF_4 + H_2O(l) \xrightarrow{\text{disproportionation}} Xe(g) + XeO_3(s) + O_2 + HF$

} Redox

→ $XeF_6 + H_2O(l) \longrightarrow XeO_3 + HF$ (Not a redox Rkn.)

THE END